

What is Reflective Practice?

Reflective practice is an ongoing and dynamic process for the teacher. It involves thinking critically and deeply about what is happening in the class room. It occurs spontaneously, immediately but, also goes in a very planned and structured way. Most importantly, reflection leads to action. Teachers need to value reflection and evaluation of their own practice and their own professional role in the class room. In addition, teachers' professional knowledge and understanding includes ways to reflect and ways to evaluate what is happening. This is part of the professional role of the teacher.

Teaching is a profession and as such in a professional setting reflection is very **deliberate**. It is also **purposeful**. There is a reason for it. It is **structured** and it **links our understanding to theory to practice** and also includes deep learning and not superficial learning. Therefore, reflective practices aimed at development becoming a reflective practitioner is becoming an excellent teacher.

What are the key words that we can take from this message?

- **Mental processing** (reflection is structured thoughtful, it is not superficial)
- **Better understanding** (we gain a clearer, thoughtful understanding of what is happening in the classroom and reflection includes the teacher in the process of teaching)

Reflective practice encourages the teachers to understand their:

- learners,
- their needs and
- their abilities.

Why Reflect?

Reflective practice leads to

- Higher quality teaching and therefore better outcomes for learners.
- Awareness for the teachers of the values of teaching and their own beliefs about teaching.
- Challenge and to change
- More inclusive environments for learners.
- Improvement
- promotes collaboration between teachers as professionals.

Why is reflective practice so important?

Research shows that reflective practice is crucial for teacher's learning and for extending their skills as professionals. It is a highly personal process for any teacher and the teacher requires the professional to take their daily class room experiences to internalize, understand them to pull them over in their mind. In this way, it is possible for the teacher to filter through what is happened to focus on what is most important, to understand their personal values and their biases as teachers before deciding how to better move forward as a practitioner.

Research also shows that teachers, who regularly reflect on their teaching, reflect on why they do things and reflect on new knowledge which is being created, use this to achieve the best outcomes for their students.

Reflective practice is a process by which you stop and think and you consciously analyze the decisions you make.

Critical analysis and evaluation for the teachers refocuses the thinking of the teachers and helps to generate the knowledge needed for good quality teaching. We can represent this visually:

- subject knowledge,
- teaching skills,
- interpersonal relationships in the school as well as
- their own personality with a focus on research and teaching.

What Is to Be Gained from Reflective Practice?

Clearly, reflective practice means best practice. It provides a real shift in focus for the teacher away from deterministic views of teaching, a shift in the power of teaching. It creates a general dialogue about what is teaching and learning.

The benefits of reflective practice

Most importantly, reflection helps us as teachers to develop our own learning and provides a teaching framework. Stephen Brookfield (1995) explained this and proposed the following framework. According to him, reflective practice

- Allows us to consciously develop our own repertoire of strategies and techniques in the classroom. We can draw upon our teaching and use this in the context and discipline of teaching.
- Helps us to take informed decisions in the classroom to take the actions which are meaningful and justified and be able to explain these to others.
- Helps us to adjust our teaching, respond to problems and issues which we have faced as teachers in the classroom. For example, rather than just reacting to a poor evaluation, we investigate and try to understand the underlying reasons for a poor evaluation and use this to take appropriate action.
- Helps us to become aware of the underlying beliefs and assumptions that we have as teachers about teaching and learning and as a consequence we can try to understand what we do and why we do.

- Helps to promote a positive learning environment. Through reflection, our teaching becomes much more responsive to the students and their feedback and through this we get good quality learning.
- Helps us to locate our teaching in the broader institutional and even social context in which teaching and learning takes place.

Advantages and Drawbacks of Reflective Practice

The key principles in relation to reflective practice and the drawbacks associated with it are:

- Lack of conceptual clarity
- Too much focus on the individual
- Failure to acknowledge the discussion, dialogue (discourse) around reflective practice in teaching.
- Failure to understand the complexities involved in teaching in the classroom.

Consequently, the concept of RP is abstract.

- It is ill-defined.
- It is open to multiple interpretations.
- It is applied in a wide variety of ways in different practice environment.

Why Is Reflective Practice Important?

Reflecting critically allows us to synthesize different perspectives to help, explain or to justify and challenge what we have encountered in our teaching. RP may provide evidence to support our

views or our practices or it may even challenge them. Critical reflection also allows us to analyze what we have learned and how we have learned it. Most importantly, it enables us to take control of our own development. It is in light of these two functions of reflective practice that a great deal of importance is now placed on critical reflection in the process of professional development for teachers.

Why it is important?

The passage discusses the concept of reflective teaching, which involves more than just casually thinking or talking about teaching experiences. While many teachers already reflect on their lessons and student behavior, this reflection can often be superficial, focusing only on immediate reactions. Reflective teaching requires a more systematic approach, involving the collection, recording, and analysis of thoughts, observations, and student feedback. This process helps teachers understand the reasons behind their successes and challenges in the classroom, allowing them to make informed changes to improve their teaching practices. Reflective teaching is crucial for continuous professional development and competence..

How RP can develop your work further

- Address student's needs
- Working with colleagues
- Your role with your students
- Planning students' learning
- Working with external agencies/parents

Principles of RP

1. **RP is problem-oriented** or in other words problem-posing or problem-solving.

2. **RP is based on action research.** Action research is a very powerful tool for change and improvement based on a cyclical process of thinking, planning and doing.
3. **RP ensures that teaching and learning are research-led**
4. **RP is data driven.** This means that there is ownership by the teacher of the data which is central to the teacher professional development.

3 specific types of reflection

1. **Technical rationality:** (Behavior and skills) This is essentially the practical skills that the teacher brings to teaching the students. It is the behaviors and the attitudes, the beliefs that the teacher has about learning that impacts on their teaching in the classroom.
2. **Reflection-on-action:** This involves description and justification. It is dialogic, in other words a discussion. It is explorative and also engages multiple perspectives and factors. Reflection-on-action happens after teaching.
3. **Reflection-in-action:** This is the teacher's thinking 'in the moment'. Thinking which occurs in the second by second, the minute by minute. Teaching that occurs between the interactions, between the teacher and the student in the classroom.

So, there are three types of critical thinking, three types of reflection, technical rationality, reflection-on-action and reflection-in-action.

- **Reflection-in-action:** It refers to the thinking the reflection that the teacher does moment to moment while teaching is happening there in the classroom. It is a reflection on what the students are doing or saying and also what the teacher is doing. Reflection-in-action allows the teacher to see clearly what is happening to consider why is happening and then to respond to do things differently in relation to that reflection.
- **Reflection-on-action:** It occurs after teaching has taken place, after learning has happened. You may think more deeply at this point about what the pupils didn't understand. What caused something to happen in the classroom and what options were valuable to you, while you were

teaching. Your responses will depend therefore, on the level of knowledge and experience and your understanding of theories and the values, the beliefs you have as a teacher in relation to this type of reflection.

Hatton and Smith (1995) identified two types of reflective practice.

1. Dialogic reflection
2. Critical reflection

Dialogic reflection: it refers to a less intensive process than critical reflection. It involves a self-discourse with the teacher. It's an attempt to explore an event internally. It involves considering the judgments and decision that the teacher has made during their teaching.

Example

Dialogic reflection is an example of a basic form of reflection proposed initially by Brockbank and McGill (2000). They said that in dialogic reflection, the individual thinks about what they are going to do in their teaching, about the information they'd like to convey to the students and how they would do this, the methods they would use to convey that information. They also consider the level of engagement of the student in the class. Afterwards they then reflect on learning. They think about how far the individual goes for the students, the objectives for the lesson of the teacher have actually been made. This can be represented in the following model.

Critical reflection: This refers to efforts from a much broader perspective, thinking about historical even political perspectives on the problems which are being encountered. This is been described by Boyd and Fales (1983) as "the core difference between whether a person repeats the same experience several times becoming highly proficient in one behavior, or learns from experience in such a way to be cognitively and affectively changed".

Therefore, critical reflection facilitates transformational learning in a way that dialogic reflection does not and transformational learning can happen gradually, slowly over a period of time or it can happen quite suddenly based on some critical situation in the classroom.

What do these considerations mean for RP?

It suggests that RP is

- Non-judgmental
- Collaborative, and
- Should be flexible and based on open-mindedness.

Critiques of Reflective Practice

Ethical concerns

The teaching and application of RP have generated these concerns over the period of time and they continue to be discussed. These ethical concerns relate to issues of confidentiality, privacy, informed consent and also in relation to the professional relationships that teachers have. Therefore, practitioners who are engaging in RP need to be aware of these concerns, and the risks involved in the process of RP. A further problem can be with the way reflection takes place. It can lead to feelings of disapproval and self-rejection because the process is so critical. Further, if an individual understands the word 'critical' to mean 'negative', that can have very serious consequences long term for the teacher.

Professional concerns

If RP is done badly or ineffectively, this has a consequence in terms of what can be missed for the teacher and the value of it goes diminished. If RP is applied in an uncritically way, then reflection

can actually reinforce prejudices and pre-conceived ideas about teaching and learning and about practice. This can actually lead to continuation of the way sudden practices are used in the classroom. RP at the level of the individual practitioner means that organizations can actually remove themselves from the responsibility of professional development, moving it to the individual rather than the organization.

Pedagogic concerns

Reflective teaching raises two main pedagogic concerns;

Developmental readiness

Are the teachers ready for this sort of developmental activities?

Do they have developmental readiness? And also, the extent to which forcing teachers to reflect may be counter-productive. Developmental readiness identifies the difference between a novice teacher and an experienced teacher. The novice teacher may not be ready to reflect critically. Therefore, developmental readiness defines a distinction between a novice teacher and the expert who has practical mastery.

Forced RP

If RP is forced on the teacher for example, as part of the course, this can actually affect the quality of the reflection and especially where it may be linked to an appraisal process for teachers, any sort of genuine and honest critical self-evaluation could actually be diminished.

Conceptual concerns

There are conceptual concerns about how ideas can be transplanted from one place to another.

Is there clarity of understanding of what teaching and learning is and what reflective practice is as a process? It can be understood in multiple different ways and actually therefore, the exercise of RP can engage in different traditions and therefore, is no uniform between teachers and organizations (Finlay, 2003). Therefore, some may see it as a primarily personal or even confessional process that deconstructs socially situated action and therefore does not have broader implications for teachers.

Informal reflection: This relies on experiential learning. Experiential learning is an approach to informal reflection is often quite dysfunctional. It is always incomplete. It is a cyclical process, one thing leading on to another and therefore, is never ending. This is based on reflection on present experience and this can therefore, lead to misinterpretation. Such an approach can be manipulated to fit our own beliefs and therefore, can be quite biased because we view the world quite selectively. What we see and what we hear, we view through our own internal lens. Informal reflection can be both planned and unplanned but will always involve a conscious awareness that learning is actually taking place. A by-product of some other activity, usually unintentional or unexamined and can actually test here our beliefs.

Formal reflection: learning from formal reflection is a little bit different. It requires becoming aware of conscious learning in non-routine situation. Learning from formal reflection also requires an awareness of learning to surface, to not be tacit (understood), to move from being tacit to being more explicit. In literature this referred to as ‘mindfulness’. Therefore, formal reflection requires models and frameworks for the teacher to use. These provide a very specific and structured approach to reflection. All formal models rely on following parameter:

Learning

Behavior	Visible behavior Function: teaching, maintaining professional relationships
Reasoning/Judgment	Student-oriented understanding Function: problem-solving

Professional thinking	Critical appraisal of literature and own practice Function: optimizing teaching based on experience
Personal reflection	Mindful awareness; exploring Function: balance

1. Behavior: There is a visible behavior and the function associated with that is teaching and maintaining professional relationships in the school.
2. Reasoning/Judgment: This relates to student-oriented understanding and also about function of problem solving.
3. Professional thinking: The teacher engaging in a critical appraisal of literature in relation to their own practice and the function is to optimize teaching based on experience.
4. Personal experience: This is a mindful awareness of what is happening in the classroom. The teacher exploring what is being going on and the function of this is to balance, what is happening to eke out (use) all the good things and to identify the problems which need to be resolved.

Teachers reflect on the

- philosophy that provides the foundation for their teaching,
- their values and beliefs about teaching and learning,
- their practice

Reflection on your philosophy

Reflecting on your philosophy is whether you gave children the opportunity to express their thoughts and their feelings and the support you give to that level of independence. It is also about

investigating your own prejudices and your own biases and understanding and appreciating your own beliefs and values about teaching and learning.

Reflecting on beliefs

You can reflect from your own perspective about yourself. For example, he might believe “I am student-oriented” ... or... “I see myself as providing a service” that would be another perfectly good belief to have.... or.... even, “I have professional expertise” if you have been teaching for many years. These are beliefs about yourself. There can also be beliefs about families. For example, families are the most important part of children’s life, or you may actually see families as part of the problem that interfere your work in the classroom.

You also have beliefs about children. You could see for example, that children are very capable as learners. Children are interested as learners from birth or you may just see children as very vulnerable and needing a lot of support and guidance.

Reflection on values

Reflecting on values is also very significant for the reflective practitioner because it informs the decisions that you make in the classroom.

These come from our life experiences, especially the early ones from young childhood, from the family and the community. They come from study and from experience. More interestingly, these can be shared in every setting in terms of how children learn and develop.

Professionals reflecting

It is very significant part of reflective practice. It is important for you to become aware of the assumptions that you hold and the values and beliefs that you have about teaching and learning, and how all these things come together to create your professional identity.

There are some assumptions that we need to consider about reflective practice.

- **Assumption 1:** The reflection takes place in the classroom and is conducted **by** the teacher and **for** the teacher. Therefore, such reflection is contextualized and very private to the individual teacher.
- **Assumption 2:** Teacher reflects and acts on what they observe. This means that reflective practice is both a process of research and action.
- **Assumption 3:** Reflective practice requires dialogue for true understanding (Kemmis & Mc Taggart, 1992; Wells, 1999, Vygotsky, 1999). This places reflective practice within the social constructivist model of learning.

Consequently, a **social constructivist model of learning** suggests that reflective practice is a process of thinking which is mediated by language or in other words potentially influenced by the language that we use to describe our understanding.

Assumptions and reflective practice

Reflective practice is therefore about becoming aware of our own assumptions and how these assumptions govern our practice (what we do in our classroom) and how these assumptions must shift in order to change and change is necessarily in teaching. There is also an understanding that there can be a resistance to an assumption change or to a shift in assumption and finally a change to assumption to support a better state of affairs improved teaching and learning. The reflective practitioner is therefore someone who lives reflection on a day-to-day basis. Therefore, becoming reflective practitioner takes a lot of time, lot of commitment and requires discipline and an understanding of the responsibility about being a reflective practitioner.

Key points about our assumptions

Reflective teaching involves discovering and researching our own's assumptions. Our own assumptions can be explored through a number of different lenses. We can explore our assumptions through

1. autobiographical writing usually things like learning journals
2. student feedback whether written or verbal
3. discussions with colleagues regularly
4. reviewing the literature on teaching and learning

Stephen Brookfield puts this very well. He says “Teaching in a critically reflective way involves teachers trying to discover the assumptions that frame how they teach. Teachers have four complementary lenses through which they can view their practice. The lens of their own autobiographies as learners,

the lens of students' eyes, the lens of colleagues' perceptions, and the lens of educational literature”.

Why are these lenses important?

Reviewing practice through these lenses helps us to admit and surface the assumptions we have about teaching, methods which we use or techniques that we feel appropriate. In so doing therefore, we can ensure the best for student learning.

Why best practice?

The best practice is important because the 21st century is a different century in which we were educated and therefore the needs of children in the 21st century are changing. We need to constantly think and rethink what we are doing in the classroom. What children need in the 21st

century is something **that is** going to help them survive better and thrive more in this new global technological environment and the reflective practitioner has a means through achieving best **practices** to help their students help children to achieve their dreams. **Moreover, children will need creativity, problem-solving abilities, a passion for learning, a dedicated work ethic and lifelong learning attributes.** The opportunity is therefore that can be given to children are dependent on the reflective practice inner achieving the best practice the best approaches that they can in the classroom.

What Does Best Practice Look like for the Reflective Practitioner?

Students are engaged and focused on their work. Well one thing to consider is that a best practice classroom is where students are on task for very much of the time. They are engage in the process of learning.

We can also think of a best practice environment as: one where **teachers are collaborating and working together** on very realistic tasks, which place students at the center of learning or in other words **students-centered learning.**

And perhaps most importantly is that the best practice environment is an **activity-based environment** and this is a phrase **that** teachers **s** use a great deal these things.

And a very important aspect of this new **21st-century** learning of the best practice environment in our classrooms which are environmentally important learning spaces or in other words **conducive to learning.** **Also, there is a joyful feeling of purposeful movement industrious thinking and a vital and vibrant atmosphere and environment.**

Core Best Practices:

- One best practice is that the teacher will **educate the whole child.** A math teacher for example: what does teach math but is aware of the child as a whole learning individual and their needs for learning in their interests for learning.

- And so, therefore **teaching is not only based on the knowledge of the subject but on the knowledge of how children develop and learn.**
- Another core best practice is to think of **preparing children for success not only in school in terms of academic achievement but also in later life.**
- And we should also consider that best practice is about **providing an environment that is enriched** in other words one where children can thrive **from** a mixture of approaches and resources.
- Another core best practice is to think of **students as active participants in the learning** process rather than as passive observers.
- And a learning environment that **provides a challenge** but also is very rigorous in terms of what students are learning.

Core Worst Practices:

Some bad practices for example:

- Worse practices are teachers who **plan and teach in isolation** without collaboration.
- **Teaches to the test “one size fits all”.** Teachers teach simply so students pass the test and teach one subject in the same way to all students.
- Teachers who assess who **teach without assessing students’ needs** so that some students may get left behind.
- Teachers who do not challenge themselves and **stay within their comfort zone**. All is also considered to be not the best practice.
- **Applies student-centered/activity-based approaches in name only.** Teachers who say in label only that they are using student-centered and activity-based practices. This is another aspect of what we were consider not best practice.

Operationalizing Best Practice:

- We can talk about operationalizing, these best practices simply by thinking of the title and status of teacher as a skillful reflective practitioner.
- The teacher therefore needs to be concerned really about the consequences of teaching, as well as what might be called technical proficiency. The long-term consequences on the child not just about passing the test or getting through the curriculum.
- They need to be prepared to experiment with the unfamiliar things in the classroom and learn from their experiences. This is another aspect of best practice.
- And the reflective practitioner who we can see is using best practice has an approach which is very open-mindedness and honest and is wholeheartedly about the student.
- The best practice practitioner is committed to professional dialogue with others, talking, collaborating with peers and sharing, understanding and acquired knowledge.
- And also the best practice practitioner is someone who focuses on action and evaluating and revising what they are doing regularly on an ongoing basis.
- And perhaps most importantly the reflective practitioner who can be seen to be one who is using best practice is themselves a lifelong learner and responsible for their ongoing professional development.

Guiding Reflection:

RP develops an awareness of how professionals come to hold their assumptions and informs reflection on how this informs and shapes practice. So reflective practice develops all of these skills potentially and so in the classroom, we can see best practices.

Implications for Achieving Best Practice

- **Reflecting in practice** is very much about thinking on your feet the moment to moment reflection that happens while you're teaching.
- **Reflection on practice** is very much about thinking back after the event after the classroom teaching has happened.

- What is important in terms of this reflecting in practice and reflecting on practice? It is about the teacher reflecting also on their own values and their own attitude about what has been happening in terms of the teacher-child interaction and children's overall learning.
- And in this way very often in the literature you'll see that the reflective practitioner needs a lot of guidance to do this in a very constructive way and therefore make a positive change.
- And to make changes teachers need to not only deepen their understanding of the subjects they're teaching. But, also an understanding that their own values and beliefs and assumptions about teaching and learning. And this is what requires a lot of formal practice and some very good quality guidance.
- Studies show that the without effective tools without a structure for reflection then what can happen is that reflection tends to be quite superficial. And so what you can use for reflective practice. Our questionnaire, or another structured techniques such as recording formats. Guided questions to guide the reflection.
- Most importantly reflection needs to be ongoing. It needs to be a normal part of the day-to-day routine for the teacher not something which is done just once in a while.
- And therefore we can break the reflective practice down the structure into certain components. And one component is about collaborating with other teachers
- This is very significant because, it indicates that teachers are part of a learning community. If there is collaboration then there is a long-term commitment to reflective practice. While, guidance and structures are important factors in engaging a professional's long-term commitment to reflective practice
- And more than anything else therefore, reflection is not only encouraged but mutually supported and that is a very significant process to occur. A professional can bring a different perspective to the reflection, and encourage a teacher to think about what values they hold, why they hold them, and what works best for children and their learning.

Regular Learning Networks:

Another structure we can give to reflective practice is regular learning networks. Reflecting regularly on other professionals' comfortable and predictable worlds has been proven to create reflective teaching "hubs". So, in other words, opportunities for teachers to get together. This does require an organizational perspective on the reflective practice process and how the school or the educational organization can support the teacher's reflection. The interesting thing about such learning networks or hubs is that they don't necessarily need to focus on the events in the classroom. But, they can include a wider dialogues-around policy in curriculum and around other issues, in teaching, pedagogy, and learning.

Time and Space for Reflection:

Research does show that a significant impact on the ability of the reflective practice to achieve best practice is that the organizations give time and space for reflection to occur in the busy school schedule. This can be difficult but whereas cool allocates five or ten minutes at the end of the day this become extremely valuable. Networks are significant in this way because networks give chance for groups to get together and for reflection to occur.

Informed by Contemporary Research:

Another structure that we can give to reflective practice is that it does need to be informed by contemporary research. Research tells us that regular reflection is not only about what happens in the classroom but linking theory to practice. Also an effective practice is regular, and it includes both reflection in and on practice. The one-off workshop has been shown to be quite ineffective in terms of teacher development. So, any sort of reflection which is ongoing and they are therefore, support best practice through development. Simply, reflective practice in this way can become a much more strategic and collaborative suite of the long term rather than just a passing on of information.

Comparison with Traditional Models of Teacher Learning:

We can contrast with the traditional model of professional development which is very much about that **transitioning** from one person to another of accepted knowledge, and accepted experience **and that is not personal or based on individual experience.** And the primary assumption there is that knowledge about teaching needs to be passed on is acquired by one person. And then, shared with other people around them rather than something which is mutually believed and understood by everyone. Therefore, theory needs to be thoroughly understood and **carefully applied** in this regard. Because, theory then means that the reflective practitioner is achieving good practice. So, the Traditional approach to professional development reflects a basic assumption that information is passed on and reflective practice is not about that. Reflective practice is about learning from acquired experience. **The traditional approach to professional development reflects on this assumption that information is a stimulus for behavioral change, and that individuals receiving knowledge will use it to improve performance.**

The Reflection Process:

The teacher needs to review,

- What happened to think about?
- Why certain things happen in a particular way?
- **How** they happen?
- **And what** were the consequences?

And most importantly how could the insights from this sort of experience be used in future lessons.

What have they learned? What is the teacher **learn? What could they do now in relation to?**

What happened before?

Collect Information: **Then** there is a part of the cycle which is about collecting information that could be about asking the students for feedback. It could be about asking colleagues to come and watch you teaching to give you feedback. It could even be about you watching other teachers in their classrooms. So that you can consider how they are approaching their teaching. This leads into **the** questioning and analyzing phase and where you are posing questions for yourself.

- What happened?
- Why did that happen?
- How did it happen?
- Who was left out in the class?
- Who succeeded most of all in the class?
- What could be improved?

This whole process then leads back into another planning phase ready for teaching, so you can see the reflective cycle. It is an iterative and it leads to a best practice approach.

- So collecting information can be talking to children, colleagues, and parents.
- It can be about observing class students in the classic could be observing other teachers.
- It can be having teachers observing you and giving your feedback.
- It could even be about video in your own teaching or taking an audiotape and listening or watching those things over and over again.
- Collecting information could include looking over the lesson plans for the lesson and looking over other people's lesson plans to see how yours relates to theirs. And certainly looking at your lesson evaluations. If the students evaluate what they say if you evaluate what were you thinking at the time when you filled out the evaluation.

Question/Analysis:

- Question and analysis is about looking for a solution to in order to correct a problem which is very often considered a single-loop learning.
- If it takes a wider perspective looking at bigger questions and looking at how theories are being used or applied in the classroom this is considered double-loop learning.
- It could be about asking questions about changing and different perspectives
- And asking why things happened.

Plan:

Planning is about developing the hypotheses and using the **reflective practice strategies** to apply in the classroom and test out the results. There are some Reflective practice strategies as follows: Journals or diaries, Meeting with Mentor or critical friend, reflective practice blog, Action research, Video, and Observation.

It's (planning) very much about asking well, So, what if I do this what's the result? If I don't do this what effect will that have?

Act/Do:

Acting and doing in the classroom is testing your assumptions do they work, don't they work. Things can go wrong and failure is not a failure. Failure is a learning opportunity and it is an opportunity for the teacher to say okay. Now, what now what is next now what can I do.

Review:

- Reviewing is very much about reflecting on practice reflecting on **what has happened** after the event. Asking the question well what did happen.
- Van Manen proposes is in terms of **hierarchical levels**.
- The first level he talks about his **technical reflection**. This is the actual practice the teaching, the skills for teaching, and the pedagogical activity that takes place in the class. At this level, our teachers are concerned with purely the technical aspect of being in the classroom How they're talking, how they are presenting, and how they're interacting with the students. Teachers are also concerned with applying knowledge to achieve instrumental outcomes, and actions taken are evaluated based on of their success or failure in the classroom.
- **Practical reflection** is the next level. It focuses on the learning experience of the student. It goes beyond that technical rationality aspect and starts to investigate into questioning a "What is going on in the classroom? What learning is taking place? And how is teaching affecting that learning?". And clarifying the end objectives and the assumptions behind teaching activities designed to achieve those objectives.

- And finally there is **critical reflection**. This is the highest level of reflection that you can achieve. It focuses on what knowledge is of value and to whom that knowledge is valuable. At this level, teachers **are** not simply concerned about the goals and the activities and the assumptions that underlie those goals and activities. But the larger context of teaching and learning **is** the bigger picture of education. And it is very much connected with the teacher's beliefs values and assumptions. Their ethical and moral issues **are** associated with teaching children in schools.

Reflexivity:

There is a term used very often in texts about reflective practice and that term is reflexivity. Reflexivity for teachers means not only engaging both in a personal reflection but also engaging with wider discussions about teaching and learning. So what's very good for us to consider what skills were talking about when we use **the** word reflexivity?

Skills Underpinning Reflexivity:

Self-awareness:

The first skill we can talk about is self-awareness. Self-awareness is very much about your thinking and your feeling about how you sense the world **you are** how you into it. What is going on and very much helps to develop your understanding of what goes on in the classroom? And how teaching and learning affects the students.

Reflection:

Another skill underpinning reflexivity is of **course reflection**. This is very much a process of learning and evaluating what is going on around us. This can be day to day can be weak too weak. It can even be career-long.

Critical Thinking:

And of course, the third component underpinning reflexivity is **critical thinking**. Critical thinking has its roots in critical theory. And is very much about the teacher identifying and challenging their assumptions about learning. Challenging the importance of the context in which teaching takes place and thinking through how **they are** teaching impacts on learning and the alternatives which are open to them in terms of how they approach their work in the classroom.

So, we can put these **three** things together and see reflexivity as being made up of self-awareness. This understanding of self-reflection is **an** ongoing activity career-long potentially. And critical thinking is very much based in an understanding that critical practice is not superficial at all.

Types of Reflexivity:

So, **where** reflexivity becomes more of introspection. We can look at reflexivity from some very specific perspectives.

- One perspective is in terms of **hyper reflexivity**. This is about the teacher deconstructing what they're doing deconstructing their teaching in very fine detail.
- **Hypo reflexivity** we can also think of system **systematic reflexivity** **which** is looking at the assumptions and beliefs that underpin the teaching practice.
- And we can also think of **epistemic reflexivity**. **This** is exploring the deeper beliefs about teaching that the teacher holds and where these three aspects of reflexivity come together then we get very deep thinking about what is going on. Therefore good practice from extensive thorough critical reflection. **Consequently**, we have to see that reflexivity is not a one-time thing, **And it is** very much based **on** day-to-day experience and practice.

Topics for Critical Reflection:

Pedagogy and how you know the teacher connects with practices in the classroom needs to be think thought about in terms of the topics which directly being considered. This leads to a number of crest questions, **How are practices and the physical environment acknowledge**. And what is

important that **we are** thinking about the autonomy that **we are** giving children the sense of empowerment the children get from being in the classroom.

We have to consider also the wider picture because there's also the empowerment that we're giving whole families **about** to what we do as teachers. **So**, we go beyond questions like well.

- Who is not listening or paying attention in the classroom?
- And why and **we are** thinking about what strategies might change that and perhaps even the physical environment.

So there are **several** questions that we need to consider and I've listed some I'd like to share with you. The questions relate to culture and communities, and the lives of the families, as well as the individual practitioner and so, they are **embedded** into certain sections. **There** are some questions I put down.

Modelling reflection with students:

Let's just go through them the first one,

- 1- **Was this activity successful while why not?** That's a very basic question. It's a good starting point for the teacher.
- 2- **The second question if we do this again what can I do differently to help you learn more?** This praises the teacher's reflection from the student's perspective rather than their perspective.
- 3- **Did this activity help you learn more than others we have done and Why?** **This** gets you to understand the student's perspective on their experience of the methodologies that you use.
- 4- **Are the relationships that I have with my students helping or hindering their ability to learn?** This will get you to think about your day-to-day interactions with students. How you're talking to them and how they're talking to you.
- 5- **Could the problems I have in my classroom be solved by pre-teaching my expectations or developing rules and procedures to deal with these issues?** This will get you to think

about the problems the issues that you may be experiencing from a very pragmatic point of view.

6- **Was my demeanor and attitude towards the class today effective for student learning?**

This gets you to think about how students may perceive your emotional state.

7- **Am I excited to go to work today?** Students will know if you're not happy in the classroom. And therefore what you need to do is be sure that you are enjoying what you're doing.

8- **Are my students excited to come to my class today?** This is connected to your attitude and whether **you are** excited about being in the classroom.

9- **What choices have I given my students lately?** Remember students are there to learn but **they also have the** potential to be empowered by what is taking place.

10- **Can I explain at least SOMETHING about each of my students' personal lives?** This gives you a focus on the individual student or maybe their interests and needs as learners.

11- **Does my assessment accurately reflect student learning?** It's all too easy to use checklists and to put ticks in boxes without really thinking **about** how much the student has **learned.**

12- **Do my assessments reflect learning, or merely task completion or even memorization skills?** This sort of question will get you thinking about the higher-order learning of the students.

13- **Why did I REALLY choose this particular lesson to cover a certain objective?** There could have been many others this will get you to think about the different approaches that you can take.

14- **What evidence do I have my students are learning?** There can be more evidence than just assessments there could be feedback or your observation of what teachers that students are doing.

15- **What new strategies have I tried likely that might benefit a student I am struggling with?** This is a good question because it will get you thinking about expanding your skills and trying out new techniques.

- 16- **In what ways am I challenging students who are clearly being successful in my classroom?** Very often teachers tend to teach to the middle and the people at the extreme end the lower end and the top end get missed. Those at the top end get missed most of all what are you doing to challenge them.
- 17- **What do I do when students are not learning in my classroom?** How do you react what do you do?
- 18- **Which students benefited from a particular activity?** It's worth knowing. What worked well and why it work well?
- 19- **Which students did not benefit from an activity?** Of course, if something goes horribly wrong this may be an opportunity to change your approach or it may be just understanding that activity didn't match with those particular students.
- 20- **In what areas can I still improve?** This is a good question to think about in terms of your reflection. Because it will take you into that cycle of constant renewal.
- 21- **What's stopping me from improving in these areas?** What are the barriers because there will always be things which will get in the way?
- 22- **In what ways can I support my colleagues in my student's learning?** This places you as a teacher in a community of practice where there is lots of support.
- 23- **Do my actions of the teacher show my belief that all students can learn at a high level?** A part of the student's learning is dependent on your approach to teaching and your expectations that they can achieve much higher than they think they can.
- 24- **Do my actions as a teachers show that I take pride in my work?** Remember teaching is a profession and as a teacher, you are a professional. Are you a professional?
- 25- **Are the relationships I have with my colleagues conducive to creating a collaborative culture focused on learning?** with collaboration comes a best practice.
- 26- **Are the relationships I have with students' parents conducive to learning?** Remember teaching is not just about kids in the classroom but it's also about whole families.

- 27- **What new ideas have I tried in my classroom lately to keep myself energized about teaching?** Don't stick to old patterns. If you're using this question then it is challenging **for** you to extend your repertoire.
- 28- **What have I done lately to relieve stress and to focus on my health, to ensure I remain an effective teacher?** Teaching is a very stressful profession. And what you must do is always check that you are in tip-top condition to be able to teach in the classroom.
- 29- **And finally what things am I currently doing that you could you know reasonably.** Do with more priority.
- 30- **How much time have spent with friends and family in the last two weeks?** Health is important and you must look after your health as a reflective practitioner. This is how you will stay in the profession for a long time.

Learning from Reflection:

- So reflection is a learning tool.
- Reflection is the process of consciously thinking through what is happening. **This** helps us again an understanding and to make meaning from what is going on in our lives.
- Therefore it's associated with learning that's occurred through experience and activity. How **do** we internalize this in terms of our roles as professionals.
- Reflection is a means to assist the teacher to think and to explore, to go beyond **what is** gone in the classroom. To understand thoughts and feelings and to work through the experience to get a new understanding as well as **self-awareness** of how things are happening and why things are happening in the **teaching-learning** situation.
- As a tool, it's there for active consideration and learning from all the actions that are taking place. These thoughts and actions come together to help us develop meaning through reflective thinking.
- Perhaps the most important aspect therefore of engaging in reflection is the **on-going** personal and professional learning that needs to happen. And that **you** can demonstrate that

to others this is all about achieving a good standard of proficiency as a teacher and a good standard of professionalism.

- This process of reflection is very often undertaken through reflective writing. You'll undoubtedly have heard of reflective journals. Reflective writing can lead to a greater understanding and a better knowledge about why things are happening. And it should also challenge your preconceptions about what is going on and the theories as they are applied in the classroom. When you reflect on a situation you not only see more of it but you start to see it differently.
- This different way of viewing things is about situating reflecting in a process where the action is the final stage of reflection.
- So what do you know about and how confident are you about certain aspects of the reflective practice process and learning from reflection. Well for example,
 - ✓ What do you know about the **roles and responsibilities of the teacher** this is worth reflecting on?
 - ✓ What do you know about **learning styles**?
 - ✓ What do you know about **planning a course** or planning a lesson?
 - ✓ What do you know about **and how people learn**, how children learn?
 - ✓ What do you know about **lifelong learning**? This is something which is talked about a lot now.
 - ✓ What do you know about **communication skills**? This is the practical aspect of the teacher in the classroom.
 - ✓ What **do** you know about **presentation skills and demonstration skills**? What's the difference between these things?
 - ✓ How confident are you with **questioning and explaining** things to students?
 - ✓ How confident are you using **different teaching methods** to teach subjects?
 - ✓ **How** comfortable are you **in** using **ICT** information and communications technology in the classroom?

- ✓ What do you know and how confident are you are with **assessing learning** and also the process of **reflection** itself and course your own **subject knowledge and the skills** used to teach that subject knowledge?

So, learning from reflection as we can see from those questions is very holistic. It covers a whole range, a whole spectrum of professional aspects. Beginning to develop a reflective behavior is about becoming a professional at least now you should start with your own cultural and historic background. As a beginning for being a best practice reflective practitioner.

The concept of **cognition** is actually quite narrow gives a narrow focus to what we're talking about here. But reflective practice enables the individual to think beyond that and to create more knowledge based on their experience of actually what's going on, **and** what they're experiencing day-to-day in the classroom.

This means that reflective practice cannot be an isolated activity it is something that has to be cooperatively generated. And therefore words like collaboration become interesting for us. As reflective practitioners collaboration helps us extend what we're doing beyond the individual learner. Facilitate a contract the relationship into something which is more interdependent between those involved the actors the students and the teacher in the classroom.

Reflection-on-Action:

- Teachers consciously review, **describe, analyze, and evaluate** in reflection on practice. It's a process of higher-order thinking, analysis, and evaluation as; well as being ongoing. It provides insight for the teacher in terms of future practice.
- Teachers in reflecting on practice, will think about their experiences **and responses as they occur**. But most importantly will consider their responses to those experiences as well.

Reflecting “in” and “on” action:

- In both types of reflection. Reflection in practice and reflection on practice. The teacher focuses on the connection between their feelings. How **they are** responding to a situation

and the theory that they understand about what is happening, what **they have** been taught about what is happening.

- Teachers are seeking new understanding in this light of; what their experience means for that theory and what that theory means for their experience.

The Core of Reflection:

- For Schon, reflection in practice was the core component of what he called professional artistry. He saw teaching as very much art, which was different from positivistic views of teaching, **where** teaching was very much a science, very rational and technical.
- **So**, in Schon's view, this technical rationality really sorts of failed to pick up the idea for teachers in terms of teaching as a very rigorous activity as opposed to what it was doing in terms of children's learning. Schon explains this by saying that teachers have to do more **than** just follow set procedures that **they are** told is the best way to approach. But to think about drawing on the experience, and drawing on practice and making experiences practical. **So**, they start to grow their theories in terms of; what **they are** doing in the classroom.
- And in this way, the core of reflection is very much intuitive. And is quite creative for teachers. Both reflections on action and reflection in action means for teachers to revise, modify, and ultimately refine themselves as professionals **or experts**.

Levels of Reflecting

There are different levels of reflecting. Grushka, Hinde-McLeod, **and Reynolds (2005)** identify **reflection on action** and **reflection in action** but, they link this to **reflection for action**. What this does is offer both technical and practical considerations of critical questions for the teacher to engage with. Under reflection for action, they say that teachers need to consider their resources as well. For example, how long will the lesson take? **These** are sort of very practical technical issues. They also talked about how to make the **resources** relevant in different learning situations **or styles**. Again, a very practical perspective on reflection. In this sense, teachers are always questioning,

why are they teaching in a particular way. Why are they using certain resources? Why do they believe, and what do they believe?

- As; Zeichner and Liston (1996) go on to differentiate between five different levels of reflection which take place during the process of teaching. For example,
 - ✓ They talk about rapid reflection.
 - ✓ This is very immediate.
 - ✓ It is ongoing
 - ✓ It is an automatic reaction to what they're experiencing with the children in the classroom

Repair:

They also talk about a level of reflection as repair. This is where the teacher is making decisions to alter what they are currently experiencing to try and change. Maybe the way children are behaving or to change activity. So that students can learn more or to even slow students down so that they are thinking more about learning.

Review: Another level of reflection; they talk about is review. This is where the teacher is thinking about and discussing or writing about some element or some key components of their teaching.

Research:

Another level of reflection is research. This is where the teacher is engaging with theory and with the dialogue in texts thinking over time about collecting data and understanding whole pictures of teaching and learning.

Re-theorising and Reformulating:

Another level of reflection that they talked about is re-theorising and reformulating. This is where the teacher is critically examining their understanding and their approaches and trying to understand how these links back to what they have been taught to link back to the theories which have been explored for years and years in the text.

Guided Reflection

Let's look at guided reflection which is a very specific form of reflective practice. John developed this concept in 1994.

- Maughan and Webb (2014) consider several key questions that can help guide reflection for example,
- They say that teachers can ask, what is most important or interesting, useful, and relevant about what they did or something that they used a resource that they used?
- They sort of asked the question, how can it be explained with theory or just through experience?
- They offer the question, how is it similar to and different from other issues, events, or experiences that the teacher had? To make connections between their teachings.
- The teacher can ask, what have I learned from this? And what does this mean for my future teaching? This is where the teacher is engaging as a learner in the teaching process.

Biggs and Tang (2007) also offer some questions for reflection they ask questions like,

- What was the problem again, a very practical technical type of question to ask what went wrong? What's the evidence of that problem?
- What were the causes of that particular problem?
- How did the teacher deal with the problem? It is a very practical consideration.
- They asked a question how did the solution to the problem relate to the theory that the teacher understands? This is helping the teacher make connections between theory and practice.

Guided Reflection and Bloom's Taxonomy:

Perhaps, the most significant form of guided reflection would be to use Bloom's taxonomy. And Bloom's taxonomy provides a framework where the teacher can ask questions of themselves and their teaching from lower-order through to higher-order thinking.

First, let's try to reflect on Bloom's taxonomy about a taxonomy of reflection. In terms of **lower ordering order thinking**, we can think of the teacher just remembering to describe **what I do**. And this goes through to understanding. What was important about what I did? Beyond this, we move into **middle-order thinking** applying and analyzing. **Where** could I use this again this particular technique? **How** can it be applied elsewhere and analyzed? **Did** I see any patterns; which I need to be aware of or any considerations? And then through to **higher-order thinking** evaluating, how well did I do **and** then also creating, how can I use all this information next time? **Or, what** can I do in the future?

A Taxonomy of Reflection:

Blooms Remembering:

Bloom's level of remembering, **Or in** other words, what did I do?

Teacher Reflection:

The teacher can reflect by asking such questions as,

- What was the lesson?
- Did it address all the content?
- Was it completed on time?
- How did students "score" on the assessment?

These are again very practical, **and** easy questions to ask, but they are the important starting point for the reflective practitioner.

Blooms understanding level:

- What was important about what I did? **Did I meet my goals?** Can be reflected on by the teacher in terms of, Can I explain the major components of the lesson? **And** Do I understand how they connect **with the previous and next lessons?**

- Where does this fit into the curriculum? What instructional strategies were used? Did I follow best practices? These questions are linked to the understanding level of Bloom's taxonomy.

Bloom's Application Level:

Bloom's application level which is in relation to questions like,

- When did I do this before?
- Where could I use this again?

Teacher Reflection: can be reflected on by the teacher in terms of questions like

- Did I build on content and processes which I have taught in previous lessons? This would underpin students' learning.
- How does this lesson scaffold the learning for the next lesson? How does it provide a foundation for what is to come?
- How could I adapt the instructional approach to another lesson?
- How could this lesson be modified for different learners? Again, this is a level of reflection which is in the middle of Bloom's taxonomy.

Blooms Analysis Level: This is now getting into higher-order thinking. Do I see any patterns or relationships in what I did? This in terms of the reflective practitioner is very much focused on questions like:

- What background knowledge and skills do I need to teach a particular lesson effectively?
- Were the instructional strategies right for that lesson? Were they effective?

Teacher Reflection:

- Do I see any patterns recurring in my teaching? Which I need to consider are goods or other destructive patterns?

- And do I see patterns in the style of the approach that I have? Do students react to me when I teach in a particular way?
- Do I need to consider how that is affecting their learning? What were the results of the approach I used? Was it effective? Or could I have eliminated or recognized steps?

Bloom's Evaluation Level:

- How well did I do?
- Do what work?
- What do I need to improve?

Teacher Reflection:

These get to be translated in terms of teacher reflection to questions like,

- Did some aspects of what I did need improvement? Could they be better?
- Where are the needs of the individual learners met in the lesson?
- What levels of mastery of good quality learning took place for the students?
- What have I learned about my strengths, weaknesses, and my areas in need of improvement? That again is a good level of analysis. And Am I progressing as a teacher? , Am I developing myself?

Teacher reflection: can further take on aspects around the incorporation of lessons for the future.

- What can you do in the future to?
- What you have learned from this?
- What changes would you make?
- And what would you keep the same?
- How can you best use your strengths in the future?
- And what steps should you take in terms of using resources or applying new sources to challenge students' learning?

Teacher reflection is the training that would help me to meet my future intentions. The teacher needs to look beyond what is happening in the classroom. In terms of, how to develop themselves as professionals, they need to think about their role development over time.

Topic 022: Action Research

Action Research

When you are thinking about reflective practice, it very much resembles action research. Action research involves systematically changing your teaching about problems or issues you are experiencing in the classroom. Therefore, it is about taking responsibility for change. It is important to understand that action research is targeted at the teacher and it is not about changing anything else.

Educational Action Research as a Critical Social Science – Carr and Kemmis:

This is put very succinctly by Carr and Kemmis who say that Action research is a form of research carried out by practitioners in their practices. In this way action research is participatory. It is a very democratic form of educational research for educational improvement. This is because action research must involve the learners in the classroom, they are part of the change action that the teacher is taking.

Action research emphasizes teaching. It emphasizes practice is committed to acting action and committed to changing practice. It is also a way to express a commitment to improvement for the teacher. It is about not standing still, and it's about trying to look at the theory and make connections with practice. In this way, action research is seen as a very practical form of inquiry and it's characterized by self-reflecting through cycles that spiral continually for the teacher to plan do review (planning, acting, observing, and reflecting).

In this way, it involves a number of critiques

- One is **reflexive critique**. This is about questioning things, questioning arguments, and trying to find new possibilities for **new** actions.
- There is also the idea of **dialectic critique**. This is very much about discussing actions and discussing theory from different perspectives that the teacher is aware of.
- And what we have to consider also that action research is very much dependent on a collaborative resource. It is very much about people working together whether these are teachers or whether these are teachers and students. The collaboration is there and it is also true to say that there is an awareness of risk in terms of action things may go wrong and that's not necessarily a bad thing as long as the teacher can learn from what goes wrong.
- Action research is very much a pluralistic approach to critical thinking. It's very much about discussing the questions and identifying the problems. But it is very much about reporting them from very different perspectives.

Donald Schon states **that** action research is very much a relationship between understanding and change action.

- He believes that the practitioner has an interest in changing the situation, in changing the status quo. And this is better than leaving things away they were. It also there develops a better understanding of the situation.
- He sees action research as his idea of reflecting **on the** action. It's very much about learning while doing and learning from doing. (Action research is something that any practitioner can do as a part of his/her everyday practice, and it does not just imply research).

Some Suggestions for Thinking about Practice:

Bryant's researcher states that there are many themes and aspects to practice that can be questioned. There is certainly a way of looking at theories what he talks about is that,

- The teacher can look **at** action research from the point view **of** a setting, where the acting activities are taking place. What is the classroom environment actually like?

- He also talks about the play you know there was looking at ideas in terms of a game. Where are the rules of the game, the regulations which are a play in the classroom, and the rules that govern the teacher?
- He also talked about the reflection as thinking in terms of script. So in other words, texts that identified but don't you know completely explain performance. This is an approach where the teacher can really monitor using a script whether that is an internal dialogue or something written down in order to uncover a problem and resolve it.

Finally, in communication, there can be different interpretations of meanings and so, the way the teacher communicates these interpretations becomes a very unique perspective.

The Critical Friend

The Critical Friend:

- The critical friend is a very useful technique to think of as a reflective practitioner. The critical friend is a process that focuses on a professional relationship and encourages reflective practice with the colleague. It's very much built on cooperation and collaboration.
- It also addresses a number of situations in which teachers find themselves. For example, working in isolation, working as independent units within the school. A critical friend can actually help address this.
- It can also address the idea that the teacher is once they're in the classroom considered to be the person that is certified to know everything. Therefore, the critical friend is a way of continually professionally learning and not just sticking with what is been taught previously.

Background:

The background to this comes from the Annenberg Institute for school reform at Brown University in the USA. It was the first to develop the critical friend model for collegial dialogue. Interestingly

it is currently used by an estimated 35,000 principals and professors across the world in about 1,500 schools and universities. So, it is a very popular approach.

Key Contributions from the Critical Friend:

- The critical friend is about providing a balance between a friendship something which is very informal and supportive with something which is very critical and analytical.
- The critical friend is somebody who asks why things have been done in a particular way. They are helping the teacher seek to gain a sense of the context in which things are working. And the way things are working and also to challenge assumptions and perceptions.
- The critical friend, therefore, is helping the teacher think creatively and to encourage others in the environment to do so.
- As a friend, a colleague in the school becomes very significant. Therefore, the critical friend is very friendly and honest but also critical and analytical. As a colleague, they can share professional reflections. And help teachers look beyond the classroom and look more to the educational horizon.
- The critical friend is using their skills in a way to ask very superficially even “naive” questions of the teacher. This is a way to promote reflection and to get the teacher to self-evaluate. What is going on simple questions like, why did you do that? And What happened? The critical friend is also there to provide advice and some guidance and hopefully some very practical suggestions for the teacher.

Critical Friend Ethos:

- So, in a way, it is like keeping or maintaining a close distance.
- The critical friend is supportive is very friendly just like friends are. But they hold on to the role of an evaluator.
- As critical friends, they are very knowledgeable about teaching and they understand the subjects that teachers are working with. But they are often very supportive as well.

- And what the critical friend does is identify the challenges that the teacher has. And they have no responsibility for overcoming those challenges instead, they support objective and reflective debate to find solutions.
- Critical friends of course within this role need to make sure that they maintain confidentiality with the teacher whilst being frank. At the same time sensitive and independent the teachers experience the problems that they are facing.

What the Critical Friends actually DOES NOT DO?

So, what do friends do and what do they not do? Well, the critical friend does not make decisions for the teacher. And reporting formally and informally to others is something that they do not tend to do unless it is with the agreement of the teacher that they are talking to.

Limitations of the Critical Friend Role:

The limitations of the Critical friend role are as follows:

- The primary limitation is in terms of their role of advisor and supporter, and provider. Because they have to provide a constructive critique it is not just about friendship.
- Critical friends are not responsible for and they of course can never contribute to the quality of teaching. That's why the teacher does it themselves.
- In particular, critical friends have to take more roles in terms of becoming a negotiator on behalf of the teacher with the school administration. If there are very significant problems to resolve. Critical friends of course in this way may add value through engagement with a number of teachers, and so sharing knowledge around the community.

Participatory Methods for Reflection

I am putting aside time in school for focused discussion maybe with a critical friend or with a group is essential in order to ensure that reflection becomes a regular practice for the teacher.

It is important to have a facilitator in this regard somebody who uses critical questions to help guide thinking and stimulate discussion. Also, to ensure that there is a deep reflection or thinking about the issues. Maybe, even to record the discussions for the teacher to reflect on at a later date.

What Can Teachers Reflect with?

Such conversations and debates can occur with:

- **Colleagues**
- They can occur with **other professionals in the teaching environment**. For example School counselors
- It could be with **other professionals working in different educational environments**.
- It could even be of course with **families, parents, and children**.

Method:

Essentially what the teacher has to consider in this regard is the method there needs to be:

- **Careful description**
- **Thoughtful listening**
- **Questioning feedback**

And for me, the two most important components here are **thoughtful listening and questioning feedback**. It is very difficult sometimes to listen to critique from others and that does take a lot of courage from the teacher.

Types of Questioning Feedback:

There are types of questioning feedback and these can be considered in terms of hot feedback warm feedback and cool feedback.

- **Warm feedback** consists of feedback that is very supportive and appreciative of what the teacher is saying.
- **Cool feedback** offers very different perspectives and questions about teaching and raises questions about, Why the teacher took particular actions?
- **Hard feedback** challenges and extends thinking. It raises the level of concern to a very critical and analytical point of view.

The Participatory Process:

- So, as a participatory process reflective practice allows colleagues and others to share ideas, and issues and to seek suggestions and practical solutions between the group members in order to positively overcome the issues and problems. This is best of all done in small groups not so well in large groups. Because there is too much interaction with smaller groups. Also, it is easier for individual teachers to feel more comfortable sharing issues and problems.
- Hard feedback challenges and extends thinking and raises concerns in this sort of small group environment.
- As a participatory process then the teacher in a small group can:
 - Share the issue
 - Also, provide the context in which the problem or the issue arose.
 - And themselves frame key questions which the group can discuss and explore in order to try and find a resolution to that particular issue or problem.
- So, what really relies on our questions which are very probing questions? Questions that help to clarify understanding. The group can ask more questions to learn about the issue that the teacher is raising.
- And this leads to then hopefully an analytical evaluative discussion. The group discusses the issue using all types of feedback both warm and cool feedback. And the group finds a way to address possible suggestions and solutions to the issue using the hard feedback

approach. So, using cool, warm, and hard feedback makes truly analytical and evaluative thinking about the issues in question.

Foundation Models 1: John Dewey

So, let's look at some models of reflective practice that will introduce in this lecture. We will look at the models in more detail. First of all, we can consider Dewey model of reflective practice.

What is Reflective Practice?

- He saw **the** reflective practice **as** something which is very dynamic, quite spontaneous and requires a lot of honesty and **deep and critical thinking** from the teacher.
- He said that reflective practice actually should **lead to action**. It is not something just to think about it is an important thing to lead for the teacher to do.

Routine Action vs Reflective Action:

Dewey model is based on a fundamental difference between the sort of traditional approach to teaching which is about instruction and imitation **(quoted in Furlong and Maynard:31)**. And really said that the teacher in terms of reflective action needs to confront perhaps some **pre-assumptions, prejudices, and not judgments proper that rest upon a survey of evidence** about teaching. **(Dewey 1910: 4-5), and Reflective action**

- Reflective action, therefore, is very spontaneous or should be persistent, very active, and needs to be carefully engaged with by the teacher. **(1910:6)**
- For Dewey, it is in problem-solving that we find. **“the steadying and guiding factor in the entire process of reflection” and without which “the course of suggestions flows on at random”.****(1910:11)** Therefore, becomes a focus for the teacher in terms of how they go about teaching it affects teaching fundamentally.
- He saw reflective action as more than just thought it that's got a wider dimension to it.

- And indeed, he actually said that reflective practice can have emotional aspects for the teacher as well. While we cannot learn or be taught to think, we do have to learn to think well, especially acquire the general habit of reflection. (Dewey 1933)
- And what this actually means is that the process of reflective practice is a rational, purposeful action, and a very complex thing to do. Not straightforward and should not be entered into lightly or superficially by the teacher.
- It includes a good level of rationality or in other words a lot of clarity of thought by the teacher about the reflection process as well as the teaching that they are doing. (Dewey 1933)
- And interestingly, from a modern educational perspective, this does actually draw links not just with the cognitive aspect of teaching. But the way the teacher is responding to what is happening in the classroom. The emotions which are being created in their interactions with students.

Dewey's 5 Stage Model:

So, let's look at the model the five-stage model.

- First of all, he says that what the teacher does is to **identify a problem** something which is vexing, perplexing to them, and needs to be fixed.
- From then the next step is to observe and **refine the identified problem** to clarify the understanding of the model.
- Beyond that the third step of the 5-stage model is for the teacher to develop a hypothesis to explain the problem, or to try and understand it from a technical point of view. Also, as its origins and possible solutions.
- And this must lead to deeper scrutiny of that problem.
- **Test the hypothesis or** a deeper understanding before leading into some sort of action. Dewey, J. (1938) **Logic: The Theory of Inquiry**. Troy, MN: Rinehart & Winston

So, we see that placed a great deal of emphasis on this reflective thought process which is why he was saying that it is not natural to do it is something **that** has to be learned and has to be based **on** experience. He believed that reflective thought began with our needs and **having an experience that raised some difficulties or dilemmas (the “felt difficulty”).**

From this experience, we actually start to ask questions. For example

- In terms of **the problem**, the problem stages the question is well, what is going on here? What is the problem, what is happening?
- And in terms of **conceptualizing the problem** we might say well, how **do** I fix this? What might I do to put it right? And that leads to experimentation about solving the problem **or acting by trying out a possible solution.**
- Finally, in terms of solving the problem, we are **exploring the avenues** that we can take to do that. **And how it might be further adapted in the future.**
- So, within professional development, **our interest is not so much in learning from a single experience, or learning in the short-term, but more in the long-term.** Because it fits into the cycle of teaching and learning.
- It is **a** very developmental process and **enables well-grounded professional knowledge and skills** in the experience of the teacher, not focused so much on theoretical applications.
- And interestingly, what Dewey said is that although it is a cycle it should really be viewed as a spiral one cycle leading into another cycle being **correctly** connected by experience.
- This spiral actually leads to the reflective aspect of the process. We can see this in a model. We can see here, that the cycle of reflection and thinking about action and then experimenting **with** the action itself through experience goes into another cycle thus creating a spiral.

Foundation Models 2: Donald Schon

Foundation Models 2: Donald Schon (1983)

Reflecting “in” and “on” Practice: Schon’s model of reflective practice is based on reflecting in practice and reflecting on practice. These are terms we have come across before they are very popular terms and approaches. We can represent these ideas of reflecting in practice and reflecting on practice in a very simple diagram.

- We can see here; that the starting point is the actual teaching and the learning that is going on in the classroom. And it is the thinking at the moment and the reflection in practice that occurs at this point.
- After the session is when the teacher reflects on practice. This is when the teacher has more time to think about what happened and to think back on what happened.

And this whole process of reflecting in practice and reflecting on practice then leads to preparing for the next teaching session with the students planning to get the resources ready. Then, this leads into the next teaching which again will have a reflection in practice. So, again you can see this repeated cycle being engaged by the reflective practitioner.

Development of Schon’s Model:

- Schon made the distinction in this model between **technical rationality**. Or in other words, the application of learned theory and what he called **tacit knowledge**. Or in other words, experiential learning is gathered from the classroom. This can be more easily understood as the “**theory-practice gap**”.
- He believed that this was in a sense of theory experience gap. Because the theories that the teacher is trying to apply in the classroom may not generate the sort of outcomes that the teacher is expecting. In other words, a different experience. Therefore, the teachers may have theoretical knowledge (technical rationality).
- And what this can actually mean is that. It changes the way they are thinking about teaching. Actually, may not even define the way they actually think about their own practice in the classroom.

- So, in terms of **the** Schon model what we are actually thinking about is the experiences in the classroom. And how these **are** combined, **and** how these add together to generate learned and acquired theory for the teacher.
- Schon saw that the real teaching environment, therefore, was the place where the teacher **applied**, tested, and evaluated **the theory** for themselves in terms of what they are actually doing.
- Theory is, therefore, not a rigid thing here. It is something **that** can change **and** grow with the teacher's experience. Although, a new teacher would be applying theory quite rigidly as their experience develops. And as they learn more about their own approach to teaching and their own beliefs and values about learning. Then, **the** theory starts to become a little bit less rigid and they start to apply things differently.
- So, reflection in action and reflection on action allows teachers to begin to continually improve their own practice to change their teaching adapt learn and evolve.
- And what we have to remember is that this does not make theory invalid at all. It simply begins to change the theory that the teacher is acting and its theory is, therefore, a valuable starting point for the teacher.
- And this is what Schon calls **theory in use** rather than just theory.

Schon's Model:

So, in this model, we can see what Schon is talking about and the teacher is attempting to make sense of, and use theories in their teaching putting into action what the books are saying, what the literature is talking about. Therefore, this leads to some sort of real-world experience. So, in other words, what they are learning from what happens in the classroom. So, the teacher moves from knowing that or **in other words**, knowing about the theory. To know how. Or in other words, **knowing** how to put the theory into **use**.

Foundation Models 3: Kolb's Reflective Cycle (1984)

Another very important model of reflective practice is Kolb's reflective cycle where active experimentation leads to a transfer of learning from the current cycle to a new cycle. It is a very popular model you probably already know about it. What Kolb is talking about is a cycle that is based on experience entirely. In terms of this model, there are four main components. And there is a diagram to illustrate this.

The four components are,

- concrete experience
- reflective observation
- abstract conceptualization
- and active experimentation

So, the experiential learning cycle is made up of these four components. And we can look at them in a little bit more detail to try and understand what each of these four components are.

Components of Kolb's Model Applied to RP.

Concrete Experience: So, concrete experience is very often seen as, the starting point when reviewing the cycle. Although because it is a cycle the teacher can enter at any particular point. For example, the reflective practitioner could start with abstract conceptualization which would lead to active experimentation and then concrete experience. But concrete experience gives a very hands-on starting point for the teacher very practical.

Although this model can be entered at any stage, it tends to be the main point of entry. The concrete experience for the teacher is a very convenient and immediate place to start the reflective practice process.

Reflective Observation:

Reflective observation is the first level of reflection. Initially, thinking why did that happen? trying to understand their concrete experience, and trying to rationalize it. It also entails stepping back

and viewing the event or incident from an objective perspective. And It also gives the teacher that point to stop and think, about what you did and why you did it. With this model there is no reflection in action and no reflection on action, there is just observation of the process that took place.

Abstract Conceptualization:

Then this leads to abstract conceptualization. A little bit of a difficult word but in principle very simple to understand. It is trying to understand what has happened in relation to theory or to what the literature is telling us should be taking place. Therefore, it is making a cognitive connection with practical experience. It is looking for connections between interpreting, theories, events, actions, feelings, and all the experiences that happen in the classroom.

Active Experimentation:

And of course, with all of these cycles we see, there has to be some action involved. Otherwise, reflective practice has no point. So, in this cycle, we can see that after abstract conceptualization the teachers move into a process of experimentation. Experimentation suggests more than action. It is about trying something to see if it works it might fail which is why the cycle leads then back into concrete experience. So, the whole process is again more of a spiral rather than just one cycle. The spiral means that the cycle is continuous and never-ending.

Foundation Models 4: Greenway's 3-Stage Model (1995)

Greenaway's 3-Stage Model (1995)

Let's look at Greenway's three-stage model. It is a very popular model and is simply called the "plan-do-review" model.

The starting point I suppose is to talk about DO having an experience in the classroom. This leads to a review of that experience and then planning. The next steps in teaching planning the approach that you will take. We can look at this in terms of a model again very simple.

Benefits of Greenaway's 3-Stage Model:

- One benefit is it is a very simple easy-to-apply model.
- Because it is focused entirely on the action. It is grounded in the teacher's everyday experience and that gives it a lot of value and gives it a lot of meaning for the teacher.
- It is a very simple model it's also very quick to do. It's not at all time-consuming and especially for the beginning teacher. It has a lot of benefits over more complex models. Because it is much easier to understand and apply.
- And what we are looking at with this model. Therefore, is something which can be practiced in schools very quickly and easily.

When to Use

'Plan –Do- Review'

- What you actually need to do with this model is to think of it in very practical terms. It is a model for continuous development of the course.
- It is especially useful for new teachers.
- Or if you are starting with a new class of students. And beginning a new term or improved approach in school.
- It is especially useful also with activities in the classroom which can be quite repetitive; where you are doing things over and over again. Because the plan-do review cycle will help you refine what you are actually doing in those lessons.
- It can be a very useful approach. If you are collecting any data for further analysis in order to verify and prioritize problems or root causes. If you are doing some action research or if you just want to gather more data about your students.
- Most importantly, it is very useful if you are trying to implement some sort of change to your teaching. It is very quick to apply you can monitor the change process very easily.

The "Plan-Do-Review" Procedure

Plan:

So, in the procedure plan what you need to do is recognize an opportunity, and actually make a change or do something in your teaching.

Review:

The reviewing is actually analyzing the results of the action that you are undertaking and identifying what you have learned. And what you need to do then is to plan new improvements, which you will then action in the DO phase of the cycle.

Questions to Ask for “Planning”

So, Some useful questions about what we can look at in terms of each of these three stages in terms of the planning stage. Some questions might be well,

- What data indicates a change is required? Rather than just initiating some sort of change for any reason. There is got to be some good foundation for the change to take place.
- What change is to be made?
- How do you know the planned change is appropriate? What other alternatives are there? And what are you actually going to do? So, there are some fundamental questions that we can look at.
- What sequence of steps is needed to implement this change?
- Who will be responsible for carrying out each step? It might not just be the teacher alone; it could be others involved in the school.
- Who will need to be consulted?
- Who will the change affect? How long will the change take? It is another consideration for the teacher. It might be actually just a space of half a lesson, or one lesson. Or it could be over a period of a week or even a month.
- How long will each step take? Is it one action in the classroom? Or is it a whole approach to a whole lesson?

- How will you know you have completed each step?
- How will you monitor the change's effectiveness and the benefits of the change?
- How will you monitor and track the progress? That's another question is worth considering in the planning phase. Because you have to be prepared for all aspects of the reflective process.
- How will you collect, review and act on the information?
- What will you do about unexpected problems?

Questions to Ask for "Acting":

- In terms of acting, how will you monitor the progress and the effectiveness of what you are doing? How you will collect the data? How will you know? and the results of what you are doing.
- And of course, How will you save the recording of observations and results? (Planned and unexpected) in comparison with the original data or the goals, measures, and objectives. Will you use video? Would you just make notes for yourself in a journal? Could you get another teacher to watch you observe and give you some feedback?
- And part of acting is, what did you achieve? And what did you learn? from the whole process of going through this cycle. So, checking goes on continually throughout the planning phase the doing phase, and the review stage. You are continuously checking, you are not just engaging in either doing or planning.

Questions to Ask for "Reviewing":

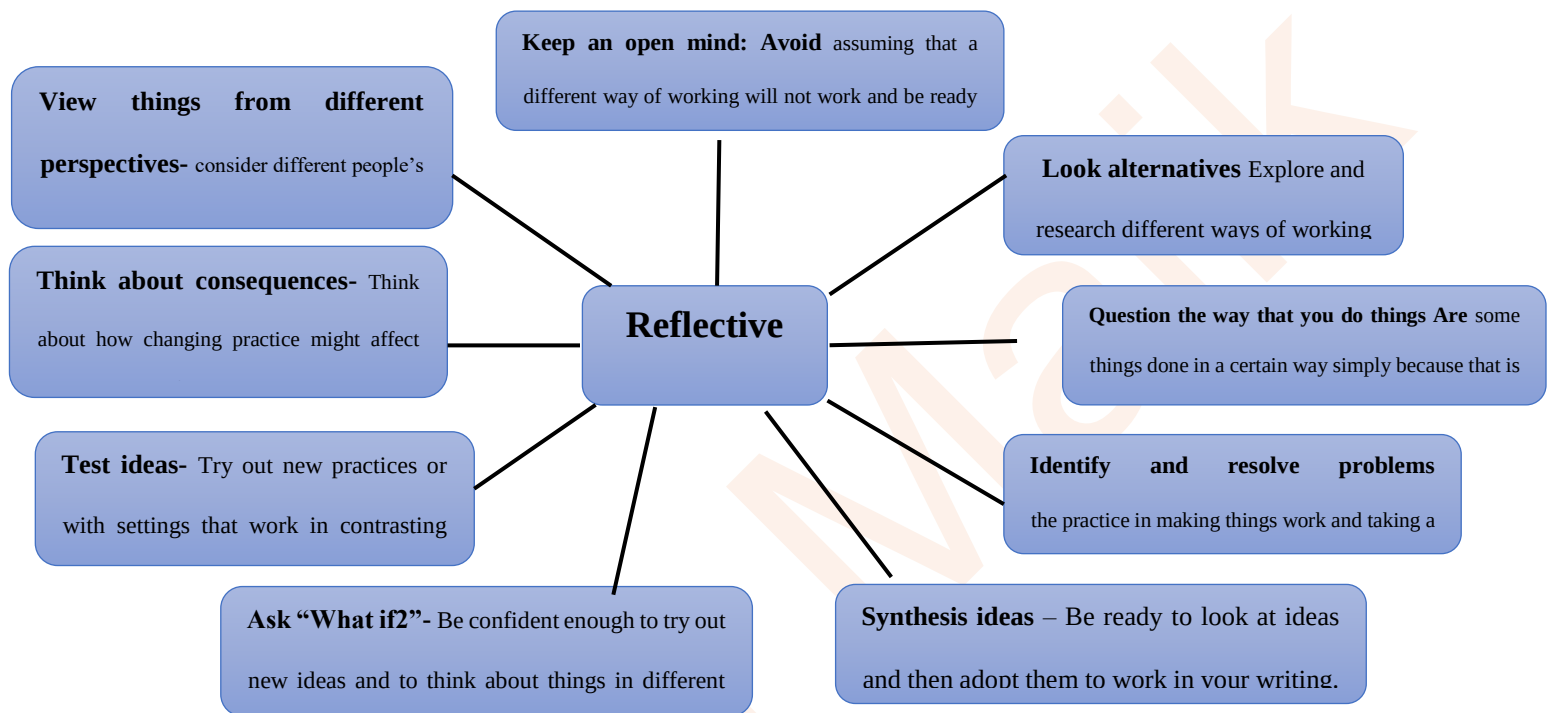
- So, what will you do with that learning is a significant question to ask during the review stage? How are you going to use the information? Or what will you do with the learning-adopt it abandon it, run it through another cycle again to test?
- What did the information that you collected? Impact in terms of change, what was the effectiveness of what you actually did?
- What can be done to further improve what you are doing in the classroom?

- How can you refine your own teaching in terms of what subject you are teaching?
- What lessons have you learned that can be applied elsewhere?
- How can these lessons be communicated to others? These are all aspects of this very simple process. There are probably many more questions that you can consider for yourself in relation to plan do and review.

Topic 029: Foundation Models 5: Roth (1989)

Roth's (1989) summaries of Reflective Practice processes as follows:

- In Roth's model of reflective practice. He very much sees it as a process of questioning asking what, why, and how as a teacher you are doing something. And also asking what, why, and how other teachers are doing what they do.
- **Seeking alternatives**
- And in this light is very much about the teacher keeping an open mind about the teaching.
- Also comparing themselves with other teachers, comparing and contrasting the approaches that different teachers have.
- In this way, it is very much about seeking a framework rather than using a framework. Although, this model is a theoretical basis. It is very much a process of action and reaction that's taking place.
- As a reflective process, it is very much then about asking questions like well, what if I did this thing in my teaching? What if that happened? And then seeking ideas looking for ideas, seeking opinions, and viewpoints from others in order to find answers to those questions.
- Therefore, it is very much using a prescriptive model of thinking, considering the consequences of actions, hypothesizing, and also synthesizing or testing the information with the theory. Also seeking, identifying, analyzing, and resolving problems

Diagrammatic Representation of Roth's Reflective Process:

We can see these questions or ideas are put into a very simple model. The **reflective practitioner at the center**. If you like conducting reflective analysis. And all these aspects surrounding the teacher everything from keeping an open mind, looking for alternatives, and comparing and contrasting. And the different questions that all these aspects of reflection bring to light.

So, Roth's foundation model describes 24 procedures for the reflective practitioner. The procedure ranges from those that correspond to cognitive models of reflection and teaching techniques to an embracing like "adapt and adjust to instability and change." And what this really means is the model is quite complex but very thorough. Because it is based on questions from the actual teaching is quite simple for the teacher to use.

- It is very much about hypothesis and synthesis, test, and quite interestingly.
- Griffith and Tans (1992) provide a very comprehensive way of categorizing reflective processes that consider both time frames and purpose. This means that it is always about higher-order thinking rather than lower-order thinking. It is not about identifying and recalling. It is very much about deep analysis or critical thinking.
- Then, ideally a process not only of reflection but, also of research and thinking about teaching, review, research, and even repair putting things right which have gone wrong.
- These labels imply how reflection occurs in the individual teacher's mind, but do not occur in isolation. They occur within a context with other teachers.
- So, you can see that this list is reminiscent it reminds us of something. It very much reminds us of the upper levels of Bloom's taxonomy. The levels of evaluating, analyzing, and also creating. And while it does seem to take into account the notion of different time frames, it does not locate the processes in the day-to-day work. So, the model fits into Bloom's taxonomy very neatly.
- And this constitutes professional practice. According to Roth, the labels that he applies within the reflective practice process, all of these questions related to these labels are about the teacher's thinking, acting, and believing themselves to be a professional.

Topic 030: Foundation Models 7: Peters (1991)

Peter (1991) described the reflective practice as a four-step model which he called DATA. DATA stands for:

- describe
- Analyze
- Theorise
- and act

Describe:

It is the first step of reflective practice. It needs to be detailed, needs to be thorough. As we have seen before the word honest comes up again it needs to be honest about what is actually happening.

The description phase, involves paying very close attention to what happened? And what is happening in the classroom. Who was involved? And does not remove that from the feelings, thoughts, or emotions but the teacher is having the way they are reacting and thinking about what has happened.

Analyse:

This leads to the analysis step of the model. This includes the assumptions and the beliefs that the teacher has about their teaching, learning, and what happened. It aims to link at this point or begins to link the theory with the practice. It is about the approaches and the underlying beliefs, rules, and assumptions that the teacher has tried to understand not just what happened but the thinking behind it.

So, here the teacher is very much exploring their own practice, and why this particular practice operates in a particular way. This is why it is quite a deep critical approach. In the analysis phase analyzing the beliefs, values, and assumptions, the source of the information that the teacher draws on what really is required in order for the teacher to make sense of what's going on in the classroom makes sense of the situation.

Theorise:

And this leads to the theorizing step of the model. This is where their practitioner is trying to understand from a cognitive, a conceptual point of view not only what has happened in the classroom and what is happening in the classroom, but what they understand from a theoretical point of view. And in the third step, practitioners theorise about alternative ways to approach learning. They take a theory developed from the previous step and create a new theory.

This requires teachers to think about new or additional sources of knowledge that they can draw on from the literature that they can go to. And even colleagues in the school that they can talk to

about what is happening. Remember sources of theory and understanding are not just what you learn on a course. But they can be much wider reaching than that.

Here, the teacher is thinking about the practice from different perspectives in order to formulate a new or different understanding and consider a new or different approach. Therefore, not just one perspective in the book but a range of books, and also from other teachers' experiences. So, the theory is very wide, very broad and the teacher is looking for as many different viewpoints as possible.

Act:

And the final step in the process is Act. In this step, the practitioner tests the new theory. As in the Greenway model, this is done, but the teacher is actually going to do it in terms of implementing actions, and theories in the classroom. The success of the process is, therefore, not only reflection but the thought is a very clear part of this model.

In the final stage, the teacher put their new thinking into practice by trying out a new or a different way of doing things in order to enhance and improve their practice. And what this would do as with all the models is lead to another cycle of reflection, acting, thinking, and doing. Teachers can use this model as part of a cycle of ongoing learning and continuous improvement. The goal is to integrate scientific and practical theory, and one must reflect on one's practical theory. Therefore, it is not although it looks quite linear DATA and the teacher can always go back and go through another cycle of the process.

Topic 031: Critical Models 1: Brookfield's Critical Lenses

We look at some foundation models for reflective practice. Now, we need to look at some critical models. Again we will be reviewing this briefly and looking at more detail in a later lecture. Perhaps, the most significant critical model is Stephen Brook's field critical lenses (1995). He said that teachers look through different classes, and different eyeglasses in order to reflect. And he identified these four different lenses as,

1. Our **own view of autobiography**.
2. He also said that we can look at what happens in the classroom from the **student's point of view**
3. And also from our **colleague's point of view**.
4. We can also look at what happens in our teaching from a **theoretical standpoint**. And he very conveniently labeled these. As the
 - **Self-lens** (our own viewpoint)
 - **The student lens** (the viewpoint of the students)
 - **The peer's lens** (in other words, the viewpoint of our colleagues)
 - **The theory lens** (which is why it's called the four lens approach.)

Despite the fact, that teacher's professional experience can often be considered as a little bit invaluable from a theoretical point of view. Stephen Brookfield makes a significant point in saying, "The experience and the autobiographies of the teachers is really the foundation point, and the most important aspect of being a teacher." (1995 p.31)

He says that as teachers we must consider this to be more important sometimes than the actual theories. Teachers also should consider whether the learning experience was inclusive and motivational. And he goes on to say that these four lenses are very important because, The most important thing that the teacher can do in terms of teaching, is trying to get inside the student's heads, what are they thinking about learning, what are they thinking about being in the classroom trying to understand how they are learning. (Brookfield, 1995 p.35)

Reflective Practice:

And therefore he made a number of significant points in terms of these four lenses.

- One point is that he says that, reflective practice and teaching actually become about identifying power and how power can support, and also hinder the educational process.

- He also goes on to say that what the teacher needs to do, therefore, is to challenge certain **untested** assumptions in order to address that power imbalance that may be in the classroom.

Applying the Four Lenses:

And in this light, he came up with a number of very useful frameworks for the reflective practitioner.

- a) One is the **critical incident questionnaire (CIQ)**. It is a series of questions that the teacher can ask themselves in order to, reflect on the teaching in the classroom.
- b) He also came up with the **Critical practice or Audit (CPA)**.
- c) Perhaps, most significantly he also identified how the critical incident questionnaire can be used as a framework for a teacher's journal writing, if they are keeping a reflective journal.

Critical Incident Questions:

So, the critical incident questionnaire has a number of significant questions. For example:

- At what moment in class did you feel most engaged with what was happening? This is a significant question for the teacher to consider.
- At what moment in class were you most **distant or** most removed from what was happening?
- What action that anyone **took** whether as a teacher **or a** student was most helpful or most productive?
- **What action that anyone took did you find most** puzzling or problematizing and **confusing?**
- And also what surprised you most? As a teacher, in terms of what you did in the classroom? What were you not expecting? **(This could be about your own reactions to what went on, something that someone did, or anything else that occurs).**

Critical Practice Audit:

- So, the critical practice audit takes a number of critical incidences say two or three in a week and uses the questionnaire to reflect on those incidences.
- The idea is that as a teacher you write a brief description about what happened in a particular incident, including details of what happened, who was involved, where and when it took place, and what made the incident critical to you. And use those questions to reflect on the incident in detail.
- Part of the audits also to reflect on your own assumptions as a teacher about what happened to think about what were the interactions that generated a particular incident. And how you as a teacher challenged what was happening both with yourself and with the students.
- The critical practice audit is also to think about the information the evidence that you have got as a teacher and to check its accuracy. Are you really seeing what really happened? Or are you looking through your own lens the self-lens that autobiographical lens which is perhaps skewing the reality? If you were not able to check them out at the time, how could you check them in the future? What sources of evidence could you consult? And what sources of evidence could you consult?
- What different perspectives could be taken on the incident? Therefore, as you think about it through the eyes of other people (the student lens, and the colleagues lens, the peer's lens, and also the theory lens) involved. Are there different ways the situation could be seen, Or that your behavior could be interpreted?
- Therefore, what we see through these four lenses is a very critical approach self-critical by the teacher and critical of the processes which are taking place looking not only within but also without the school.

Topic 032: Critical Models 2: John Model for Structured Reflection (2000)

John's model for structured reflection is another critical model that we can consider for the reflective practitioner. It is quite a simple model. It has two phases to it.

The **first phase** is “**looking in on the situation**”. This involves the teacher focusing on particular aspects of themselves, and paying attention to **their thoughts, emotions, and** what was happening in the classroom. So, looking **into** the situation is the first part of this process.

The **second phase** is quite simply “**looking out of the situation**” and trying to describe it so that others might actually understand it from your perspective and even from a theoretical perspective. And as such, it is situated then in **the** five sources of knowledge **that** Johns describes and each of these sources of knowledge have very particular cues that we can use as teachers to guide our thinking and reflection. For example:

Look In-

Looking in he said very important that you need to find a place where you can actually focus on yourself. He saw this as a very solitary activity in the initial stages because you need to concentrate and have space and time for reflection. Pay attention to not only your thoughts but the feelings, the emotions which have been generated by the experience. And then of course you need to record the experiences that you **have** writing them down or maybe even using an audio tape.

Looking – Out:

Looking out then requires **writing that description** within the sort of situational surrounding of the thoughts, and the emotions and then trying to find what is significant in that. This is called looking out because in a way, it is like trying to describe it for somebody else to understand, and in this way you get a clearer understanding **of** yourself.

This particular **description of the looking out process can be focused in the following ways.**

Aesthetic:

And we can actually talk about it from an **aesthetic point of view**. Or in other words, the teacher can reflect on what they were trying to do? What they were trying to achieve in their teaching?

And why did they react the way they did? to say some poor behavior or some extremely good behavior in the classroom.

Personal:

There is also a **personal perspective** to this. Or in other words, the teacher's reaction to the experience in the classroom. A good reaction, a bad reaction. How do they feel? what are the emotions associated with that?

Ethics:

We can also look at it from an **ethical perspective**. Or in other words, the teacher should think well. If I did something was it in the best interests of the students or the best interest of the school? And what were the influencing factors on the teacher in terms of their own reaction?

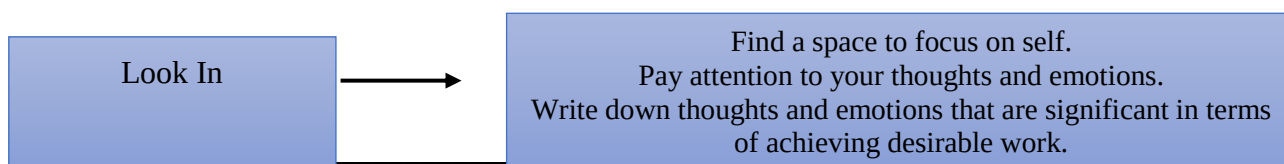
Empirics:

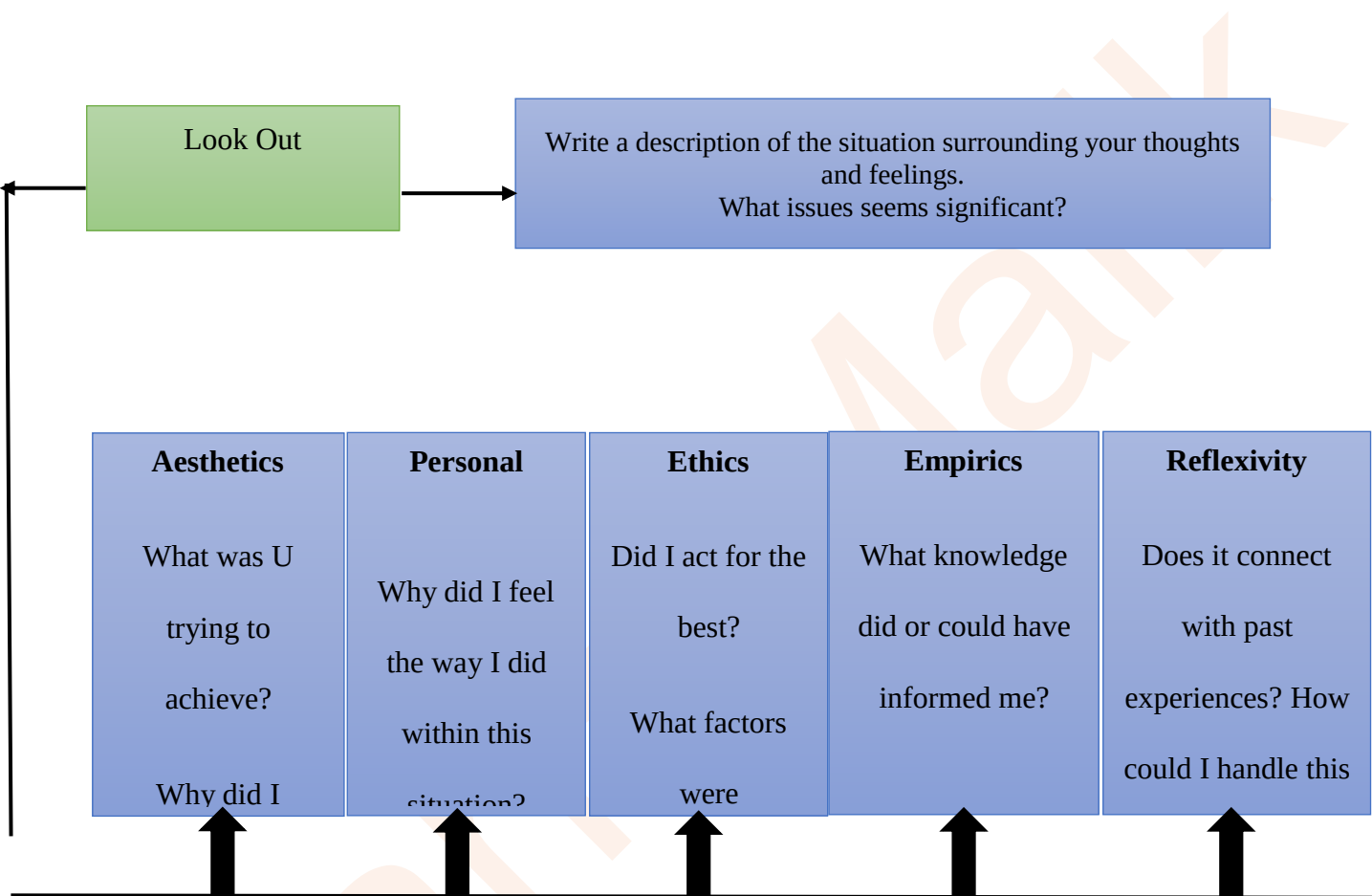
Another perspective identified in this model is the **empiric perspective**. This is very much starting to relate the experience with what the literature says. Or in other words, Is there information or is there some knowledge that could have actually helped me deal with this situation in the classroom?

Reflexivity:

A very significant part of this model, and the particular perspective is about **reflexivity**. This is a term that will come across in future lectures. Reflexivity is about making connections with our understanding. Does it reflect previous experiences that we have in the classroom? Therefore, Can we handle the situation better? And what's very good about these perspectives is that we can represent them in a model which is very easy to understand.

John's Model of Structured Reflection





In one component, we have **looked in** finding the space: describing what happened. That very solitary component but, then **looking out**: describing it for somebody else for our own purposes, and within the looking out we get those different perspectives. From the **aesthetic perspective**, What was the teacher trying to achieve? From a **personal perspective**, how do they feel? The ethics, what factors were influencing them? The empirics the knowledge that they are drawing on, And then **reflecting back** on previous experiences and making cognitive connections.

Topic 033: Critical Model 3: Rolfe, Freshwater and Jasper (2001) – The “What Model”

Another critical model to look at is simply called the “What” model. It is called the “what” model because it uses three very simple questions.

- What?
- So what?
- Now what?

Levels:

Let's see what these questions actually referred to.

- In the **first level**, the individual reflects on the situation in order to describe it. It is very much about **reflecting on the situation**. What happened? What was going on describing? As we have seen in other models.
- The next level, **So What** is reflecting on what that really means in terms of the literature that we have read and our theoretical understanding of the concept that we understand?
- And finally that leads into **Now What**. In other words, their action step of the process of the reflective cycle. At this level, they reflect on the action, what can be done to improve the incident, and the consequences of such action. So, we have, What, So What, and Now What.

Level 1: The Descriptive Level of Reflection:

What

We can very easily think of these in terms of very specific questions and for example, in the first level though what level is? What is the problem? What reason is there for us being stuck at this particular point? And what was my role in the situation? What action did I take?

So, it is always using that keyword what to describe? What actually happened? And what were the consequences for me? What were the consequences for the students? These are just sample

questions. The idea is that the teacher him or herself will use the key **what** in order to frame the questions, and as part of the reflective process.

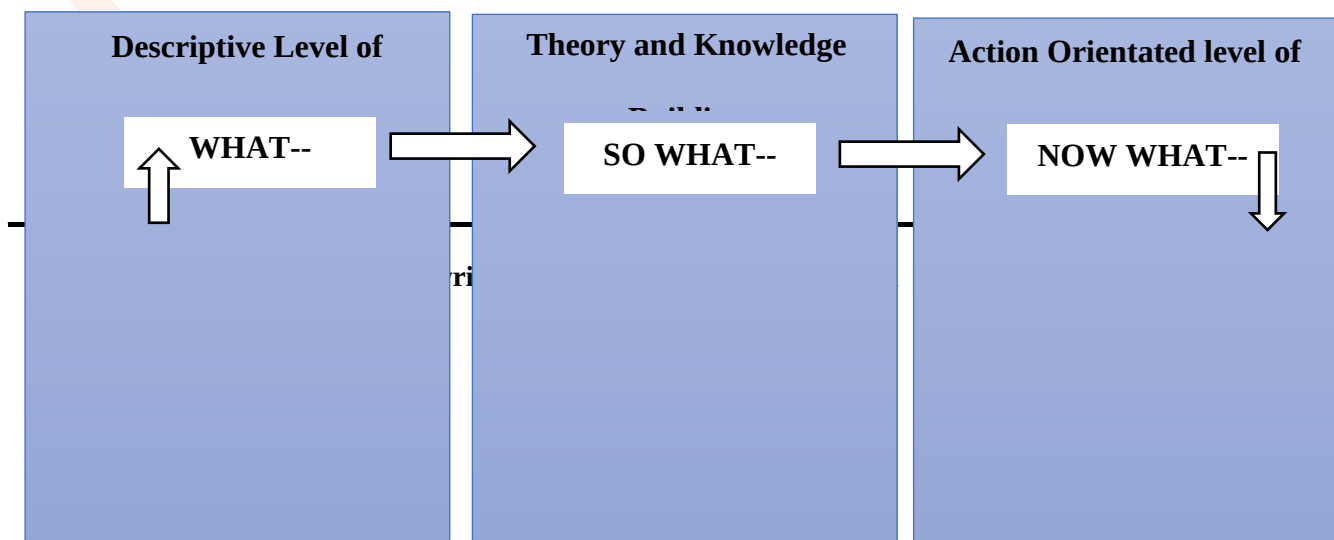
So –what:

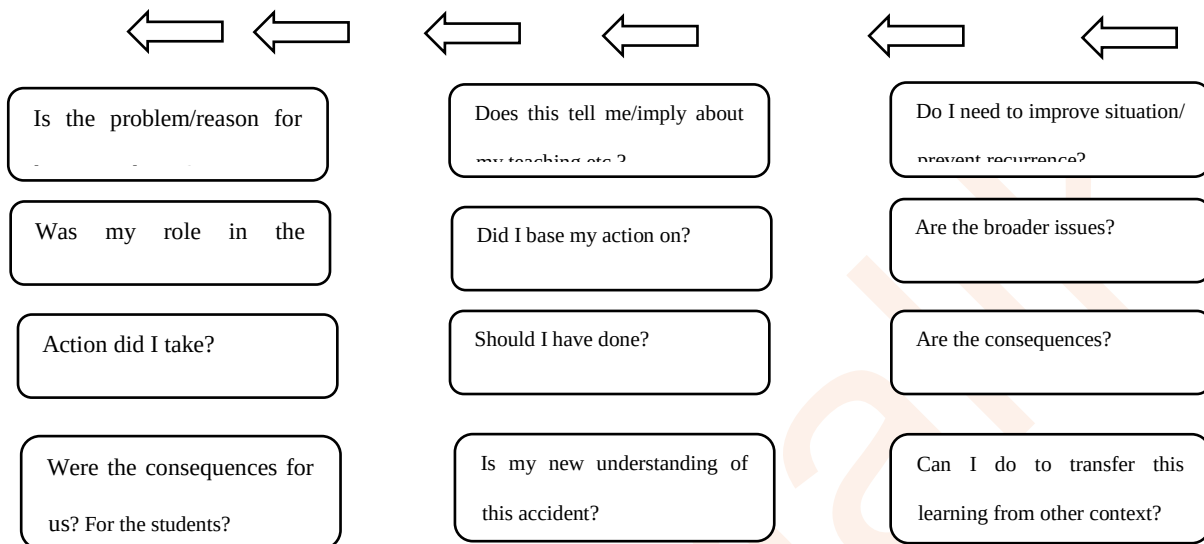
The next level the **so what** theorizing, the knowledge-building component. So what, does this tell me is a good question to ask? So what did I base my actions on? In other words, why I did **do that**? So what should I have done in that particular situation? You can start to see how you can frame particular questions using that cue so what? Another questions is my understanding, so what is my understanding of the incident? So, these questions **are** the first, second, and third level questions you can phrase them in any way you choose, and you can ask any questions to help you think the way you want to think about your teaching.

Now-what:

The action phase now what question. Now-what do I need to do **to improve the situation/prevent recurrence?** Now what are the broader issues at stake here? And now what are the consequences for me or for the students? And now what can I do to transfer this learning to my next teaching session? Again the questions that you can ask here a person you can identify **as** your own they are not fixed. **As** in the other models, we have seen this gives a lot of freedom to the teacher. These **are** very useful frameworks **so what to know is** what gives you the starting point for the questions you can ask. And what seems important with all reflective practice is that whatever process you take, whatever model you choose to use that it is actually part of a cycle. So, these questions these what questions, **what, so what, now what** needs to be iterated a number of times not just once but again and again.

Overview of Rolfe's Reflective Model





We can look at these in terms of again a nice straightforward model of the **what, so what, and now what** components going through repeatedly a number of iterations. Whatever questions you ask at those particular levels is up to you as a practitioner.

- But at the **What level**, it is very much about description, whatever viewpoint you take is entirely up to you. But the **What** question will help you describe the things happening in the classroom.
- The **So What** is where you will take that description and start to link it to your own understandings and theories.
- And this then will lead into the **Now What**. In other words, the action-oriented component of the model.

Critical Models 4: Rodgers (2002)

Rodgers's model of critical reflection very much encompasses Dewey's idea of reflective thought. In this model, Rodgers illustrates reflection as a cyclic process. He has broken down the reflective process into **four phases**.

- The first phase is described as **Presence in experience**. This is different from any other models we have seen presence in experience is about the teacher learning to see really, what is happening, seeing clearly and honestly.
- The second phase is **describing that experience**, describing what has been seen and trying to understand it to differentiate the different components.
- The third aspect is **an analysis of that experience** trying to link to deconstruct, link it to theory and to literature
- And then of course the final stage is **experimentation**. Or in other words, learning to take action from the analysis of the experience.

According to **Rodgers**, interestingly in this model, the idea is that the teacher although going through a cycle can actually move backward, and forwards in the cycle at any particular point. So, it is not necessarily a smooth process but something which can go backward and forwards.

Presence in Experience:

This is because the first element is the teacher being present in the experience and how they perceive it. Because in the model, the idea is that the more the teacher can actually see, the more the teacher can perceive which will affect the rest of the reflective practice process. Therefore, the teacher may need to move backward through the model in order to keep perceiving clearly what is really happening to question their version of reality. So, it is very much about learning to see, differentiating the different parts, giving them meaning, and responding intelligently to what happens in the classroom.

Description of Experience:

It is the differentiation and naming of an experience's diverse, and quite complex elements. So, that it can be looked at seen, and told from as many different perspectives as possible. Because you won't just describe the experience once you would describe it potentially a number of times. Because you'd be moving backward and forwards through the model. There are a number of description that you could make, and you would be constantly trying to refine the description, and

refine your perception of what is happening. And what this is very much about is the teacher being very clear about their own perception and their own apprehension of the way they see. This is the cornerstone of this model. It is quite different from other models, and is present in the description.

Analysis of Experience:

The analysis involves generating a number of different explanations. Therefore, there could be explanations from previous descriptions, as well as later descriptions in order to get that very critical in-depth, and honest perspective of what's been going on in the classroom. It is the phase where meaning-making happens.

Although the analysis comes from the description, there is often a dialectical relationship between the two. The model says that the teacher can move backward. Therefore, the analysis and the description can become very closely linked each depending on the other. The description can become a reaction to the analysis as much as the analysis can be a reaction to the description.

Experimentation:

Experimentation is the final as well as the initial phase of the reflective practice because it doubles as the next experience. This final phase of reflective practice is about action. In this model action is not the word that is used but clearly experimentation. It is very much focused on the teacher trying things out, and testing things in the classroom. And not being afraid of the consequences but learning from the consequences in order ultimately to improve their teaching.

Therefore, what this clearly says about this model is that knowledge from experience acquired knowledge is not something that is constructed in isolation, but it is very much linked to the students that the teacher is working with, and colleagues that the teacher interacts within the school.

Topic 035: Critical Models 5: Gore and Zeichner (1991)

The Gore and Zeichner model of reflective practice is another critical model to just briefly consider. What they suggest is that there are four major aspects of the work of the teacher, professional

work, and what they say. Therefore, the teacher needs to ask and reflect on particular questions in relation to these four aspects.

The purpose of this is to put it for the teacher to gain a thorough whole and holistic understanding of their teaching and their role in teaching, and their role in the classroom. The four components are very clear.

Academic Reflection:

The first component is an **academic reflection**. There are certain questions that the teacher can ask themselves in relation to academic reflection. Do I know my content really well? For example; Am I using the right strategies for teaching and my students' needs? Am I well-organized for my lessons? These are all focusing on the professional role from a very technical and practical point of view. Other questions that can be asked about the sequencing of the content. Was the lesson at the right sort of pace for the students? Was it too fast or was it too slow? Have I completed the planning cycle with suitable, relevant assessment strategies to evaluate student learning?

Also, it is useful to think of this aspect of the model in terms of a creative or innovative you are being in the classroom. And are you trying out new approaches not only to excite your students but to excite yourself or are you just going through a routine. So, it is good to reflect on the academic aspect.

Developmental Reflection: They also talked about developmental reflection. Am I providing teaching and learning contexts that give students a chance to develop their own perspectives on their learning? I like to think of this part of the model in terms of; how well the teacher is helping the students learn, and how to learn. And for me, that is a very significant part of being a teacher in the classroom.

Some more questions to ask in terms of developmental reflection. Have I evaluated the students' skills and thinking to determine the stages at which each of them is able to engage in different learning contexts? So, that you can actually determine how much they are learning.

And what about changes to the instructional approaches you can reflect on that try things out can you extend the use of a particular approach? Or Have I planned suitable instructional and task modifications to accommodate the differences in the students' thinking, emotional and physical capacities? Have I designed teaching and learning activities for individuals as well as groups? Have I taken into account and effectively utilized students' various interests to design lessons and curriculum? For example, if you are using group work, how are the students learning in a group? How are they supporting each other? Or how are they contradicting each other?

Social Reconstructionist (Critical) Reflection:

The most complex aspect of this model is the critical reflection or the social Reconstructionist approach. This takes a very much more holistic view of reflective practice. Actually, looking at what the teacher believes about teaching? What the purpose of education is? And, what the value of education is in society? And so, questions you can ask are things like, Do I have particular philosophical beliefs, and values about the values, purposes, and function of teaching which will affect the technical approach that I take in the class?

Have I critically evaluated certain aspects of teaching from the school's perspective? How do you articulate those and who helps you articulate those? Have I considered who determines the curriculum? And what does that mean in terms of; how you teach and how the students are learning and indeed what they are learning? Have I considered in what way a curriculum supports or neglects the individual learning needs of students from different social, cultural, and individual groups? This is taking you beyond just pure description and is being very critical about the imposing factors that can affect you as a teacher and affect the students as learners.

Are there ways in which I can implement components of the curriculum or different curricula in the school in a more unique and innovative way? So, you are actually meeting the needs of both worlds both the school education system but also the individual learning needs.

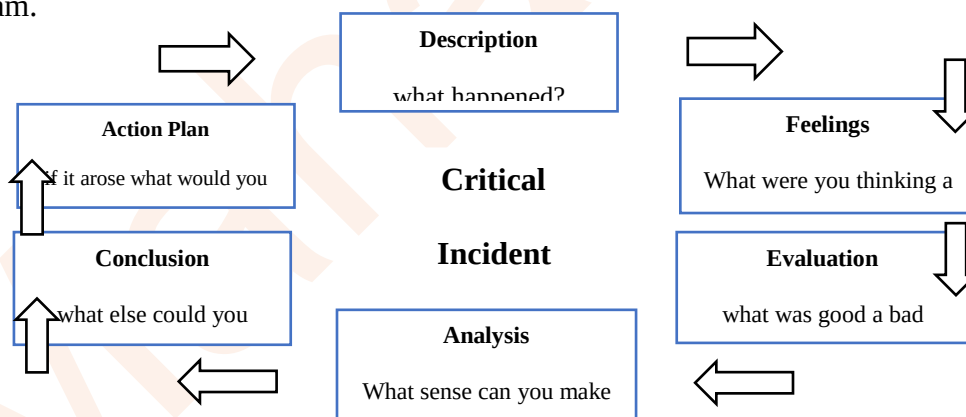
How can I overcome short come shortcomings or problems with particular systems in schools or particular issues in curricula? Therefore, make teaching and learning a much more interesting and creative process.

Have I analyzed the assessment processes may be no compulsory tests are they really testing learning or is it just about testing memory? There are some fundamental questions for the teacher to ask here from this broader critical perspective about their role as a teacher.

Critical Model 6: Gibbs Reflective Cycle (1988)

A very popular critical model is Gibb's reflective cycle. This is a way for the teacher to think about the different aspects of something which has happened in the classroom. Again, it is in the form of a cycle. It is very much about understanding something an event, or a situation describing it, evaluating, and establishing a course of action in relation to that reflection.

This model is very much about helping the teacher understand a particular situation from a very detailed point of view. Because there are quite a few steps in this cycle and we can see this in the diagram.



The starting point here is the **description** of what actually happened in the model. This leads to **feelings and evaluation**. It is important to consider the aspects of the teachers' feelings who knows what were you thinking and feeling about that particular incident? And then **evaluate** what was good about it and what was bad about it. After the evaluation comes, the **analysis** what sense

can you make from this particular model? What sense can you make from what happened? again linking it to the real concept. And from this, the teacher can make some conclusions and come to some sort of decision that will lead to action. Of course, those from the action there will be another chance for a description. So again, we get that spiral of thinking.

So, a great Gibbs reflective cycle has a lot of components to it but again it's all focused on asking particular questions, a particular point in the cycle so. For example:

Description:

In the description phase “What happened?”. It is very much about the teacher describing in detail the event that took place and including particular focus points. In other words, where were you; who else was there; what the students were doing; why were you there; why were other people there; what was the context of the event; what happened? Asking very simple questions but quite probing questions.

In terms of what happened, what was your part in any particular incident that happened in the classroom; what parts did the other people play; and what was the result? Again, with previous models the questions here are not fixed they are very much for the teacher to think about and design for themselves. It's the individual components that add up to make the reflective process.

Feelings:

The feeling component is very much about the teacher trying to recall the reaction to incidents in the classroom. How were they feeling about it? What were the emotions that were involved? How does this feeling make you feel about and the event; when it started; and during; what you were thinking about at the time; how did it make you feel? This is very much about linking the teacher to the activity in the classroom, the actions which are taking place. How did you feel about the outcome of the event? So, in terms of feelings, you are thinking before, during, and after what happens in the classroom?

Evaluation:

The next phase of the cycle is evaluation. What was good and what was bad about the experience? This is for the teacher to make a judgment for themselves about the teaching in the classroom and the actual incident. This leads to further questions about what was good about the experience and what was bad about the experience or what went well; what didn't go well and perhaps to start to consider why.

Analysis: What sense can you make of the situation?

Because from this phase the teacher moves into analysis after the evaluation. What went well; what didn't go well the good experience the bad experience; How does this link to what we know about education about teaching and about learning from all the theories which were exposed to? So, it is very much about breaking down the analysis into component parts and could include questions like well what did you do; what did others do that was good; how does this actually relate to; what we know about theory in what way; did others contribute to what happened in the classroom?

Conclusion: What else could you have done?

This ultimately leads to the point where some judgment is made by the teacher. Some conclusion what else could you have done? For example this is quite different from the evaluation and the analysis stage. Because it is very much about putting a stop to the reflection that's taking place. It has to end at some point and there have to be some decisions. So, it forms part of the reflective cycle. Even though it does actually stop thinking. Because after the conclusion phase, when you are thinking about; how things could be done differently; what were the results; and how this will impact the future.

Action Plan: If it arose again, what would you do?

In the Gibbs cycle, the action plan that the teacher comes up with is a result of all that previous thinking including the conclusion, and the judgments that were made about what was happening in the classroom. With the action plan, you have to consider how you would do things; how you

would do things differently based on the reflection. Perhaps most important to remember is that again from action comes another cycle of reflection. So again, it is continuous even though there is that point of stopping conclusion built into the cycle. Here the cycle is tentatively completed and suggests that should the event occur again it will be the focus of another reflective cycle.

From Theory to Practice: Boud's Triangular Representation

From Theory to Practice

So, we've been introduced to some models of reflective practice. And the value of these models is in helping the reflective practitioner. The teacher begins to deconstruct their experiences and ensuring develop a deeper understanding of the questions and the issues involved in the work that they do.

Boud's Triangular Representation

David Boud (1987) has defined reflection as:

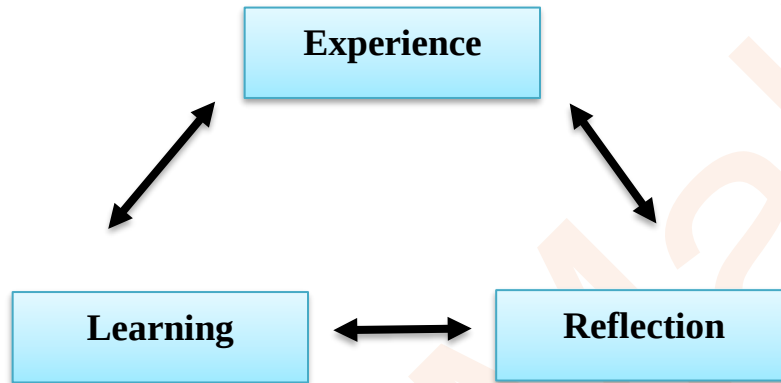
“A conscious activity in which we engage to explore our experiences and develop new understandings and conceptualisation.”

Boud, 1987

The thing about the models of reflection is that they're all cyclical in nature, representing the idea that reflection leads to learning. Cyclic learning is never complicated nor transferable without reflection further and what Boud's adds to this discussion is something quite significant and that's the idea that reflection needs to happen in different contexts without any reflection on application of learning in one context in relation to another means that actually reflection is a waste of time.

And so, this model is an attempt to actually start to address that imperfection and we concede in the diagram here it's a very simple representation.

Boud's Triangular Representation



Boud's idea is that the reflective cycle is there and it can be clockwise or counter clockwise as a reflective process. But clearly, we can see that expression experience and reflection are part of the cycle and what is added here is learning by the teacher learning which can be taken to another context. So, this model attempts to capture the essentials (that experience and reflection lead to learning), the essence of what reflection is really about and what it is aimed at doing is making a clear connection a proper translation in terms of experience and learning.

So let's think a little bit more about this triangular representation.

1. What the reflective practitioner to do is return to the incident the event upon, which they're reflecting and to not only try to describe it for themselves.
2. But also, to understand their emotional response to it the feelings that they have in relation to their teaching.

3. And from this very personal reflection comes an evaluation a revaluation of the teaching in light of experience, knowledge and experimentation, trying to seek a deeper understanding for it.

4. And from this reflection comes some sort of thinking about action in relation to change so there is a planning phase to the cycle.

Boud's recognizes that this is a very individualistic process of learning and this can be criticized because reflective practice is very often seen as a collaborative activity.

So Boud's poses three alternatives:

1) should we reject models a reflection

Altogether this is just adding to the discussion about reflection and reflective practice.

2) Should we rehabilitate them to capture their previous potential

we should try to think differently about what reflection really is for the teacher and what reflective practice really is.

3) Should we move to new ways of regarding reflection that are in keeping with what we know about the context of practice?

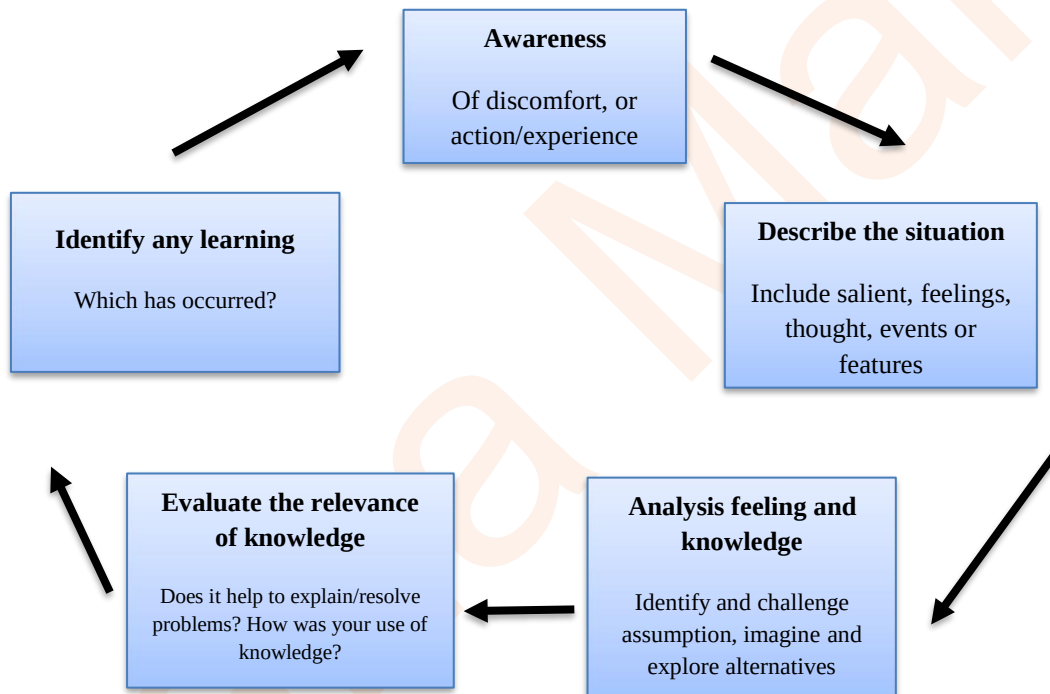
He starts talking about whether the reflection can actually be applied in new contexts and what a context adds to the process of reflection.

From Theory to Practice: The Atkins and Murphy Model

Another model of reflective practice which is aiming to move reflective practice from theory into action is the Atkins and Murphy model. Atkins and Murphy (1993) address many of the criticisms identified by Boud with their own cyclical model:

We can see the model here it has five components.

- Awareness
- Describing the situation
- Analyzing the feelings and the knowledge associated with that situation
- Evaluating the relevance of the knowledge within that situation
- Identifying the learning which comes from that



And we'll look at these in some detail Atkins and Murphy model has five stages to it and the stage is linked together in a cycle.

Stage 1: Awareness of uncomfortable feelings(usually due to new, unfamiliar or negative situations)

Stage one is very much about the teacher becoming aware of some need for change. Because of some uncomfortable feelings due to the situation that they're confronted by.

Stage 2: Examination of components of the situation

Stage two then comes from stage 1 and stage 2 is about the teacher examining those feelings and the components in the situation which generate those feelings.

Stage 3: Analysis of the components of the situation

Stage three leads then into analysis of the situation based on the evaluation

Stage 4: Summary of outcomes of reflection

Then into stage four there is some sort of conclusion a summary of what all those feelings and all that thinking generates.

Stage 5: Recognise learning from the reflection

And then finally into a stage five which is the learning stage and what the teacher has actually gathered from that experience.

So, this model attempts to provide a deep level of support for the reflective practitioner more so than other models tried to do.

And we must remember that this is not to say that the other models that exist about reflective practices are redundant or a waste of time, **but** this is just another way for the teacher to think about their role in the classroom. **It is important to remain alert to the potential to provide superficial responses as the critical, questioning and challenging...** What is important is to realize that as a reflective practitioner part of your role, part of your duty is to critique not only yourself and what is happening in the classroom but also, to add to the critique on the models which can be used for reflective practice.

Atkins **and** Murphy suggest that **for reflection to make a real difference to practice, we need to follow reflection with a commitment to action.** Any sort of reflection is not useful unless the teacher has a real commitment to action. Action is what reflective practice is all about and they

make an interesting connection in relation to action here which is action can happen after the event but it can also happen before the event in terms of planning and preparation, the problem with such models of reflection is that they do not take account of the importance of reflection before action. So, we must remember in this model that it is actually trying to make a connection not only with the reflective process but the actions that the teacher needs to take in the classroom.

Self-Awareness (mindfulness)

- Insight into personal sensations, emotions, thoughts and actions.
- Acceptance/ownership of response
- Recognition/acceptance of emotions, sensations thoughts in others involved.

This involves honest examination of how situation has affected the individual and how the individual has affected the situation.

Description

- Comprehensive account of the situation either verbally or in writing
- Involves the ability to recognise and recollect accurately salient events and key features of an experience.

Critical Analysis

- Involves examining the components of a situation
- Identify personal cognitive and emotional schema

- Identify existing knowledge
- Identify biases and assumptions
- Imagine/explore alternatives
- Integrating new understanding
- Formulate generalisations
- Predict likely consequences of actions

Evaluation

- Validation and appropriation of knowledge
- Along with synthesis this is crucial to the development of a new perspective.

From Theory to Practice: John Smyth's Framework for Reflection on Action (1993)

Smyth's Framework takes the teacher through a number of phases:

- A Descriptive Phase
- Reflective Phase
- An Emancipator Phase

1) The Descriptive Phase

- A Description of Practice events
- Examination of Descriptions for genuineness and comprehensiveness

2) The Reflective Phase.

Reflective analysis against adopted theories:

- Reflective analysis of the situation
- Reflective analysis of intentions

The Reflection phase is again the analysis of what's taking place, analysis not only of events but also the **situation of the context in which they occur** and trying to understanding the intentions behind them

The Reflective Phase Includes:

- Knowledge about practice processes and applications
- Self-awareness

And this model does not remove the teacher from this there is a sense that is the part of the process. Is the teacher developing a much stronger Self-awareness of themselves as practitioners?

3) The Emancipator Phase

- Critique of Practice regarding conflicts, distortions and inconsistencies

Now the interesting part of this model is the Emancipator phase this is where the teacher is very much critiquing their own practice and looking at inconsistencies in terms of what they're doing in relation to what they know they should be doing ought to be doing.

The Emancipator Phase Involves:

- Engagement in Emancipatory and change processes learning and change in practice
- Self-critique and emancipation

This is not to say that therefore the teacher is behaving incorrectly, it is very much about liberation. It's about the teacher beginning to realize that things can get change and those things are not static and that the self-critique is an opportunity for growth and development.

Smyth operationalizes these phases of Reflection for teachers as a set of activities developed from question cues:

Activity: Describe

Cue: what did I do?

That is very simple question, of course you could come up with any number of questions as part of the description

Smyth proposes Reflection for teacher as a set of activities developed from question cues:

Activity: inform (analysis)

Cue: what does this mean?

Another question could be well that what does this actually mean in terms of me as a teacher or

What I'm trying to do within a school?

Smyth propose Reflection for teachers as a set of activities developed from question cues:

Activity: confront (Self-awareness)

Cue: How did I come to be like this?

For example in term of developing a self-awareness that needs some sort of confrontation personal confrontation by the practitioner and cue could be how did I come to be like this in the classroom? , how did I come to feel these things?

So, you can see that the teacher is not extracted from this process at all and very much is within the action and the question help the teacher to focus on this aspect of their practice

Activity: Reconstruct (evaluation and synthesis)

So, in terms of the teachers starting to synthesize and learn from what is taking place there is a sort of reconstruction process and there are particular cues which the reflective practitioner can use.

Cues:

- What do my practices say about my assumptions, values and beliefs?
- Where did these ideas come from?
- What is it that causes me to maintain my theories?

From Theory to Practice: Agyris and Schon (1978)

Perhaps the most well-known model of reflective practice which attempts to link theory to practice is the model by a Garrison. Agyris and Schon pioneered the idea about two types of learning from reflection.

Those reflections are:

1. Single Loop Learning
2. Double Loop Learning

The theory is built around the recognition and amendment of a perceived fault or error.

Single Loop Learning

Single loop learning is when a practitioner or organization, even after an error has occurred and a correction is made, continues to rely on current strategies, techniques or policies when a situation again comes to light.

Single loop learning is about the practitioner reflecting and then acting in relation to something which they have experience in the classroom. It relies on standardized techniques practices in order to solve a problem and does not change those techniques or practices at any particular point.

Double Loop Learning

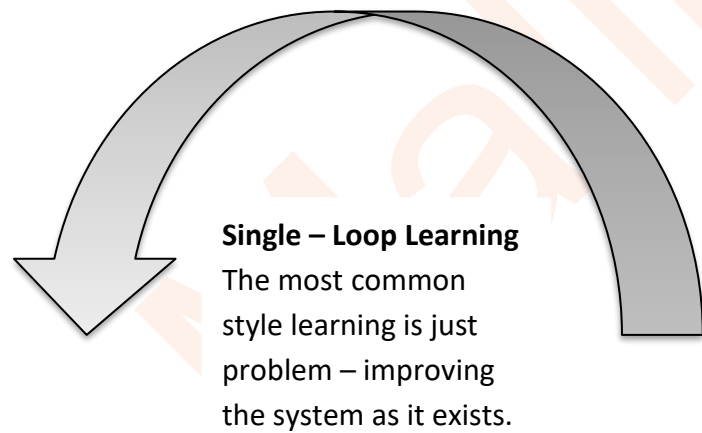
Double loop learning involves the modification of personal objectives, strategies or policies so that when a similar situation arises a new framing system is employed (i.e. new approaches are used to solve it).

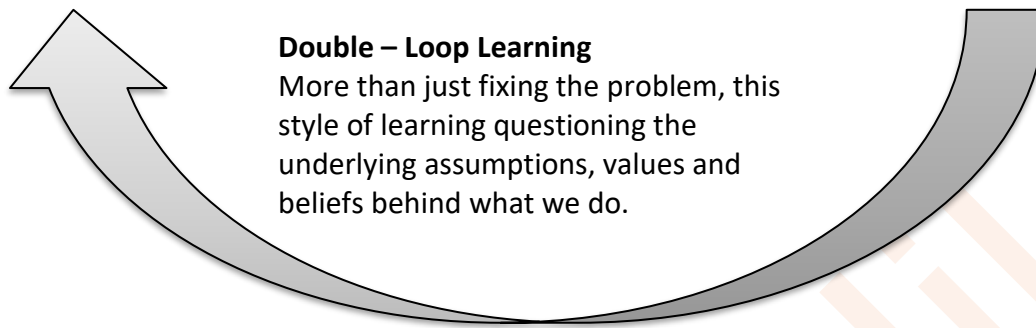
Single loop learning is in contrast to double loop learning where the learning can impact on the practices or the approaches which the teacher uses in order to change them.

So therefore, a reaction to an event can actually change with time with single loop learning their reaction would not change.

Explanation Through a Diagram

It is possible for us to look at this in very simple diagram (i.e., given below) and we can see here that single loop learning just takes one step back from the results, the event and the experience that the teacher has and through to strategizing approaches to deal with that event in the future. They can be very standardized approaches.





With double loop learning, it is a little bit deeper in the sense that the teacher is thinking about their assumptions about the event why things happen, their beliefs around why things happened which they can inform the strategies and therefore can possibly change the strategies and then these strategies are used to deal with the event in future.

So, single loop learning and double loop learning are two approaches within this model to change practice but one is quite fixed and other is very flexible.

So, single loop learning then does rely on fixed practices single ideas and repetition of approaches rather than trying to extend practice.

Single Loop Learning involves connecting a strategy for action with a result. For example, if an action we take yields results that are different to what we expected is through single-loop learning. Therefore, there is a certain amount of **automaticity** about how the teacher approaches their work in the classroom. There is less thinking and trying out new ideas rather than just relying on old practices.

Double – Loop Learning is the cyclical process of applying a new strategy to achieve an outcome may occur several times and we may never succeed. Running out of strategies may push us to re-evaluate the deeper governing variables that make us behave the way we do.

Double loop learning is much more creative as a process in a way somewhat experimental for the teacher changing assumptions means that the approaches will change, the approaches may work

or the approaches may not work and therefore this leads to a new experience and therefore a new double loop.

Complex Ways of Processing Information are:

- Re-evaluating our goals
- Reframing our goals, values and beliefs

The above ways are more complex way of processing information and involves a more sophisticated way of engaging with an experience.

The key point of **double loop** learning is about the teacher re-evaluating their own perceptions of what they are doing by looking at their goals and trying to reframe their assumptions and beliefs about teaching. This is called double-loop learning and looks at consequences from a wider perspective. Whereas single loop learning focuses very much on a much smaller, a much more mechanistic perspective.

In situations where teacher employ double loop learning, **not only will we evaluate our own goals and beliefs, but also those of the people we interact with.**

Double loop learning is very much about the teacher looking much more deeply at their own role where single loop learning actually is becoming quite instrumental in terms of how the teacher is approaching teaching and both can have a constructive outcome.

To have a constructive outcome, we may need to modify and adapt some of these variables to create an attitude that is open to many cultural values and application methods. Double loop learning is considered to be much more constructive potentially because of the opportunity for change and the opportunity for growth.

Mezirow's Transformational Learning

Mezirow (2000) argues that reflection only leads to learning if it leads to transformation. Mezirow suggests that transformation occurs where the original starting point causes a dilemma that needs to be addressed. and then suggests that there are a series of possible phases which may be gone through:

Mezirow suggests that there are a series of possible phases which may be gone through:

- A disorienting dilemma includes loss of job, divorce, marriage, back to school or moving to a new culture.
- Self-examination with feelings of fear, anger, guilt or shame.

In this way there are a number of phases for the teacher to go through. Mezirow talks about a disorienting phase. Something which causes some issues, some problems for the teacher and because of that issue or problem, a teacher needs to go through a process of self-examination. For instance, what are their feelings in relation to the experiences? how they are thinking about it? And how they are trying to rationalize what happened?

- A critical assessment of assumptions
- Recognition that one's discontent and the process of transformation are shared.

This then leads into an exploration of the assumptions that the teacher has and the value of those assumptions about their own role as a practitioner. And from this, then there can only be a process of transformation taking place.

- Exploration of options for new roles, relationships and actions
- Planning a course of action
- Acquiring knowledge and skills for implementing one's plans

The exploration of self that Mezirow is talking about here is very much a process of transformation. In the sense that it's about exploring, planning and developing new knowledge and understanding about the role that's what transformation is about in this model. It's about the teacher developing a much clearer perception, much deeper understanding and more clarity in the knowledge about what they do.

- Provisional testing of new roles

- Building competence and self-confidence in new roles and relationships

So, part of transformational learning needs to be about testing out the different roles that the teacher can take on. And within those roles, building up a level of confidence in relation to the role in the classroom. With time and with transformation, the teacher can become more confident as a practitioner.

- A reintegration into one 's life on the basis of conditions dictated by one 's new perspective

In order for that confidence to occur, that self-confidence to develop, feelings and emotions along with assumptions and beliefs have to be integrated into the teacher's perception of themselves. And this starts to develop different perspectives for the teacher on their role in the classroom.

What Is Particularly Striking about the Mezirow's Ideas?

- Particularly Striking About the Mezirow's Ideas is the emphasis on conscious, self-managed learning.
- Learning is not a direct result of the experience; it happens because the individual takes charge of their critical reflection and explicitly plans and carries out steps to learn from it.

Therefore, learning is not a direct experience. It is learned experience. It is acquired experience in the classroom. And it is about linking knowledge that is acquired from training, literature or research with the experience that is had in the classroom.

The following is a sequence for reflectivity draw from Mezirow 's model:

- A disorientating dilemma
 - Picture the event
- Critical Assessment of Assumptions:
 - What does it mean to you to feel this?
 - What advice are you giving yourself in the picture?
 - How do you interpret what is happening?
 - What is your intention?

And, in terms of that critical reflection, it's very important here for the teacher to think about what they were actually trying to do rather than externalizing the process of teaching. It is about internalizing teaching and understanding what their purposes were.

- **Exploration of new roles**

- How would you prefer this to be different?
- When this begins to occur for you, what will be different about you?
- How do you interpret what is happening?
- What is your intention?

The exploration of roles becomes part of this transformative learning and really is the underpinning concern in relation to this model. How does the teacher interpret? What is a teacher interpreting? How critical is the teacher being? It is only with questions like these that the level of reflection developed transformation. If the reflection is quite superficial then learning is actually not going to be transformational, change is not going to happen. So, it does require a certain level of criticality.

- **Planning a course of action**

- What are you aware of that keeps this from happening?
- Dangers to change?
- Benefits to staying the same?
- Does it happen a little bit sometimes already? What's different at those times?

The transformational model has to lead to some sort of action. It's not just about thinking and it's also about doing and really then the teacher needs to think about the action in terms of change as a change process. And what are the consequences of that change? Could there be some dangers involved in it? How will it affect students' learning? What are the negative effects as well as the positive? And, this is very much then as a transformational process something which through a cycle is repeated.

- **Acquiring knowledge and skills for implementation**

- What will you need to know/accomplish/overcome for this to occur?

The interesting thing about this because it's focused on learning. It's very much about acquiring knowledge from experience rather than knowledge from texts. And, a lot of people say that this is perhaps the most valuable knowledge for the reflective practitioner to acquire.

- **Trying out new roles**

- **How will you know when you are more on track?**

So, it is about trying out new roles, experimenting and acquiring knowledge through that experimentation to become a better teacher.

Modelling Reflective Practice: Criticisms of Reflective Practice

- A number of writers discuss key concerns of RP in terms of its problematic side: Brookfield (1994, 1995) explains the cultural and personal risks involved.

You can see by now that there are many models of reflective practice and now, we should also look at some criticisms of reflective practice. And, perhaps the most significant figure to look at in terms of critique of reflective practice is Stephen Brookfield. He actually identifies that for the reflective practitioner, there can be a significant number of personal risks involved both in terms of their confidence and also in terms of reflecting critically on their role.

- Ash and Moore (2002) say that new teachers can choose not to reflect constructively and critically, falling back on pre-conceived understandings.

Ash and Moore actually say that there are certain preconceived ideas and preconceived understandings about reflective practice which can interfere and, in a way, deconstruct the actual process.

There are four areas of concern about reflective practice:

- Ethical
- Professional
- Pedagogic
- Conceptual

Whatever criticisms we look at, they tend to fall into four categories. And we can see the categories related to the ethical considerations around reflective practice as well as professional

considerations. There are of course the considerations around teaching (that pedagogic considerations) and of course, there is the conceptualization of reflective practice itself.

Ethical Criticism of RP

- Concerns related to confidentiality, rights to privacy, informed consent and professional relationships.
- Practitioners engaging in RP need to be aware of the risks and the potential for conflicts of interest

Professional Concerns

- Relevant when RP is done badly or inappropriately. In this case the value of RP is missed and it simply rationalizes existing practice
- If applied uncritically, it can reinforce prejudices and bad practice
- The inappropriate use of RP may devalue teachers' professional work instead of promoting it.
- Where teachers follow RP models in mechanical, routinized or instrumental ways, they fall into the trap of engaging neither critical analysis or their emotions.
- At the level of the individual teacher, it can be means for organizations to ignore their responsibilities so that fostering good practice is a matter for the teacher rather than the organization.
- Two key concerns relate to 'developmental readiness' and the extent to which forcing student teachers to reflect may prove counter-productive.

Pedagogic Concerns: Developmental Readiness

- Teachers need to be developmentally ready to engage in critical reflection and some individual may not be capable (ready) for reflection.
- The respective abilities of 'novice' and 'expert' are relevant here. Novices tend to follow models mechanically and experts tend not to rely on models.
- As compulsory element of teacher education courses or of organizational requirements, RP ends up being superficial, strategic and guarded.

- As part of an assessment process, self-examination may will be discouraged.
 - Reflection and assessment are incompatible.

Conceptual Concerns

- Problems are posed with the way teachers embrace RP in uncritical and reductionist ways
- Often ideas are transplanted without sufficient care across philosophical, disciplinary and cultural boundaries
- Reflexivity is a concept which is easily miscommunicated; it can be understood in many different ways according to the aims and function of the RP activity.

Criticism of RP

- Regarding these issues therefore, care needs to be taken to recognize complexity and problematize what is involved in RP rather than accepting it as ‘fact’ or as ‘self-evident’.
- The ethical, professional, pedagogic and conceptual concerns presented here neither condemn RP nor suggest that it should not be used.
- What these concerns indicate is that RP should be applied selectively, taught sensitivity and generally used with care.
- Practitioners need to be critical and reflexive about the tools they are being given and not use them blindly.

A Critical Thinking Framework: What Is Critical Reflection?

Critical thinking involves:

- Analyzing
- Conceptualizing
- Defining
- Examining
- Inferring
- Listening
- Questioning
- Reasoning
- Synthesizing

What is Critical reflection?

- With a purpose to evaluate information in a disciplined way to help us evaluate and refine our thinking.
- As reflective practitioners we must want to be better at thinking, to seek out and be guided by knowledge and evidence that fits with reality:
 1. Belief
 2. If shown to be unfounded
 3. Leads to a change of position and an appropriate response

We can look at this from a very simple point of view that the teacher, the reflective practitioner has a set of core beliefs which guide the actions in the classroom. And through reflection, they can decide whether these beliefs improve or support what is taking place or indeed can actually detract can cancel out the effectiveness of what is happening. And this process leads into some sort of change in understanding, some sort of change in position for the teacher in relation to the way they are teaching.

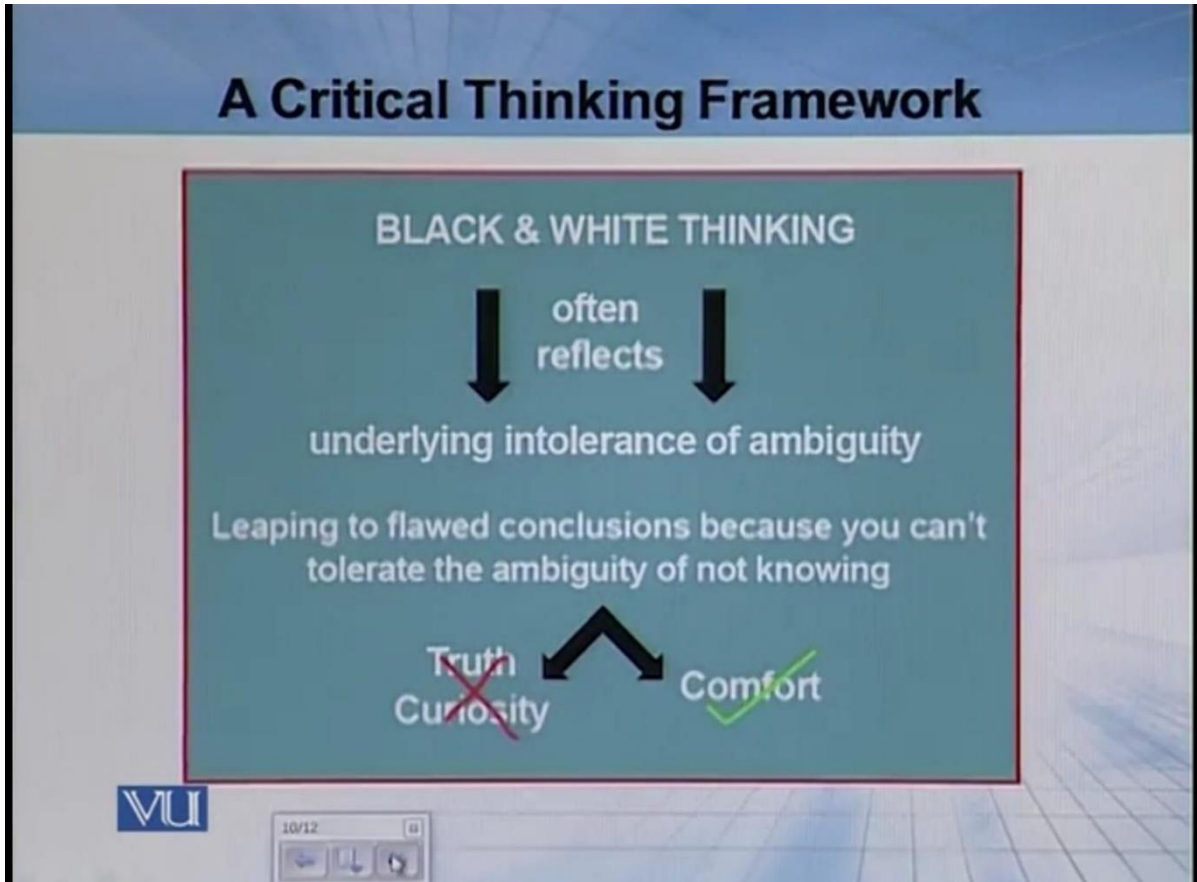
Critical Thinking Involves Reasoning

- It based on sound and consistent logic
- Not on emotions or social pressure
- Truth is not determined by the emotions that accompany them nor on the beliefs of social groups.

This can actually get in the way of critical thought; it can actually prevent critical thought from happening. And so, the teacher's role here is not about dismissing reason but really about looking honestly at themselves and perhaps having to be honest about the flaws that they have as teachers.

- **Critical Thinking Involves Understanding**

So, it is very clearly based in black and white thinking, right and wrong thinking about what is taking place and where the flaws are not accepted. This can actually move the teacher to a point of comfort which is not good necessarily if bad practices are being continued rather than accepting truth and then becoming curious about what is actually happening.



The difference between truth, curiosity and comfort here is quite significant. Because with comfort, there can be complacency and with complacency, people can get into bad habits in the classroom. Where there is an honest reflection and critical thinking approach to reflective practice then there is truth and with truth comes the opportunity for change.

Critical thinking moves us away from:

- **Rash conclusions**
- **Mystification**
- **Reluctance to question**

So, critical thinking then is moving away from the emotionality of the role of teaching and becoming much more rational about it. And it's moving away from making quick decisions and it's moving away from superficial understandings and beliefs and it's moving away from being scared of criticism.

Critical thinking moves us towards:

- Intellectual discipline
- Clear expression of ideas
- Acceptance of personal responsibility for own thinking

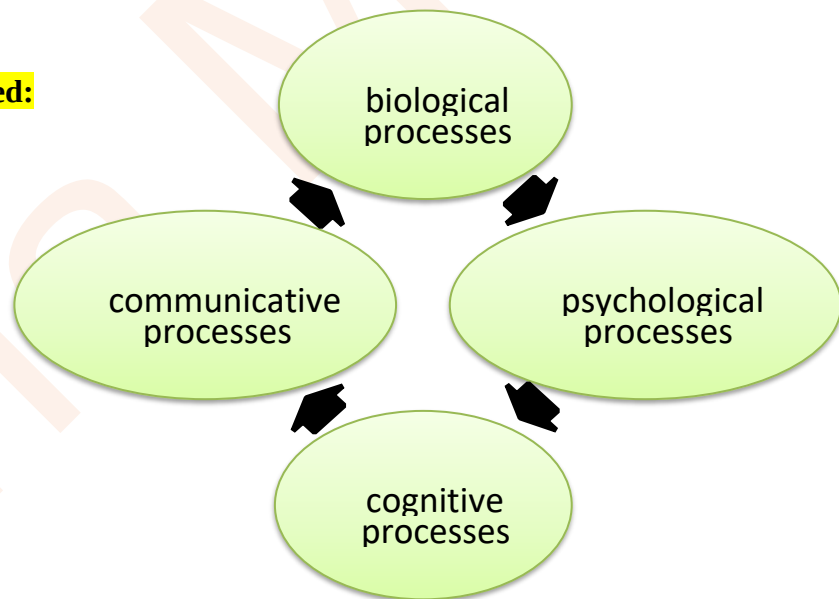
It is actually moving towards a much more intellectual perspective in terms of the discipline of teaching and an acceptance therefore that the teacher has a particular responsibility in terms of what is happening in the classroom.

A Critical Thinking Framework: What Is Critical Reflection?

Critical thinking is part of the reflective practice process. We need to consider some of the mechanisms involved in critical thinking.

The thinking process is as followed:

- Biological process
- Psychological process
- Cognitive process
- Communication process



Other types of thinking or “non- critical” thinking

- Habitual Thinking
- Brainstorming
- Creative Thinking
- Prejudicial Thinking
- Emotive Thinking

There is habitual thinking, the ways that we regularly think. Brainstorming, creative thinking, there's also prejudicial thinking and emotive thinking.

Let's look at these each in turn:

Habitual thinking

- Thinking based on past practices without consideration of current data.

Habitual thinking is what we will do regularly, in a way sort of unthinking and we will rely very much on that in everyday practice.

Brainstorming

- Saying what comes to minds without edit, judgement or evaluation.

Brainstorming is very much about making judgments and evaluating thinking creatively and innovatively about a situation.

Creative Thinking

- Putting facts, concepts, ideas and principles together in new and original ways.

Creative thinking is making links between facts and concepts, looking at the principles of what we know and questioning those principles.

Emotive Thinking

- Responding to the emotions of a message rather than the actual information presented.

With any experience that we have, there is always an emotional response and part of critical thinking is to understand that aspect of the process.

We are thinking critically when we are:

- Thinking rationally so that we rely on reason rather than emotions
- Requires evidence, ignore no known evidence, and follow evidence where it leads and are concerned more with finding, the best explanation than being right.
- Analyzing apparent confusion and asking questions.

We are thinking critically when we are self-aware so that we:

- Weigh the influences of motives and bias, and
- Recognize our own assumptions, prejudices, biases, and point of view.

We think critically when we are honest so that we:

- Recognize emotional impulses, selfish motives, immoral purposes or other modes of self-deception.

We also of course need to be honest because the word critical in critical thinking assumes for us that there will be no a cover-up of reality.

We are thinking critically when we are:

- Being open-minded so that we evaluate all reasonable inferences.
We also as critical thinkers need to be open minded because without that mindfulness, it would
- Consider a variety of possible viewpoints or perspectives, remain open to alternative interpretations.
- We're thinking critically when we're being open-minded so that we accept a new explanation, model, or paradigm because it explains the evidence better, is simpler, or has fewer inconsistencies or covers more data.
- By being open-minded so that we can accept new priorities in response to a re-evaluation of the evidence or re-assessment of our real interests and do not reject unpopular views out of hand.

By being open minded:

- we're accepting new visions
- new versions
- new priorities about teaching and learning

What we are clearly doing is not necessarily rejecting particular views but trying to understand all the different views which we're being confronted with.

We are thinking critically when we are disciplined so that we:

- Are precise, meticulous, comprehensive and exhaustive.
- Resist manipulation and irrational appeals, and
- Avoid snap judgements.

We need to be very disciplined. Critical thinking is not something which can happen superficially. It is very meticulous, very comprehensive, very thorough and therefore, what the

teacher cannot do is just jump to conclusions very quickly. Conclusions need to be arrived at overtime very carefully.

We are thinking critically when we are able to make good judgments so that we:

- Recognize the relevance and/or merit of alternative assumptions and perspectives.
- Recognize the extent and weight of evidence.

Critical thinker needs to be able to make good Judgments in relation to their role as teachers. And what this requires then is a look at the evidence, the value of the evidence, the weight of that evidence. If we react too strongly to something which is perhaps inconsequential not so important then that's actually not a good judgement, we need to be able to understand the weight of the evidence.

A Non- Critical Thinker

- Sees things in Black and white
- Sees things as “yes” or “no” with no subtitles.
- Fails to see linkages and complexities in information and situations.
- Fails to recognize related elements.

We can look at the opposite side of this or what is a non-critical thinker? A non-critical thinker is somebody who sees something very plainly. For example, they see questions as yes or no not as a maybe or it could be. And of course, somebody who is a non-critical thinker, is not able to see the connections and the linkages between things which they are experiencing. A non-critical thinker fails to recognize all the many elements involved in teaching.

Nine Attributes for Teachers to Become Critical Thinkers

- Be open-minded to new ideas
- Know that people have different ideas about the meaning of words.
- Separate emotional and logical thinking.
- Question things that don't make sense.
- Avoid common mistakes in your own reasoning.
- Don't argue about something that you know nothing about.
- Build a string vocabulary to better share and understand ideas.

- Know when you need more information.
- Know the difference between conclusions that could/must be true.

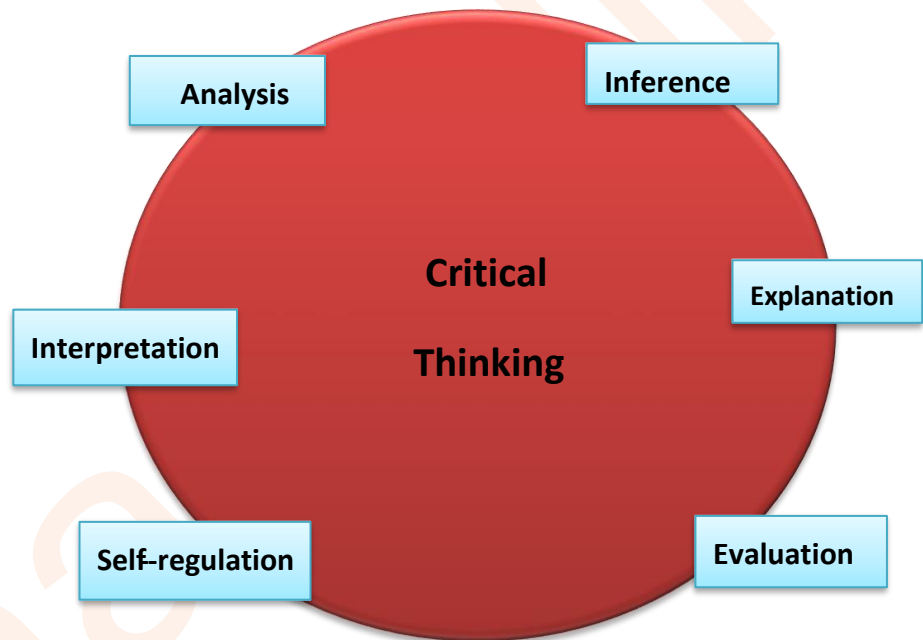
A Critical Thinking Framework: Core Ideas About Thinking Skills

- Critical thinking skills are extremely important in developing as a successful reflective practitioner.

Core Critical Thinking

Skills

1. Analysis
2. Inference
3. Explanation
4. Interpretation
5. Self-regulation
6. Evaluation
7. Interpretation



Interpretation

What it means: Having the ability to understand the information you are being presented with and being able to communicate the meaning of that information to others. As a teacher you will be presented with a variety of information in many different types of situations. Interpretation skills will enable you to better decode information and add clarity to what you have discovered which in turn will help you better understand your students.

Analysis

What it means: Having the ability to connect pieces of information together in order to determine what the intended meaning of the information was meant to represent. Having this skill will better

provide you with the ability to ‘read between the lines’ and help you understand how things you do will impact on your students and your organization.

Inference

- What it means: Having the ability to understand and recognize what elements you will need in order to determine an accurate conclusion or hypothesis from the information you have at your disposal.
- It is important for you to be able to understand what additional information you may need in order to determine what happens. The ability to understand the information you already have and determine what you may still need to find is an important skill.

Evaluation

- What it means: Being able to evaluate the credibility of statements or description of a person’s experience, judgment or opinion in order to measure the validity of the information being presented.
- You will need this skill to evaluate your students’ responses, so you can determine if the information you receive is valid and whether or not it needs to be further looked into.

Explanation

- What it means: Having the ability to not only restate information, but add clarity and perspective to the information, so it can be fully understood by anyone you are sharing it with.
- Your ability to clearly explain your ideas while keeping in mind who you interact with is important for making sure the information is understood and well received.

As a teacher, you try to look at the information and **phrase** it in a different way, so you get clarity about what it actually means. It’s a restating of the problem.

Self - Regulation

- What it means: Having the awareness of your own thinking abilities and the elements that you are using to find results.
- Your ability to understand the extent of your knowledge and that referring to a colleague with more knowledge on the subject is the best way to develop your skills and knowledge.

The final core **component** of critical thinking is the Self –regulation. This is the component about self-awareness of the teacher. The teachers’ own ability to think through a problem, to come to conclusions and to make those conclusions lead to actions

So, your ability to be able to think through, to rationalize and to come to conclusions is the core process of critical thinking and these ideas are significant in terms of that.

Critical Thinking Development: A Stage Theory

For critical thinking Development we can look at a staged approach and there are a number of stages that are at an individual can go through as a critical thinker.

Stage 1- The Unreflective Thinker

Stage 2 – The Challenged Thinker

Stage 3 - Is the Beginning Thinker

Stage 4 – The Practicing Thinker

Stage 5 - The Advanced Thinker

Stage 6- The Accomplished Thinker

Stage 1

- Unreflective thinkers are largely unaware of the determining role that thinking is playing in their lives and of the many ways that problems in thinking are causing problems in their lives.
- Unreflective thinkers lack the knowledge that high quality tinkering requires regular practice in taking thinking apart, accurately assessing it, and actively improve it. In fact, unreflective thinkers are largely unaware of thinking as such, hence fail to recognize thinking as involving concepts, assumptions, inferences, implications, points of view, etc.

Stage 2

- Thinkers move to the “challenged” stage when they become initially aware of the determining role that thinking is playing in their lives. And of the fact that problem in their thinking are causing them serious and significant problems.

From being an unreflective practitioner the unknown reflective thinker, the reflective practitioner can move to being challenged thinker. This is where they’re starting to think a little bit more and understand the role that thinking plays in their lives. At this point the thinker the teacher is starting to identify problems and trying to understand the causes of problems.

Again, it's involving standards for the assessment of thinking, clarity, accuracy, precision, relevance, logicity etc., a level of clarity and which the unreflective thinker wouldn't have.

Through the process of thinking that the reflective practitioner is starting to get a feel for certain standards of teaching and to understand that they are not meeting those standards.

Challenge thinker is also developing some understanding of the role of self-perception and self-concept for themselves as a teacher and the impact that this has on teaching.

At this stage the thinker develops some reflective awareness of how thinking operates for good or ill. At stage two also the reflective practitioner is not really thinking about the impacts of the teaching process on the students.

Stage 3

Those who moves to the beginning thinker stage are actively taking up the challenge to begin to take explicit command of their thinking across multiple domains of their lives. Thinkers at this stage recognise that they have basic problems in their thinking and make initial attempts to better understand how they can take charge of and improve it.

- At stage three we have got the beginning thinker, this is where the reflective practitioner is starting to develop a level of competence as a thinker.

Stage 4

- Stage 4's aspect of thinking is much more developed where the reflective practitioner is taking more charge more responsibility of the thinking process itself
- They not only recognize that problems exist, but they're beginning to think about those problems within a bigger context not just from a personal point of view but from a whole school point of view.
- Based on their need to practice regularly, they are actively analysing their thinking. However, since practicing thinkers are only beginning to approach the improvement of their thinking in a systematic way. They still have limited insight into deeper levels of thought and thus into deeper level of

Stage 5

Thinkers at this stage have now established good habits of thought which are "paying off."

Based on these habits, advanced thinkers not only actively analyse their thinking in all the significant domains of their lives, but also have significant insight into problems at deeper levels of thought.

Stage 6

- The accomplished thinker, this is where thinking is very technical it is actually very natural for this level of thinking.
- They are able to monitor and revise and adapt very quickly in relation to experiences in the classroom.

- This leads the reflective practitioner to be able to strategize very quickly in order for continual improvement to occur.
- It is conscious, but it is also very intuitive, it feels very natural for the thinker at this point.
- This level can only be achieved with experience and you would not expect a new teacher to be able to reach this level of critical criticality until they've had some number of years of experience.
- Therefore, it's linked very much to their lives as teachers, the experiences of teachers and the range of experiences that they have the problems, that they are confronted by.

Critical Thinking

- Is not a matter of accumulating knowledge- a person with a great memory and a good memory and who knows a lot of facts is not necessary good at critical thinking.
- Is being able to deduce consequences from what is known, to solve problems.
- It's not a matter of just building up knowledge of acquiring knowledge over time it's very much about the teacher starting to make connections and to develop an understanding of consequences from actions in order to solve problems.
- It's not about having a good memory it's about having a good ability to reason and to understand.

Someone with critical thinking skills is able to do the following:

- Understand the logical connection between ideas.
- Identify, construct and evaluate arguments
- Detect inconsistencies and common mistakes in reasoning.

- Solve problems systematically.
- Identify the importance and relevance of ideas
- Reflect on the justification of one's own beliefs and values
- Is very important in the new global knowledge economy

We can see then that critical thinking is actually a very important thing, the 21st century needs people who can think much more globally and with a greater understanding and application of knowledge, and the teacher is somebody who has a significant impact in this new global thinking.

- The teacher has to be able to deal with changes quickly and effectively
- The modern era places demands on flexible intellectual skills and integrate diverse sources of knowledge

This new 21st century global perspective about critical thinking and therefore about reflective practice, does suggest that what we need as teachers are very flexible thinking strategies, flexible approaches to understanding things and intellectual skills which can adapt and change, integrate and also be very diverse in terms of using sources.

- Enhances language and presentation skills
- Thinking clearly and systematically can improve the way we express our ideas

Critical thinking also enhances language skills, presentation skills, being able to think clearly and systematically does involve the need to express ideas to ourselves and to others.

Critical thinking:

- Promote creativity
- Critical thinking plays a crucial role in evaluating new ideas, selecting the best ones and modifying them if necessary
- Critical thinking is crucial to self-reflection.
 - Teachers need to justify and reflect on their values and decisions.

➤ Critical thinking provides the tools for this process of self-evaluation.

- Critical thinking is a rational and intentional cognitive skill

We use critical thinking to take a decision and solve a problem

The key characteristics therefore about rationality, about integrity, about reflection and about being able to evaluate.

- Critical thinking follows rules of logical and scientific reasoning
- It is about scientific reasoning to an extent, it's also very logical and part of being intentional then is about seeing how things can combine to make sense for us as thinkers how do all the connections come together for us as a critical thinker.

A Scheme for Critical Thinking and RP

Venn Diagrams and The Reflective Practitioner

- It is possible for the reflective practitioner to frame Venn Diagrams as a tool for thinking about practice.
- A simple Venn diagram for RP only has 2 circles. This represents the structure of professional learning.
- The diagram presents a concrete experience and the teacher conceptualizing the meaning of that experience.
- We can see in the diagram above, on the left-hand side the concrete experience and there can be a number of elements that the teacher could include in this part of the Venn diagram.
- The circle on the left represents “self-structure”. We can more easily understand this as the learning already held by the teacher from previous experience and even perhaps from formal professional learning. This area represents the entire perceptual experience of the teacher.

- It also of course represents that their values and their beliefs and the assumptions that the teacher has about self, others and society and about teaching and their role as a teacher.
- in this field, we can see that, for example, there are three elements of internal understanding- labeled here as '1', '2', and '3'. Of course. in reality there would be more than 3. And for any specific aspect of critical thinking and reflective activity, these would be very focused on only certain aspects of the individual teacher's self-concept.

For example:

- Element 1 may be the teacher's own competence with algebra, element 2 could be the teacher's own feelings of low self-confidence doing algebra. And element 3 may be the teacher's ability to communicate simply the ideas and mechanics of algebra.

The teacher, if they were teaching elementary level mathematics could be reflecting on elements of their own competence in terms of the subject that they are teaching. I was considering algebra at this level and maybe the teacher needs to think about themselves and their own knowledge about algebra.

Element two could be the teachers own feelings of confidence in relation to that subject area. Element 3 would refer to the teachers' feelings and ability about communicating that knowledge to the students so very much based in the concrete experience.

Next consider the right field

- The circle on the right represents 'experience' which is something the reflective teacher draws from action in the classroom.

On the right-hand side, we can actually see a number of elements and these refer to aspects of how the teacher is conceptualizing the process drawing on action in the classroom.

- It could be about the teacher trying out new things with students, or teaching something for the first time.
- It represents the immediate field of sensory experience-direct observations of occurrences as well as the feelings generated by those occurrences as such it represents all that is expected by the teacher and it will be very fluid in content because experience is ever-changing.

It represents a lot of immediacy and therefore, generates a lot of feelings associated with that and as such it can be very fluid, it can of course contain many more elements.

- In this field there are three elements of experience for this example- of course, there could be more. For example element 4 might relate to pupil's misbehaviors during the algebra lesson, element 5 could be the effectiveness of the use of hands-on manipulatives in the lesson, and element 6 could be the pupil's inability to understand the technical language used in the lesson.

In this field there were there could be i.e., elements **about** the student's behavior or about how the teacher is using materials, resources to teach the concepts in algebra or it could be about the students' inability to understand the technical language which is included in this subject.

Now consider the overlap field:

- where the left circle, relating to self structure, and the right circle, relating to experience, come together and interact
- It is important to be aware that experience will impose on us and affect us, but also our own internal frame of reference, our values, beliefs and assumptions will impact on our perceptions and processing of the experience.

Where the two circles intersect, where the concrete experience and the conceptualization come together is another field for the teacher to explore. In this area, this is where the conceptualization the teachers thinking about themselves how they working, comes together with the experience that they have in the classroom. So, it's where the teachers values and beliefs about what they are doing and about their own role as a teacher links to the concrete experience of the students.

- We had a number of elements 7 in this part of the diagram and these can relate to an increased lack of confidence with algebra from the experience due to their own feelings, because of the reactions of the pupils element 8 could be about an increased understanding of how to use resources to teach subjects as a positive attribute.
- Understanding of how manipulatives could aid the teaching of algebra and make the subject easier to access and more fun for pupils, and element 9 could be improved ability to communicate algebraic understanding based on the experience of which mathematical language pupil's understood most easily.
- Venn diagram represents the process of reflection and provides a scematic format for critical thinking.
- It is important to note here that this diagram is only an example. There could be situations where you might have more than 2 intersecting circles.
- So, Venn diagrams give a very individual approach for the teacher to think about their practice of teaching and to deconstruct it in a very clinical and scientific way.
- Venn diagrams can be a very useful graphic organizer which orders complex relationships visually; they allow what is abstract to be more visible.

Bloom's Taxonomy and Critical Thinking

How does Critical thinking fit into Bloom's Taxonomy?

The cognitive Domain comprises

- Knowledge
- Comprehension
- Application
- Analysis
- Synthesis
- Evaluation

Knowledge

- Knowledge of specifics
- Knowledge of terminology
- Knowledge of specific facts
- Knowledge of ways and means of dealing with specifics
- Knowledge of conventions
- Knowledge of trends and sequences
- Knowledge of classifications and categories
- Knowledge of criteria
- Knowledge of methodology
- Knowledge of principles and generalizations
- Knowledge of theories and structures

Comprehension

- Translation
- Interpretation

- Extrapolation

Application

- The use of abstractions in particular and concrete situations.
- The abstractions may be in the form of general ideas, rule or methods.

Analysis

- Analysis of elements
- Analysis of relationships
- Analysis of organizational principals

Synthesis

- Production of unique communication
- Production of plan or proposed set of objectives
- Derivation of a set of abstract relations

Using Bloom's Taxonomy for Critical Thinking

- The Cognitive Process Characterized as Essential to Higher-Order Thinking (that is, analysis, synthesis and evaluation in the unrevised taxonomy) are integrative of the basic critical thinking concepts.

A critical thinker can think for following aspects

- Assumptions
- Facts
- Concept
- Value
- Conclusion

- Relevance
- Evidence
- Premise
- Irrelevance
- Inconsistency
- Fallacy
- Inference
- Prejudice
- Bias
- Prejudice
- Consistency
- Implications
- hypothesis

Techniques for Critical Thinking

Here are 16 basics techniques of critical thinking:

1. Clarity

- State one point at a time. Elaborate, give examples, ask others to clarify or give examples.
- If you are not sure what you are talking about you cannot address it.

For example:

Unclear: “How can we fix education?”

Clear: “How can teachers better prepare students for the future?”

2. Be Accurate

It’s important for the critical thinker to be accurate.

For example:

Inaccurate: “Most students don’t know how to learn”.

Accurate: “**Most** Students in Pakistan need to develop thinking skill”.

3. Be Relevant**4. Know Your Purpose**

It's important for the critical thinker to know the purpose of what you are trying to accomplish?

What is the most important thing here? Distinguish your purpose from related purposes?

5. Identify Assumptions**6. Check Your Emotions****7. Empathise**

- Try to see things from others’ perspective, imagine how they feel. Imagine how you sound to them.

8. Know Your Ignorance

- You still might be wrong. Educate yourself as much as possible, but still be humble.

9. Be Independent

- Think critically about important issues for yourself. Don’t believe everything you read.
- Don’t confirm to the priorities, values and perspectives of others.

10. Think Through Implications

Not thinking through Implications: “Not planning my lessons in a particularly good fashion”.

Thinking Through Implications: “If the lessons are unplanned then there are implications to that the implication is that the lesson won't go very well and perhaps the students won't learn in that lesson and therefore not achieve the outcomes in the curriculum”.

11. Know your own Biases

Another thing for the critical thinker to consider is a particular policy. It is that they have about education and about teaching and learning.

For Example:

Biased: “I believe all students should play sport”

Unbiased: “The evidence linking physical activity to academic achievement is compelling. I will have to think it through before I can say whether it has an impact on the way I teach.”

Bias might be that you believe that students should play sport whereas it would be much better from an unbiased point of view to consider how sports activities may in fact have some influence over academic achievement

12. Suspend Judgment

- Do not make a decision and then use critical thinking to back it up.

Immediate judgement: “We are here to promote the school's plan for curriculum reform. What arguments can we construct in its favor?”

Suspended judgement: “What do we want from our curriculum? Let us use critical thinking to find the best ways to do this.”

13. Consider Other Point of View

14. Recognize Cultural Assumptions

15. Be fair, not selfish**The 5-Step Model of Critical Thinking**

There are five step model of critical thinking. Let's look at each of these steps in turn.

Step 1

- Determine goals/objectives
- Define behaviours/skills you need to exhibit.
- Target those behaviour

Step 2

- Develop appropriate questions
- Engage in discussion with colleagues

Step 3

- Practice before you assess
- Choose activities that promote active learning.
- Utilize all components of active learning.

Step 4

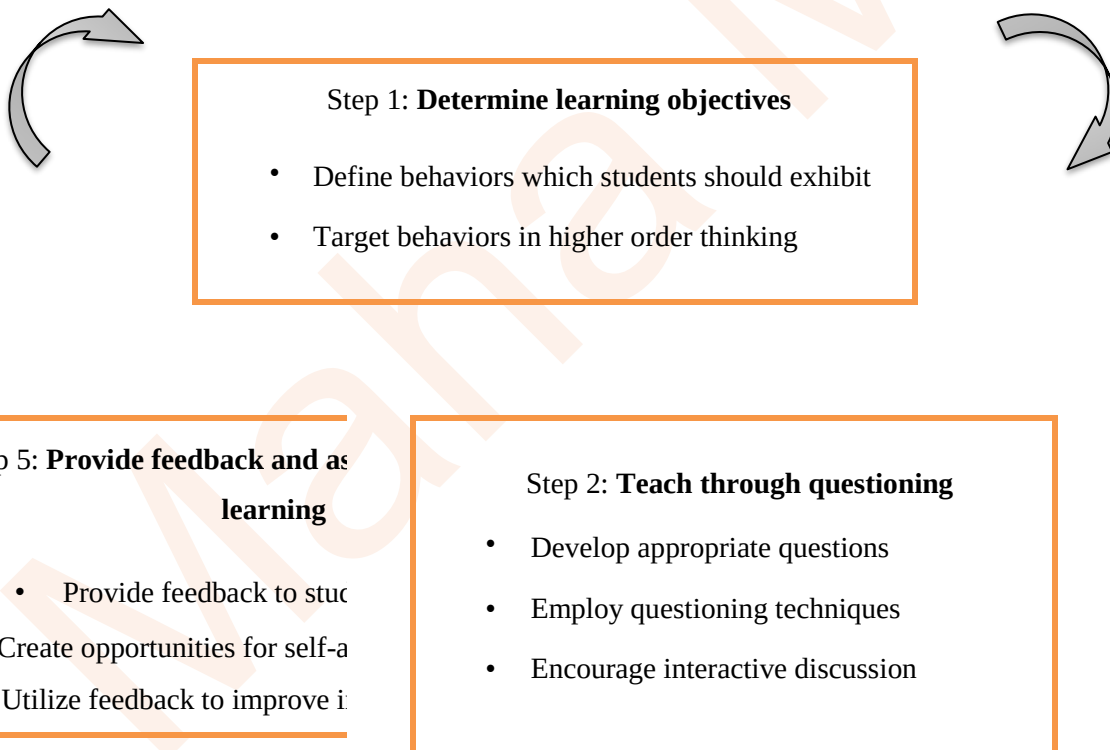
- Review, refine and improve

- Monitor activities in the classroom
- Collect feedback from students and other

Step 5

Gather and review feedback and assessment of learning.

- Review feedback
- Create opportunities for self-assessment
- Utilize feedback to improve instruction



**Step 4: Review. Refine and improve**

- Monitor class activities
- Collect feedback from students

**Step 3: Practice before you assess**

- Choose activities that promote active learning
- Utilize all components of active learning

Halpern's Framework For CT

Halpern proposes a framework based on different kinds of critical thinking:

- Thinking and language
- Argument analysis
- Hypothesis testing

Looking at particular theories around why things happen and finding ways to test them out and see whether they are true.

- Dealing with probability and uncertainty
- Decision-making
- Problem-solving
- Creative thinking
- Memory skills

Thinking and Language

- **Recognising persuasive language**
- **Mis-use of Definitions**
- **Leading Questions**
- **Paraphrasing**
- **Using multiple representations**

Argument Analysis

- **Identifying premises, counter-arguments and conclusions**
- **Judging credibility of information**
- **Difference between opinion, judgment and fact**
- **Avoiding common fallacies**

Creative Thinking

- **Re-defining a problem**
- **Generating possibilities**

- Seeing multiple perspectives

- Taking risks

Hypothesis Testing

- Predicting and confirming hypothesis

- Operational definitions

- Controlling variables

- Sampling and generalizations

- Limits of correlational reasoning

Dealing with Probability and Uncertainty

- Estimating probabilities

- Risk assessments

- Using baseline data

Decision-Making :

- Seeking disproving evidence

- Effects of memory on decision-making

Problem Solving

- Defining and redefining problem situations
- Setting goals
- Generating solutions
- Using alternatives representations
- Recognizing constraints
- Evaluating solutions

Critical Incident Analysis

CIA helps teachers focus on the extent to which critical analysis is evident in their practice. The term Critical incident refers to any work teachers do that involves analyzing situations, reflecting on past experience making judgement and decisions, and taking actions without the benefit of a standard protocol or uniform response that takes care of each and every problem they encounter.

So, let's take a look at some of this particular step:

- As a teacher you should ideally do a weekly audit to understand more about your practice.

- In particular ,it will help you to understand the assumptions that underpin how you analyse situations, make decision and take actions.

Think back over the last week:

Identify the critical incidents that have happened during that time.

A critical incident is an event that can be called to mind easily and quickly because you remember it vividly.

We usually consider events to be critical when they are unexpected a surprise .

They can be positive highs or demoralizing Lows,

Examining Your Personal Filtering System

The meaning we attribute to our experience is influenced by various factors that effectively screen out some responses while letting others through.

- The screening process leads to differing perceptions of circumstances and events ,resulting in different interpretations, and subsequently in different responses.

SITUATION

The way we respond to circumstances and situations is determined by this personal filtering.

Conceptual Learning: Learning through Conceptual Innovation

It involves:

- Seeing farther

- Understanding deeper
- Making the right decisions

It's not about:

- Working harder
- Getting more resources
- Increasing our collection of methods and tools

Its more about the reflective practitioner

- Re-thinking what is already known
- Renewing the concepts we use for understanding
- Getting wiser

For the reflective practitioner, conceptual learning is a process of rethinking or in other words looking beyond what is already known and then renewing concepts in relation to that thinking and we use concepts in order to understand situations with greater clarity. For the rich reflective practitioner this is very much then about becoming wiser in relation to the job role.

Benefits of a Concept-Based Model

So, what we can do is think about the benefits of conceptual learning or learning based on a concept-based model.

- **Facilitates “synergistic thinking” – the interplay between factual and conceptual levels of thinking.** Synergistic thinking is about linking facts with concepts rather than keeping them isolated)
- **Requires deeper intellectual processing as students relate the facts to key concepts and principles.** It requires intellectualizing situations, thinking much more deeply about processes that students are going through in terms of learning facts and relating facts and key concepts and principles.
- **Develops conceptual structures in the brain to relate new knowledge to prior knowledge.** It's very much about developing a structure for thinking conceptual frameworks in order to explore and understand what we're actually seeing.
- **Illuminate the patterns and connections of knowledge.** It is based on an approach which attempts to illuminate or in other words give us much more clarity and to understand the patterns and the Connections between pieces of knowledge.
- **Facilitates the transfer of knowledge at the conceptual level.** A concept-based approach, concept learning helps the reflective practitioner to facilitate the transfer of their knowledge that they are acquiring into other contexts and other situations in which they work.
- **Provides opportunities for personal meaning- making through processes of thinking, creating and reflecting.** This provides opportunities for the reflective practitioner to create meaning to develop a personal understanding of our experiences in relation to the theories that we understand and the practices that we employ

Technical and Conceptual Learning

We are more trained in learning new methods and techniques than for learning new concepts. In terms of conceptual learning as a practitioner, we tend to be more focused on the technical aspects, the techniques and the methods that we use rather than acquiring and understanding new concepts. One reason for this is that new methods impact on productivity almost immediately. This is primarily because of teachers, if we're focused on the technical aspect of our role as teachers then we can see the immediate impact of what we're doing and there's a sort of productivity in relation to the actions that we take.

But by limiting our learning to technical issues:

- **We are not flexible enough when we are confronted by complex situations.** From the critical thinking point of view, we're actually limiting ability to act in very flexible, adaptable and highly productive ways because we're focused on techniques and methods rather than whether those techniques and methods are actually productive.
- **We and our techniques, get out of date in a short period of time.** Therefore in terms of the techniques we can, as teachers become quite outdated or stuck in the ways of teaching year after year without change.
- **We get 'over-learned' - in a few years, our curiosity decreases.** What this actually means is that the quality of the teaching actually begins to reduce in a way the techniques, the approaches over learned and become innocence meaningless.

According to Dewey, Schon and others' concepts are **cognitive tools for coping with the world, and for solving problems.**

What Is Conceptual Learning?

Conceptual learning is a process by which the reflective practitioner needs to learn how to organize **information in logical mental structures** and organize knowledge rather than just acquire and store knowledge.

- Conceptual learning therefore focuses on organizing that knowledge and organizing principles into meaningful whole
- **Conceptual learning is a catalyst for challenging ourselves to think at more advance levels** and it is very much a way that the reflective practitioner can challenge their assumptions and find ways to change and innovate.

Inductive Vs Deductive

Conceptual learning then can move in two directions it can be inductive or it can be deductive.

Inductive Learning

Inductive learning is the process of learning and reasoning from detailed facts to general principles.

Deductive Learning

Deductive learning is a process of learning and reasoning general principles (concepts) towards detailed facts.

Conceptual Learning: Concepts and Frames

Conceptual learning relies on concepts and frames. We are more trained in learning new methods and techniques rather than learning new concepts and this is essentially about productivity rather than immediacy in learning.

Concepts determine the way we **perceive** something and the way situation is **framed** rather than just the facts of what we're actually seeing.

Donald Schon

This is exemplified by Schon, who says “while the given situation can be conceived in a variety of ways it is always **concept-structured** situation”

What is saying is that although we can see facts, we can observe the world. What we actually perceive is the based in the concepts that we hold. So, concepts enable **perception and understanding**. It moves beyond observations and facts.

The process of concept formation has three significant phases:

Perception

Experiences or learning in any form is the starting point of the process of concept formation. The experience of learning that we have and in terms of the concepts that we hold what are we actually seeing? how do we see the world? and how is that information going to be viewed by us?

Abstraction

The mind analyses the perceived images and synthesis what is common to all, neglecting what is not needed. Beyond perception is abstractions. Our minds attempt to analyze the message that is perceived and tries to make sense in relation to the experience and the knowledge that we already have. It attempts to synthesize the new information coming in with the old information that we already have.

Generalization

After making such observation in the form of abstraction for a number of times we are able to form a general idea about common. what we have to do as reflective practitioners is make sense of that information in terms of what we already know and what we have to do is to make generalizations of information we have because what that will do is to develop new concepts. The concept that we already hold.

Conceptual Learning: New Realities

Every time, we're confronted by new situations, we have to deal with new experiences. This is an opportunity for us to make sense of them and we make sense of them based on old concepts that we hold.

- Once we detect something new, some new information what we tried to do is make sense of it in relation to the concepts - concepts that we already have. What we do is, we make analogies with other things that we already know and we do this by thinking back through the experiences we have and finding points of comparison and points of contrast.
- Gradually as a reflective practitioner, we go through a process of extending and changing the concepts that we have. They start to evolve with the experiences that we have.

- This means is that new concepts begin to emerge and new concepts can actually trigger new learning processes. For example: the assimilation or the incorporation of new concepts will begin to transform the way we think and the way we behave in any situation and what this means is that this will begin to have an impact on us and our world, the people around us and the whole world around us. With these new learning process and what emerges are new concepts. Concepts which have changed and therefore, we begin to perceive and understand things very differently.
- New concepts emerge as new questions are made as we start to question differently the experiences.
- Conceptual innovation is like reframing, but deeper in other words, we start to look at a situation from a very different perspective.
- This re-framing is part of the new reality what we do then:
 - we improve our understanding of a situation.
 - We see things more deeply more clearly and it is also then an opportunity to solve the conflicts that we had previously
 - begin to find new strategies in order to dealing with situations that we face.
- Conceptual innovations and these changes give a chance to re-think the whole approach that we have and also our values beliefs and assumptions as teachers
- This actually does for us - is lead us to a point where there are many new opportunities many new possibilities that we can begin to engage with as reflective practitioners because
- This new reality will provide for us is an opportunity for a **Paradigm Shift**.

What Is Paradigm Shift?

Paradigm shift is a complete change in perspective on how the world is viewed on how situations are perceived and our roles and actions within those situations as they occur and after they occur.

A Critical Thinking Framework

Conceptual Learning: Paradigm Shift

Paradigm Shift

A paradigm shift brings a new set of concepts:

- From things to people
- From product to process

The paradigm shift brings about new set of concepts for us; concepts **from** things to people and concepts from products to process. A paradigm shift is a way to view things differently; from **one point of view** to a different point of view, from one perspective to another perspective and therefore, leads to a very different **frame of** perception.

A paradigm shift could be

- From teacher to student
- From standardizing to managing diversity

Paradigm shift should be from teacher to student **or** from standardized approaches to managing diversity in the classroom rather than looking at a situation from your own point of view as a teacher. A paradigm shift would be looking at a situation from the students' perspective. Or it could be instead of looking from a school point of view at what's happening in the classroom to a very unique individualist perspective from yourself and your students in the classroom in terms of what's learning.

A paradigm shift could be

- From replication to inspiration
- From controlling to managing change and uncertainty

It could be moving from replication. In other words, just repeating strategies which you have been taught in which everybody else is using to some sort of different approach, perhaps inspirational, testing different ways of doing things in the classroom.

It could be about moving from controlling **the students in the classroom** to actually managing change and managing uncertainty which is very different way of working with things.

We can represent the paradigm shift for the reflective practitioner in the following diagram and it's a very simple diagram to review. It has three components:

- New understanding
- Paradigm Shift
- Emergent knowing

There's not necessarily a starting point but we can think of the paradigm shift as the change in perspective, the different point of view as leading to new understanding for the reflective practitioner. A new way of doing things, a new way of looking at things. And, this new understanding, this new way of seeing and doing can actually lead to a different way of knowing what things are about or in other words, emergent knowing. Because this is conceptual learning what that can do as it builds up overtime and lead to another paradigm shift which can of course lead to new perspective, new ways of seeing and doing in the classroom. So, it becomes a cycle.



Different kinds of learning

The paradigm shift leads to different kinds of learning, and we can think of three different kinds of learning.

Single-Loop Learning (following the rules)

- Rigid strategies
- Spend time detecting and correcting deviations from the 'rules'

Double-Loop Learning (changing the rules)

- Reflect on whether the rules themselves need to change
- Requires 'thinking outside the box'

Triple-Loop Learning (learning about learning)

- Learning how to learn
- Searching for meaning and understanding

A Critical Thinking Framework: Conceptual Learning: Theories-in-Use or Espoused Theories**Theory –in-Use**

- Theory-in-use is embedded in the logic of the action, it is the theory that commands the thinking of the action.

Espoused Theory

- Espoused theory is that theory used for arguing about the actions, it is the theory used for explaining the action to others but not necessarily for conducting the action.

- Espoused theories are explicit but most theories in use are tacit.
- How to combine the espoused theories and experience for creating solutions?
- Reflective practice is about ownership of knowledge
- Reflective practice is about awareness of the knowledge we use, how we use it and how we can improve our action in real time.

Reflective practice helps us do this. Reflective practice is very much about ownership of the knowledge that we generate from **theory-in-use and an** espoused theory. And, it's about awareness of that knowledge and how we use it in order to improve what we're doing in the classroom and to improve on the actions which we are using in order to teach.

- Reflective practice is about how our minds work and how we use and create theories in practical situations.
- It is about invisible and visible, tacit and explicit, blindness and sight.
- Reflective practice is about flexibility, adaptation and effectiveness.

Reflective Practice = Evolving Brains

- If our theories-in-use are stagnant so are our brains
- If we reduce our learning to the methods and techniques, our knowledge goes out of date in a short period of time.
- If we can renew our theories (espoused and in-use) and the conceptual frameworks that are behind them, we will be able to dialogue with the disruptive changes that the future will bring to our roles in the classroom.

Frameworks for Reflection: Reflective Writing

What Is Reflective Writing?

- Reflective writing is evidence of reflective thinking:
 - Looking back at something (something that happened, or an idea or a project)
 - Analyzing what happened (thinking in depth and from different perspectives)

- Thinking carefully about what it means for you as a professional

It is therefore an aid for the teacher for the reflective practitioner to think. It's a thinking tool and this is very much about helping the reflective practitioner develop a professional stance as a teacher in the classroom.

Reflective writing is much more personal

- Reflecting writing is more personal than other kinds of academic writing.
- We all think reflectively in everyday life but not to the same depth as that expected in good reflective writing for a reflective practitioner.

We already know that reflective writing:

- Describes
- Links theory to practice
- Questions experience
- Has a focus on improvement

Example of Basic Reflective Writing

Specific tasks were shared out amongst members of my team. Initially, however the tasks were not seen as equally difficult by all team members. Cooperation between group members was at risk because of this perception of unfairness. Social interdependence theory recognizes a type of group interaction called 'positive interdependence' meaning cooperation (Johnson and Johnson 1993, cited by Maughan & Webb, 2001). And many studies have demonstrated that 'Cooperative learning experiences encourage higher achievement' (Maughan & Webb, 2001). Ultimately, our group achieved a successful outcome, but to improve the process, we perhaps needed a chairperson to help encourage cooperation when tasks were being shared out. In future group work, on the course and at work, I would probably suggest this.

For example, social interdependence theory recognizes a type of group interaction called positive interdependence. And at this point, you can see the reflective practitioner is beginning to analyze and to think

deeply about the occurrence. This is the interpretation aspect of **reflective** writing and the writing goes on at the end, ultimately our group achieved a successful outcome but to improve the process, we perhaps needed a chairperson to help encourage **cooperation**. And so, at this point in the reflective writing you can see where the teacher is actually looking at outcomes from the reflection. So, in this example, you can see the three components of the reflective writing process and these components (Description, interpretation, outcome) are very important to consider when you're actually using writing as a tool for reflective practice.

- So, you can see that reflective writing is a far greater focused activity than simply writing in a journal.

- Description
- Interpretation
- Outcome

Frameworks for Reflection: A Structure for Reflective Writing

- Reflective thinking especially if done in discussion with others can be very free and unstructured and still very useful.
- Even reflective writing can be unstructured, for example, when it is done in a personal diary.
- Reflective writing, however, is normally carefully structured.

Stages of Reflective Writing

Description

- What happened?
- What is being examined?
- Keep this part SHORT!

Interpretation

- What is most important/interesting/useful/relevant about the object/event/idea?

- How can it be explained? (e.g., with theory? from literature on the subject?)
- How is it similar to and different from others?

Outcome

- What I have learned from this?
- What does this mean for my future?

Remember the following 4 key points

- Reflection is an exploration and an explanation of events not just a description of them.
- Genuinely reflective writing often involves revealing anxieties, errors and weaknesses as well as strengths and successes. This is fine (in fact, it is often essential!), as long as you show some understanding of possible causes and explain how you plan to improve.
- It is necessary to select only the most significant parts of the event or idea on which you are reflecting.
- Do not try to ‘tell the whole story’, this would be description rather than interpretation.
- It is often useful to ‘reflect forward’ to the future as well as ‘reflecting back’ on the past.

Frameworks for Reflection: Vocabulary for Reflective Writing

- The following are a few suggestions for words and phrases that might be useful in reflective writing.

Description

- Because the range of events, ideas and objects can be so great it is not possible to suggest specific vocabulary to help you reflect.

Remember:

- We tend to talk of events using the past tense
- We tend to discuss theories or models using the present tense

WARNING!

- Using only these words and phrases will not make you a good reflective writer.

So, reflective writing can be quite straight forward, if you give yourself some nice language than simple language structures to use and not to get bogged down too much in the spelling of the grammar. What's important is the fact that writing is reflective and by using particular vocabulary, particular words, that can help your thinking **processes** much more.

- What I would say is that these are just examples and do not just rely on these only. As I said earlier, use them as a starting point and develop your own writing style as a reflective practitioner. The important thing is to remember that what you will be doing is supporting your thinking and not just writing a diary.

Vocabulary for Reflective Writing

The following are a few suggestions for words and phrases that might be useful in reflective writing.

WARNING!

Using only these words and phrases will not make you a good reflective writer!

This is just a guide.

Whatever language you use, make sure that you (1) describe; (2) interpret and (3) reflect on outcomes.

There are three stages to reflective writing:

- Description
- Interpret
- Outcome

Reflective Writing: Limitations of Self-Reflection

Limitations of self-Reflection

- Hebermas (1974) suggested that reflection requires a level of detachment and objectivity which can be confused by self-deception.

Let us look at an example of this to put it into perspective.

Dr. Brown is worried about his class. He knows that if he doesn't stick to the time allocated for each section he will run out of time and the material at the end would not be covered.

- What can happen in terms of reflection is that as a there's a difference between the teacher's perception of what happened and what really happened in the classroom.

	Perception	Reality
Problem	Poor time management	Can't relinquish control
Reflection	Question reduce time to cover material	Time management becomes the problem & the disguise
Solution	Better lesson plans: don't allow question	Not discovered via reflection

- The concern is, then, that there may be much that you keep from yourself (consciously or otherwise),...
- A further concern relates to 'memory bias'

Reflective Writing: Strategies for Overcoming Barriers

- There are always barriers before, during and after any reflective practice.
- Some of these barriers are self-imposed. Others originate within an institution.

Lack of Time and Space to reflect

- Use the journey to and from your workplace is a chance to reflect.
- Using the first 20 minutes after your learners have left for the day while events are still fresh in your mind.
- Jotting down notes in a journal to reflect on when you have time later.
- Talking with a trusted colleague over lunch.
- A group discussion with several colleagues at the beginning or end of week – perhaps this could be scheduled into the timetable?

Negative Ideas about Reflection

- It is common to find negative preconceptions about reflective practice.
- It can be a difficult and time-consuming process.

Organizational Culture

- Some organizations are supportive of reflective practice and your management might model reflective behavior. Others may not.
- At the level of the individual practitioner the aim of RP is not about changing the culture of your organization but about changing your own practice and skills. Do not be put off by a culture which does not embrace this.
- Be resilient!

Fear

- RP can be an intimidating idea, as it requires a critical, honest and open view of yourself. This can be difficult.

Lack of Knowledge and Experience

- This can be a problem for new and experienced practitioners alike.
- You need practice to be able to do it well.
- Re-read about some of the theories discussed about RP and use some of the techniques to practice.
- See what works for you.

Seeing it a success or failure

- This can be a problem for new and
- You should not see the work you do with learners as either a success or a failure.

Anxiety about introspection

- Reflective practice is a critical process, and looking for an area to develop and improve on can be disturbing.

RP as an uncritical routine

- RP should never be done as a simple ‘tick box’ activity.
- Real change comes from hard work and determination to improve.
- It would be better for you to reflect less and do it right than go through the motions every day without thinking about it.

Frameworks for Reflection: Developing Competences: Professional Competences

Professional competence can be categorized into three broad headings;

- Professional values and practice
- Professional knowledge and understanding
- Professional skills and application

1. Professional Values and Practice

- The reflective practitioner should demonstrate a commitment to a personal code of values.
- What would you consider to be core values for the teaching profession?
 - Trust
 - Respect
 - Integrity
 - Honesty
 - Fairness
 - Tolerance
 - Commitment
 - Equality
 - Service
- A commitment to serve lies at the heart of professional behavior.
- Teachers need to exemplify these values in their relationships with their students.

In applying their values in a commitment to learners, reflective practitioners must:

- Exemplify these values in their relationships with their students.
- Motivate and inspire students with a view to helping each realize his or her potential.

And therefore, there is a particular commitment that we need to consider here. There is a **commitment to the learners** by the reflective practitioner. Because the reflective practitioner must be somebody who can motivate and inspire students and these competencies which we've just listed; integrity, honesty will help us with this need to motivate and inspire other people.

In applying their values in a commitment to colleagues, reflective practitioners must:

- Work with colleagues to create a professional community.
- Ensure that relationships with parents are characterized by trust and respect.

There's also a **commitment to colleagues** as well as learners as well as students. And in relation to this commitment to our colleagues, what we need to think of in terms of professional competence is our

commitment to the educational community and the relationships that we have with people within that community. This is where the most significant competencies of things like trust and honesty, I think become more important than any others.

In applying their values in a commitment to profession, reflective practitioners must:

- Contribute to the continual review of policies and practices.
- Assume responsibility for on-going professional development

Frameworks for Reflection Developing Competences: Professional Competences (Part 2)

Professional Knowledge and Understanding

- The reflective practitioner must enhance knowledge and understanding of the subjects they teach.

So, we've looked at developing professional values as a **competence**. Another competence, we need to look at is that of knowledge and that of understanding. And in this **regard**, the reflective practitioner needs to always be enhancing, building on their knowledge and understanding of the subjects they teach. There's a particular commitment here for the teacher.

- The reflective practitioner must develop an understanding and knowledge of social and policy contexts for education.
- The reflective practitioner must develop knowledge and understanding of a range of strategies to promote and maintain positive behavior.
- The reflective practitioner must develop knowledge and understanding of factors that promote and hinder effective learning.
- The reflective practitioner must enhance skills for communicating effectively with students and colleagues.
- The reflective practitioner must develop knowledge and understanding of how to use technology to support student learning.

- The reflective practitioner must understand the interrelationship between schools and the communities they serve.
- The reflective practitioner must develop a knowledge and understanding of students social and community contexts and address the implications for learning arising from these.

Developing Competences: Professional Competences

Professional Skills and Application

In planning and leading the reflective practitioner must set appropriate learning objectives/outcomes intentions,taking account of what students know,understand and can do.

- **Manage workloads effectively and efficiently and maintain a work/life balance.**

In terms of planning and leading we can also think in terms of how the reflective practitioner is achieving the curriculum aims. Any teaching has a particular workload associated with it. And therefore, the reflective practitioner needs to think about how they are managing to get through the curriculum and manage all the work of the school the administrative duties and also have a personal life. The reflective practitioner is somebody who manages to get a good work-life balance.

- **In terms of this professional competence relating to skills and applications we can also look specifically at teaching and learning.**
- **The reflective practitioner also needs to use a broad range of teaching strategies in their teaching.**
- **In terms of teaching and learning the reflective practitioner also needs to consider the strategies that they using throughout the course within every single lesson to motivate the students.**
- **The reflective practitioner needs to set very clear standards of behavior and needs to promote these in the classroom in order to make sure that the psychological the emotional space is conducive to learning.**

- In terms of professional skills and application
- The reflective practitioner's role at holistically in terms of assessment
- A reflective practitioner is consider which assessment procedures you will use .
- A reflective practitioner in terms of this competence area of skills and application

Developing Competences: Professional Competences

Dimensions of Development

- The reflective practitioner needs to consider are different dimensions of development. And we need to think about the progress of careers as professionals.

The challenge is that can entail the opportunities which can go along with that and our own expectations associated with all of this. This means that as we progress through this professional life but our confidence as reflective practitioners will grow and we will continue to learn from experience and use that experience to help us develop as quality practitioner and say what we needed to do is think about these dimensions or what are these particular dimensions of development that we can consider?

- The reflective practitioner becomes much more sophisticated at reflective practice.
- Reflective practice and with using a wider range of teaching strategies these will help us build up our experience and actually help us as reflective practitioners.
- The reflective practitioner the developed experienced practitioner somebody being very sophisticated is somebody who can expand. They're learning and expand their experiences and therefore expand their impact beyond the immediate classroom.

- **This** ultimately will lead to much fuller and complex participation within the whole school. And of course, we've growing competence then there is increased autonomy and as a reflective practitioner autonomy means that we're much more self-assured and we have a much greater self-esteem as teachers.
- **This** then gives us a much greater capacity for self-criticism. And also, self-improvement simultaneously.
- **Then** if we're looking from a whole school protect perspective will help us consider the impact that this will have on our colleagues especially where we as reflective practitioners can enter into a mentoring situation or a coaching partnership with others.
- **This is** where good practice gets shared in the school environment. And of course, there is also the opportunity for the reflective practitioner to model good practice. And then this helps to enhance the quality of teaching across the whole school. So, at this point what we need to do is think about well what are the core values?
- **Which** underpin professional practice and encourage attitudes and conduct also in terms of what we're doing in the school. So, the competencies that we've been talking about the competencies which have been described. About values and about understanding and knowledge or will help to build a professional persona professional identity.

Developing Competences: Provisional Model

The Provisional Model 1

This model attempts to combine particular strengths as well as approaches for the reflective practitioner and at the heart of the provisional model are four key components.

Knowledge/ Cognitive Competence one component is knowledge and cognitive competence or in other words the work-based knowledge and experience that we built up the acquired knowledge that we build from experience.

Functional Competence

There's also functional competence, this is in relation to the reflective practitioner performing specific work-related tasks.

Personal Behavioral Competence

We can and also look at personal behavioral competence within the provisional model. This is about how the teacher is adapting to a work-related situation and the behaviors that they bring as a professional into the school environment.

Values/Ethical Competence

The model with values and ethical competence that we need to consider. this is about the teacher making appropriate correct judgments about their role and responsibilities within the school.

Metta competence

- **Communication**
- **Self-development**
- **Creativity**
- **Analysis**
- **Problem solving**

Meta Competencies

Communication, Self-development Creativity, Analysis, Problem solving

Knowledge/ Cognitive Competence	Functional Competence	Personal Behavioral Competence	Ethical Values/ Competence
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Developing Competences: The Provisional Model

The Provisional Model 2

Developing competencies, we've looked at the provisional model. The provisional model made up of Metta competencies and core competencies. What we need to do is consider the components the individual components of these core competencies. And what we can do is look at each of these in turn.

The Provisional Model 1

Knowledge/ Cognitive Competence

Tacit/practical- this knowledge linked closely to functional personal competences; linked closely to Schon's 'reflection- Inaction'	Technical/theoretical- this relates to underlying knowledge bases including their application, transfer synthesis etc.	Procedural – this consists of the how, what, when etc. Of the more routine tasks of teacher	Contextual – this is general back ground knowledge about education / teaching
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Reflecting Effectively

The reflective practitioner to be effective. Must have the right mental attitude for reflection. And if we look closely at Dewey's paradigm of reflective practice, what we can do is draw out three key attitudes, which make reflective thinking effective and these three attitudes are:

- **Open-Mindedness:** This is the ability of the teacher, the reflective practitioner to have an attitude which is questioning and not closed.
- **Responsibility:** It drives the process of reflection for the reflective practice
- **Whole-Heartedness:** The desire to improve and to change and to manage ongoing professional learning.

Perhaps the most important attitude that we can draw out from these three is open-mindedness. It refers to the ability to remain open to multiple alternatives possibilities, with an idea that we can always learn and

improve and develop our skills as practitioners that the reflective process becomes truly effective. This means that the open-minded teacher continuously questions routines and practices their validity and value.

In order for these three attitudes to come together and become very powerful to improve effectiveness. What **teacher** need to do is actually look at what he/she know, what are their values and beliefs as reflective practitioners. And once we begin to think along these lines, it actually starts to underpin particular ways of knowing for the reflective practitioner, our values and beliefs are what, direct us as teachers and actually give us a focus for our approach and the ways of knowing actually then how pursue bring together all these elements, these attitudes, and values and beliefs to be effective. **Therefore the reflective practitioner does not believe in one single truth or in one right way to teach and develops 'ways of knowing' and 'patterns of reasoning'.**

And we can look at different ways of knowing. For example, we can talk about **absolute knowing**. This is perhaps the most simplistic level where knowing is either about right or wrong, correct or incorrect. We can also talk about **transitional knowing** this is where we know that some knowledge is correct and other types of knowledge we need to explore or discovered.

Frameworks for Reflection		
Reflecting Effectively		
Ways of Knowing	Patterns of Reasoning	
Absolute Knowing Knowledge is either correct or incorrect. Correct knowledge obtained from authority.	Mastery Peers debate and quiz each other for mastery.	Receiving Peers talk to create a relaxed atmosphere.
Transitional Knowing Some knowledge is correct; other types of knowledge can never be discovered. Learner should understand rather than memorize facts. Not all knowledge comes from authority.	Impersonal Learner exchanges views with peers and express their opinions.	Interpersonal Learners collect ideas from others who provide exposure to new ideas.
Independent Knowing Most knowledge is uncertain so learner thinks independently. Values knowledge that does not come from authority.	Individual Learners focus primarily in thinking independently; exchange with others is of secondary importance, though appreciated.	Interindividual Learners place equal value on hearing others' perspectives and thinking for themselves.
Contextual Knowing	Peers enhance learning via quality contributions. Learners exchange and compare perspectives, then integrate and apply knowledge.	

And, this actually comes from the reflective practitioner, developing in confidence, becoming more authoritative as a teacher. Beyond this, we can talk about independent knowing this is where the reflective practitioner and the authoritative, they have as a teacher start to really take control and become very powerful.

All of these ways of knowing is something that we call **contextual knowing**. This is where the reflective practitioner is bringing knowledge together from theory, from experience into a particular context, into a particular setting, into their own classroom and therefore, synthesizing that knowledge. This is where reflection becomes extremely effective within ways of knowing there are different patterns that we can apply.

In terms of independent knowing we can talk about individual patterns of knowing and also inter individual patterns of knowing. This really underpins the idea that the reflective practitioner is not only somebody who reflects independently, but also reflects with others. As a community of practice with colleagues to create a

very powerful understandings and it's through these perspectives that we can begin to integrate and also create knowledge as reflective practitioners.

Reflecting Effectively: Professional Development Journal (PDJ)

Professional Development Journal (PDJ)

For effective reflection we can consider a number of tools perhaps the most prominent, the most used method of reflecting is the professional development journal. This is a written record of experiences in the classroom and their response that the teacher has to those experiences in the classroom, feelings about planning, preparing and delivering your lessons.

Because the reflective practitioner is writing in a journal:

- The professional development journal is quite subjective - it is written by you and for you, so it provides a dialogue with yourself. It is the individual's response, the individual's perception of what has actually taken place.
- It is very important in the professional development journal to take time to relate theory, which has been explored in the literature with the experience of practice.
- One of the most valuable functions of the professional development journal is to help you identify development points for action planning. Once the critical incidents, once the experience has been noted down, once the feelings have been identified, is for there to be some indication of action in relation to the experience.
- A professional development journal is not a diary of what happened. It does follow a structure in terms of, reflection and action.

So, what you can be doing is to be looking for themes longitudinally through the journal, to see if things recur, and then starting to develop some ideas about actions that you can take in terms of your experience.

- It's very important to get into a very good habit of writing in the journal, try to do it as soon as possible after teaching and as regularly as possible as you go through your teaching.
- The most difficult thing to consider as a reflective practitioner is writing in a journal. When do you get the time? the important thing to realize is that, it is a very critical process, but try not to think too hard about it that can actually stop you from writing in the journal. What you want to do is to let the writing flow and to get the experience, capture the experience and some critical incidents on the paper.
- A very good and simple rule to follow is little and often in the professional development journal. You don't need to write pages and pages, you simply need to note down the key incidences, the key approaches.
- A very good thing to do with the journal is to share your reflections with your colleagues in school - it can be very helpful to find that colleague are experiencing the same uncertainties or difficulties. So, you can get their perspectives on your reflections. And of course, it doesn't have to be just a word. You can actually put diagrams or drawings in there.
- Therefore writing in the professional development journal is a very structured process. It's not free form. And what you need to make sure is that you include **description** under the words (what happened?) you need to add **analysis** (how, why?) and **evaluation** (how effective was it?) and also **conclusions** (suggestions for future practice) that you can draw. The conclusions are important because these will help you identify action points.

Reflecting Effectively: Professional Development Journal (PDJ)

Reflecting effectively using the professional development journal. What is useful to consider here is, a framework that can support the process of writing. Driscoll and Teh (2001) provide a very simple but useful framework for reflection based on three questions:

- **What?** Start off writing in the journal by talking about what happened, or in other words, a description of the event, what was it that, occurred in the classroom that warranted the reflection.

- **So what?** In other words, what's your reaction to what happened? What's the actual analysis of the event that occurred? What's the analysis and evaluation of the experience of teaching on that particular occasion.
- **Now what?** or in other words, what's the next step? What will you do? What are their actions, which will come from your reflection?

What?

So, for example, in describing the occasion, describing the event in the classroom, the reflective practitioner can use what questions or in other words, what happened? What did I see? What did I do? What did my students do? These are very simple trigger questions to the first part of the frame. They also provide trigger questions for the other components in the framework as well.

So What?

These are part of the analysis and evaluation aspect of the reflection. And so, the trigger questions need to reflect analysis and evaluation, or in other words, how did I feel at the time? what were the effects of this particular thing happening? why did I react, to this particular student in this particular way? this is where the critical reflection, has most power.

Now What?

And then the trigger questions for now what, or in other words, the action planning component of the framework: what are the implications of the analysis? And then what are the next steps? For example, it could be, how can I modify this aspect of my teaching?

You can design your own template for your PDJ using the following categories (headings) can help:

- Description

- Analysis and evaluation
- Conclusion for future practice

Reflecting Effectively: Critical Incidents

Critical Incidents

What we need to do as reflective practitioners is be continual in terms of reflection, and we can use tools like professional development journals to do that. And we know that a professional development journal can have, a useful structure that can help reflection.

However, if we just telling a story in the classroom, the effectiveness of reflection can be reduced. What many reflective practitioners do is. Actually just focus on critical incidents that occur in their teaching and only reflect on these in their professional development journal.

Critical incidents are those specific occurrences within teaching and learning which you consider significant or important.

Reflective Action Planning

- **Select a specific target** (e.g. effective rules and routines or widening your range of strategies)
- It certainly involves **planning a program**, not only of reflection, but the actions in relation to that reflection. This could include discussing with colleagues, notes in the reflective journal and your actions which you're planning. It could also be a process of observing other teachers or having other teachers observe you to give you feedback and for you to acquire more knowledge about your approach as a teacher.
- It's very much about not only action planning, but focusing on the **effectiveness of quality** of what you're doing as a teacher.

The active action planning process is about:

- Identifying the opportunities that you have in discussing these opportunities. And also using this process to reinforce what you're doing well and to identify where you need to make improvements, where you need to make changes.
- Deciding what evidence of planning and practice goes into professional development.

An effective action plan then is something that we need to think about. It's not just about putting action points on paper. The effective action plan is a very useful document for the reflective practitioner.

- It helps you build up a profile of your work and your development over time.
- It's certainly something that is a professional document and therefore should not be just for your consumption, but also it should be shared with your line manager, the principal in the school and others.
- The action plan needs to be very precise in terms of how it's written, not vague.

And so, an action plan answers certain questions for the reflective practitioner.

And really the success criteria, what we'll **demonstrate** through the evidence that you collect, that you've been successful with the action planning process.

Indicators of a Reflective Practitioner

There are some key teachings and learning practices which contribute to what we do as reflective practitioners and to effective pedagogies. And what we can do is identify particular attributes, which will indicate when a teacher is being a reflective practitioner.

Intellectual Quality

Key attribute really is intellectual quality.

- The reflective practitioner is somebody who is engaged in **higher order thinking** rather than common sense thinking this is very critical thinking.

- In terms of intellectual quality, the reflective practitioner is somebody who has a **deep knowledge** and a **deep understanding** of the subject they're teaching and also the pedagogy of the subject that they're teaching.
- The intellectual quality of the reflective practitioner is exhibited through the way they **communicate** with other professionals, or how they talk about teaching, how they talk about language or in other words, the **meta-language** that they use to talk about the profession.

Supportive Classroom Environments

Another indicator of an effective reflective practitioner is that the classroom environment is supportive of teaching and learning.

- There is very clear **student direction** in terms of activities and learning, which is taking place.
- There is **social support** for our students within the entire room.
- Their classroom environment is very **active** rather than passive and therefore students engage much more with the teaching and learning which is taking place.
- And what is key to identify here where you have effective reflective practice is that the students are great degree **self-regulated**, or in other words, learning happens almost without direction from the teacher.

Framework for Reflection: Five Key Features of Reflection

Five Key Features of Reflection

- Teaching is a complex activity, in which decision are made in complex contexts.
- In addition, there are theoretical perspectives to consider, and the process of reflection brings all these aspects together.
- There are some key features of reflection that are widely considered:

1. Reflection Results in Learning

- Reflection results in learning- through changing ideas and your understanding of the situation.

2. Reflection is an Active Process

- Reflection is an active process of learning and is more than thinking or thoughtful action.

3. Reflection Involves Problematizing

- Reflection involves problematizing teaching by recognizing that practice is not without dilemmas and issues.

4. Reflection is not a Linear Process

- Reflection is not a linear process, but a cyclical one where reflection leads to the development of new ideas which are then used to plan the next stages of learning.

5. Reflection Encourages Looking at Issues from Different Perspectives

- Reflection encourages looking at issues from different perspectives, which helps you to understand the issue and scrutinize your own values, assumptions and perspectives.
- Therefore, when the term ‘critical reflection’ is used, it refers to a combination of the analytical, questioning (or critical thinking) and reflective approaches.

Framework for Reflection: Ensuring Reflection Leads to Learning

Ensuring Reflection Leads to Learning

- There is a distinction between surface level reflection (routinized, bland and unthinking) and a deeper level (conscious control, self-analysis, metacognition).
- Many theorists make a distinction between “**common sense thinkers**” and those who are “**pedagogical thinkers**”. This distinction gets to the heart of how to ensure reflection is an effective learning tool.

Common Sense Thinkers

- Laboskey defines “**common sense thinkers**” as those who reflect in an unconscious way, suggesting they are happy to use a “trial and error” approach to learning to address short-term issues that are context specific.

Pedagogical Thinkers

- “**Pedagogical thinkers**” are more conscious of their actions, asking the questions ‘what is my intuitive response to this, and why am I feeling or acting this way? Laboskey suggests that it is by taking conscious control that pedagogical thinkers ‘take a long-term view of how to solve problems ... remaining open to learning ... recognize there are no simple answers and the conclusions they reach are likely to be tentative’. (Laboskey in Cartwright, 2011).

Four Key Principles for Learning for Pedagogical Thinker

And what we can do then is look at four key principles of learning for the pedagogical thinker and these define the process of pedagogical thinking. **We have continuous reflection which is ongoing, connected reflection, challenging reflection, and also contextualized or situated reflection.**

1. **Continuous Reflection**
2. **Connected Reflection**
3. **Challenging Reflection**
4. **Contextualized Reflection**

Continuous Reflection

- This ensure that reflection is an ongoing component in learning. Often short-term experiences can lead to more complex long-term involvement which gives the teacher extensive material for observation, reflection and experimentation.

Connected Reflection

- This is essentially the component that links experimental learning to formal training.

Challenging Reflection

- This is the teacher being prepared to pose questions and propose unfamiliar or even uncomfortable ideas for making changes.

Contextualized Reflection

- This ensures that the reflection activities or topic are appropriate and meaningful in relation to the experiences of the teacher. For example, should the reflection be more formal or informal?
- Contextualized reflection takes into consideration all these components, which makes for more meaningful and educational reflection.

Framework for Reflection: Tools for Reflection

Tools for Reflection

Reflective Journal

- Most frequently used.
- Involves teachers in self-assessment, collaborative critique, self-reflection and goal setting.
- May become a shared dialogue with a colleague.

- May be an individual journey of professional self-reflection.
- May be supported by a framework (list of questions/checklist etc.).

Types of Reflective Journals

- **Reading response Journal**

- Teachers reads recommended content and then record responses in the journal.

And there are many different types of journals that we can use. The **reading response journal** is where the teacher reads, it could be some recommended text or has access to texts in the school or at home and uses the journal to reflect on the content of what they've read.

- **Double entry Journal**

- Involve a statement of context on the left-hand side and reflections and proposed actions on the right-hand side.

We also can get something called a **double entry journal**. This is a very popular approach for the professional development journal where one side of the page, the left-hand side describes the context and the situation and on the right-hand side is where the reflection, the analysis, the evaluation takes place.

- Based on Smyth's (1989) 4 phase model of "describe, inform, content, confront and reconstruct".

Based on Smyth's **four phase model** of describe, inform, confront and reconstruct, this is where journals become quite powerful.

- **Interactive Journal**

- The teacher shares the journal writings with an 'audience'.
- Provides encouragement and deepening reflection.

In **interactive journal**, this is where the teacher writes regularly in a professional development journal. But these writings are regularly as a matter of process shared with an audience that could be one individual or it could be a group of individuals. This is a means for encouragement and support of the teacher.

- **Dialogue Journal**

- Supports the process of deconstructing and reconstructing experience.
- Helps teachers make sense of the conceptual and theoretical understandings about teaching.

And you can also get a **dialogue journal** or sometimes called a **dialogic journal**. This supports the process for the teacher of deconstructing and reconstructing the experience very methodically. It helps the teacher makes sense of the conceptual and theoretical understandings that they have in the classroom by critically taking them apart, looking at the components and then reconstructing them in a meaningful way.

- **Narrative Journal**

- The teacher reflects on the values and beliefs which form the essence of teaching.
- Teachers can draw inferences from experience.

There's also something called the **narrative journal**. This is where the teacher uses the writing process to reflect on values and beliefs about their role as a teacher, what is the essence of teaching? And it is very much here where the teacher can draw on inferences, make inferences about their teaching.

- **Practicum Journal**

- Provides guiding questions about teaching.
- Enables teachers to critique and modify their teaching.

There's also a **practicum journal**. This is perhaps something that teachers will experience during training, where there would be a sequence of guiding questions which need to be answered by the teacher. This gives a very strong structure and helps the reflective practitioner critique and modify their teaching in a very processed and sequential way.

Self-Assessment

- Self- assessment framework help to develop reflective skills.

- There are many different self- models which teachers can use.

Portfolios

- A collection of structured professional ‘artifacts’ that demonstrate accumulated knowledge, skills and practice.
- Paper-based or electronic.
- Teachers can portray higher level cognitive thinking and self-reflective growth.
- Examples of artifacts include: lesson plans, students work, feedback, recordings, photos, formal appraisals, etc.

Group Reflection

- Collective reflection with colleagues in a small group.
- Provides a richer experience.
- Generates safe reflection and greater feelings of confidence.

Mentoring

- Can play a crucial rule.
- Mentors can model reflection on own practice and by directly challenging and affirming the critical thinking process.

And mentoring is another significant process for the reflective practitioner. This is a **dual** process where the teacher is coached and supported as part of reflection.

Framework for Reflection: Social Criticality

Social Criticality

- Being socially critical means starting with reality, with seeing injustices and contradictions, and beginning to overturn reality by reasserting the importance of learning.
- Only when teachers take an active reflective stance that they are able to challenge the dominant “factory” metaphor of the way many schools are conceived, organized and enacted.
- In theorizing practice and practicing theory, teachers are able to see reflection as embedded deeply in our teaching, rather than something that we do to it afterwards

Self-Development is based on:

(a) **Received knowledge**- the ‘intellectual content’ of the profession

(b) **Experiential knowledge**- classroom experience PLUS reflection.

- Dialogue and discussion are central to development.
- Articulation of ideas is crucial to shaping pedagogical thinking.

And what is also involved in that is a process of dialogue and discussion. Social criticality recognizes that as reflective practitioners, we're working within the context, with other teachers and therefore development is very much a process of shared understanding and dialogue and not an isolated individual activity as such, and therefore, it's very much about articulating our understandings and using that language to shape our pedagogical thinking.

- Cooperative development involves another.
- Self-development is based on social constructivism:

(a) The mind is mediated by language.

(b) Understanding is jointly constructed through dialogue.

(c) Teachers are guided to higher planes of understanding through dialogues they have with other professionals

(d) Knowledge is socially derived.

(e) Learning takes place in ZPD (zone of proximal development).

And what we have to consider then is that there is a **zone of proximal development** as a reflective practitioner. There are those that who are closer to us, can have much more impact and can support us. Those who are not so close, they can offer less support and less criticality in terms of our reflection.

- The **process of reflection allows reflective practitioners to develop an “inquiry stance” on their practice** and construct their own knowledge base and their own professional skills.
- Teachers advance their effectiveness through reflective practice by challenging uncritically accepted assumptions.
- Critical reflection within a group/social setting gives teachers multiple opportunities for reflection.

Framework for Reflection: Hegarty's Reflective Framework

Hegarty's Reflective Framework

- The framework consists of a template that you can follow to structure your reflective writing in three steps. This encourages you to move beyond basic description about your experience to analyze your actions, learning and emotional reactions and thus examine your practice more critically and from different perspectives
- **Take notice and describe the experience-** pay attention and record what happened; What did you do, know, feel, think, need? What decisions were made?

So, the first step is about ‘**taking notice and describing the experience**’; pay attention and record it. The journal here is the place where you can use this framework to go through each of the three steps. The first step is where you're describing. What did you do? How did you feel about it? **How** do you think the students experienced that particular lesson?

Analyze the experience- what were the actions and decisions, and why these actions and decision? What was your reaction?

The second step is ‘**analysis of the experience**’. What were the actions and decisions, and why were these actions and decisions taken? what was your reaction at this particular point to the teaching or the experience that you had? Analyzing and evaluating is the second step where the reflective practitioner is beginning to understand what needs to be learned.

- **Reflect, take action-** what did you learn? How will you use this learning? plan how you will apply this learning

And then the final step is ‘**reflect and take action**’. What did you learn? How will you use this learning and, how do you plan to apply this learning?

Framework for Reflection: Personal Practice Theory (PPT)

Personal Practice Theory (PPT)

- Researchers exploring the relationship between teacher beliefs and their classroom practices found that teachers’ experiences impact what they believe teaching should be like and that teachers form their own theories in teaching.
- PPT is the systematic set of beliefs (**theories**) guiding teachers’ practices (**practical**) that are based on their prior life experiences (**personal**).

Personal practice theory then is very much about a personal, a systematic set of beliefs that a teacher has, which describe theories for the individual teacher, and how the teacher operationalizes these theories, or in other words the theories which guide their practice. So, there is a practical component to personal practice theory. And then this is all brought together through the experiences, the work experiences as well as the life experiences that the teacher has. Personal practice theory is something that all teachers will develop in time through the experiences that they have and their evolving values and beliefs.

- What does this mean for the reflective practitioner?

- Surfacing their PPTs enables teachers to be more aware of their rationale for on-going decision-making and empowers them to become reflective practitioners.
- **Developing Your Own PPT**
 - You can use the following chart.
 - There are no right or wrong answers.

Understanding Self and Others: Professional Identity

Professional Identity

- What is it?
- A more or less conscious apprehension of what constitutes good and professional teaching and learning (Huggen 2005)
- Develop through participation in professional communities in cooperation with others on concrete tasks (Wenger 1998)
- Knowledge and skills are transmitted through language and communication (Vygotsky 1978)
- These definitions imply that
Theoretical perspectives and the habit to reflect must be experienced by teachers as normal teaching activities.

Reflection in Teacher Development

Practice Oriented

- **Practice oriented reflection (practice as training)**
- Based on experiences and what works
- Theory is not important

One premise is that, professional identity is **practice oriented** or based in practice-oriented reflection. This means that the teacher uses their experience, finds out what works and what doesn't work and uses this in order to create that their own identity as a teacher in this sense theory is not important. It is the experience which has more value.

Practice as Education

- **Professional reflection (practice as education)**
- **Critical reflection, self-reflection**
- **Theory is an important tool.**
- **Professional reflection is necessary to educate authoritative independent teachers**

But there is another premise about professional identity, which is that it is based in **professional reflection** or in other words **practice as education** in this sense theory is an important component of professional identity and actually professional identity is developed through critical reflection and self-reflection. So professional reflection is necessary to the educator in terms of being authoritative as a professional and in terms of being independent and confident as a competent teacher.

Reflection in Advance

- **Reflection in advance**
- **Central to intellectual and professional growth**
- **Reflection is mostly instrumental/practice oriented, focusing on “technical” aspects and what the students could have done better.**

Understanding Self and Others: Developing Professional Identity

Developing Professional Identity

- Meaning is constructed through communication and cooperation in concrete situations with others.
- By participating in normal teacher activities together with competent others as legitimate peripheral members.
- It is by participating in this communication and cooperation that teachers start to develop a professional identity.

Finding A Mentor

- Mentors are competent teachers serving as models for the reflective practitioner.
- Professional growth, thus implies that reflective practitioners meet mentors who focus on professional aspects of teachers' work.
- Experience critical reflection/thinking and pedagogical argumentation as part of professional teaching practice.
- Assumption are verbalized and challenged.
- Experiences with theory based practice will inform the reflective practitioner that critical reflection and theoretical perspectives are normal parts of professional competence.
- Such experiences will, thus be expressions of competence and good teaching (professional identity)

Constructing your Identity

- The construction of a professional identity and its development are individual maturation processes.
- It begin when one enters training for the professional as a novice and these processes evolve through many development stages, Until finally one perceives one's professional identity.

Understanding Self and Others: Multiple Intelligence Theory and the Reflective Practitioner

Multiple Intelligence Theory and the Reflective Practitioner

The benefits of utilizing (MI) theory in educator professional are multifold

- It emphasizes the process of learning in delivering desired outcomes.
- It promotes thoughtful consideration of learning experienced in a variety of diverse modalities.
- It generates a broad survey of experiences across levels and disciplines.
- It encourages development of strategies beyond the strengths and interests of the teacher
- It provides a varied pacing of activities throughout the development experience.
- It fosters authentic, real-world experiences that promote deeper understanding.

Understanding Self and Others: Multiple Intelligence Theory and the Reflective Practitioner

Multiple Intelligence Theory and the Reflective Practitioner

- There are now nine MIs Multiple Intelligence Theory and the reflective practitioner
- How does the reflective practitioner's approach their professional identity growth when applying MI theory.

And we can also diagram this for ourselves as well. What we can see is that for the reflective practitioner. (MI) theory can be broken into three areas, **the analytic** component of reflection. **The interactive** component of reflection and **the introspective** component of reflection.

In terms of analysis therefore, the reflective practitioner something the reflective practitioner engages with all the time then the intelligences which are based in a **logical and rhythmic and naturalistic, learning**, become very useful.

Where the reflective practitioner is being more introspective or in other words, thinking about the experience, reflecting on the experience and the interpersonal existential and visual components of (MI) theory become very useful. And what the reflective practitioner can apply from (MI) theory in terms of how they action their reflection, are based in **linguistic interpersonal and the kinesthetic** modes of learning.

- Let's Consider each path to Growing professional identity.

Visual/Spatial

- Learning visually and organizing ideas spatially. Seeing concepts in action in order to understand them.
- The ability to “see” things in one’s mind in planning to create a product or solve a problem.

The visual spatial component is about learning visually and organizing ideas, **visually spatially**, seeing what concepts, come together in action and how we understand them. This might be translated into diagrams. For example, in a learning journal. It's the ability to, to see things very clearly to get a picture of things in the mind. how are things working out?

Verbal/Linguistic

- Learning through the spoken and written word. This intelligence was always valued in the traditional classroom and in traditional assessment of intelligence and achievement.

Mathematical/Logical

- Learning through reasoning and problem solving. Also highly valued in the traditional classroom where students were asked to adapt to logically sequenced delivery of instruction.

Bodily/Kinesthetic

- Bodily/kinaesthetic learning through interaction with one’s environment. This intelligence is not the domain of overly active learners. It promotes understanding through concrete

experience.

Musical/Rhythmic

- Musical/rhythmic learning through patterns, rhythms and music. This includes not only auditory learning but also the identification of patterns through all the senses.

Intra-Personal

- Intrapersonal learning through feeling, values and attitudes. This is a decidedly affective component of learning through which we place value on what we learn and take ownership for our learning.

Inter-Personal

- Interpersonal learning through interaction with others. Not being simply talkative or overly social. This intelligence promotes collaboration and working cooperative with others.

Naturalistic

- Naturalistic learning through classification categories. The naturalist intelligence picks up on subtle differences in meaning. It is not simply the study of nature. It can be used in all areas of study.

Existential

- Existential learning by seeing the big picture. Why are we here? What is my place in my family, school and community? This intelligence seeks connections to real world understanding and application of new learning.

Understanding Self and Others: Multiple Intelligence Theory and the Reflective Practitioner

Verbal Intelligence

Verbal

Lecture, reading, writing, reporting, presenting, reciting, discussing, online discussion

Logical Intelligence

Logical

Problem-solving, brainstorming, hypothesizing, investigating, experimenting, Socratic method, online searching.

Visual Intelligence

Visual

Observing, symbolizing, drawing, outlining, conceiving, imagining, visualizing, video.

Kinesthetic Intelligence

Kinesthetic

Building, imitating, performing.

Musical Intelligence

Musical

Listening, patterning, mirroring, repeating

The intrapersonal Intelligence

Intrapersonal

Supporting, advising, advocating, characterizing, defending, evaluating, judging, challenging.

The interpersonal Intelligence

Interpersonal

Sharing, leading, helping, managing, collaborating, influencing, team-building

Naturalistic

Naturalistic

Categorizing, contrasting, classifying, organizing, semantic/concept mapping, graphing.

- For any sort of identity-building activity, you should focus on at least three intelligence but no more than five intelligence.

Understanding Self and Others: Making Professional Identities through a Mentor

- Finding a mentor in your work place can have a significant effect on both the mentor and mentee regarding professional identity and development.
- Under a formal mentoring process the mentee not only reaches the competency level of the mentor but grows beyond the established baseline along with the mentor.
- The mentor may share materials but also goes beyond the sharing and moves into the development of materials within a collaborative relationship. This requires reflective practice and collaborative planning coupled with a joint action plan by the mentor and mentee.
- Through this process both mentor and mentee (the reflective practitioner) can soar to new heights of professional growth and competence.

Functions of Mentoring

What do you do when you are seeking a mentor? Well. What you want to do is actually understand what the functions of a mentor are and try to identify or find somebody who can meet those functions and who can provide that facility for you.

Seeking a Mentor? Know the functions of Mentoring

Teaching

- Teaching the skills that teachers practice on a daily

Well, one function of course is **teaching**. The mentor is somebody who you would regard as highly experienced and highly proficient at teaching. Somebody who you could watch, learn from, and experience with.

Sponsoring

- Sponsoring someone who can identify your strengths and advise as to what activities would be most successful. Sponsoring then requires the mentor to support you when you attempt a new practice.

Another function of the mentor is that they are somebody who can **sponsor** another person. Sponsoring means that they support and guide the growth of that individual. They can identify the strengths of the individual and help the individual build on those strengths, identify weaknesses and build their weaknesses. There is somebody who may be will even advocate in the school environment for somebody else.

Encouraging

- Encouraging is a key mentor function. By helping you see the positive side of your teaching practice and building on those reflections, the mentor is supporting and encouraging your growth.

Another function of a mentor is to **encourage**. Encouraging is very important because teaching is a very complex and very difficult profession. And therefore, encouragement especially for the reflective practitioner is a vital component of building confidence and developing professional identity.

Counseling

- Counseling is the fourth function of mentoring. You need sound advice regarding teaching practice, professional conduct and the culture of the school and community.

And also, the mentor needs to be a good **counselor**. Somebody with good listening skills, somebody with good communication skills, somebody who is non-judgmental perhaps unbiased, and somebody who and we can confide in knowing that anything we say is confidential and will not be judged.

Briefing

- The fifth function of mentoring is befriending, you need someone to whom you can speak freely.

And finally, we need to think about in terms of looking for a mentor is somebody who in a sense can become a friend. Most importantly a critical friend. We are looking for somebody who can **befriend** us as reflective practitioner and, in a while, look after and guide us as teachers as we build our identity.

The Importance of Self-Awareness: How does RP Support Professional Practice?

How does RP support professional practice?

The importance of self-awareness to teacher identity and what we have to ask is how reflective practice in this sense supports professional practice. Well, what we can consider is a model called the **conscious competence model**.

The Conscious Competence Model

2. Conscious Incompetence (Awareness of the existence and relevance of the skill)	3. Conscious Competence (Can perform a skill at will)
1. Unconscious Incompetence (not aware of the skill (and lack of skills) and its relevance)	4. Unconscious Competence (the skill is so practiced that it becomes second nature)

We can see that the **conscious competence model has four quadrants**. The first quadrant is **unconscious in competence**. This is where the reflective practitioner is unconscious of what they do not know.

The second quadrant is **conscious in competence**. This is the reflective practitioner growing in awareness of what they do not know.

The third quadrant is **conscious competence**. This is where the reflective practitioner becomes more aware of the skills that they do have.

And finally, the fourth quadrant is **unconscious competence**. This is where reflective practice seems very easy where the teacher (the reflective practitioner) is unaware of what they do know. So, we can see that this model demonstrates how a teacher how a reflective practitioner moves through different levels of knowing.

The model is very useful for the reflective practitioner to consider because it is important and useful also to know where we are at any particular point in terms of our learning. And what we have to remember is that we can be at any stage in the model at any time indeed. For example: if you become a student on a new course, you could be in the unconscious incompetence phase of learning rather than any other phase.

So, the model highlights two key factors here. These factors are about the teachers' awareness of what they know and what they do not know. **The consciousness aspect (awareness) and also the competence aspect (skill).** The skill level that the reflective practitioner is at any particular point.

Understanding the Conscious-Competence Model

According to the model, we move through these different levels as we learn and we will always start at level one, which is **unconsciously unskilled**, or in other words, **we don't know what we don't have this skill.** This clearly relates to entering into a new situation where learning is new for us and of course for teachers that could be just starting out in teaching.

The next level is **consciously unskilled- we know that we don't have this skill.** This is where a teacher becomes aware of the skills, the knowledge, and the abilities that we don't have. Therefore, what we need to develop.

The next level is **consciously skilled- we know that we have this skill.** This is where we realize that we have learned the skills. We are aware that we have those skills. We are actively thinking about using them and actively regarding the knowledge that we have.

The fourth level is the level that we aspire to this is **unconsciously skilled- we don't know that we have this skill.** This is where teaching seems very easy. This is where we have learned sufficiently to develop a professional identity to point that everything seems very easy. It is an application of skills in a way that we do not even think about. It seems very natural. And instead of the matrix which we have just looked at. We can also consider it, therefore, in terms of a ladder from level one to level four and here we can see the learning ladder. The conscious competence model is applied to a ladder of learning. Level one is unconsciously unskilled. Three to level four which is unconsciously skilled. We can refer to this model as a reflective practitioner in many different ways.

The Importance of Self-Awareness: Applying the Conscious Competence Model Level 1- Unconsciously Unskilled

Let's look at how we might apply their conscious competence model as reflective practitioners. First of all, we can start at level 1: unconsciously unskilled. For the reflective practitioner, this is where we are ignorant of what we do not know, what we do not know is invisible to us, we have not yet uncovered what is missing from our practice and what we need to develop. Your confidence therefor far exceeds your abilities.

To move out of level one. to move away from being unconsciously unskilled, it can be very useful for the reflective practitioner to do a personal SWOT analysis. Looking at strengths, weaknesses, opportunities, and threats in other words to conduct a personal training needs assessment to identify those areas where you are unskilled and where you need to develop.

As part of this, it is very useful to ask other people, to ask colleagues for their input to your training needs assessment. This can help you uncover weaknesses that you are not aware of because you are unconsciously unskilled. There will be areas that you do not know are lacking and therefore input from outside is incredibly valuable. Also, it is important to make sure that you are very clear about their learning objectives that you have as a teacher because this will give direction to the skills that you need to develop. There's no point learning skills in areas that don't align with your personal or work goals.

Applying the Conscious Competence Model Level 2- Consciously Unskilled

At level 2, you are consciously unskilled, in other words by this stage, you discovered those areas which are weak, those areas which need to be developed and you are aware that you need to learn new skills and acquire new practical ways of doing things. You also realize that you are perhaps much less competent in the role than others that you see around you. It is very important at this point to remember that you are in a development process and not to lose confidence because it is very easy when you are at level 2 when you are consciously unskilled and when you are aware that you have many weaknesses and you have a lot of growth to go through. It is very easy to become unconfident. This level can be demoralising, causing you to lose confidence or even give up on your learning efforts. Therefore, it's important to stay positive at this stage

Applying the Conscious Competence Model Level 3- Consciously Skilled

At level 3, you become consciously skilled. This is where you realize that you have learned and you **are able to** apply skills in practice. It is where you are starting to put your learning into practice in the classroom and you can see the benefits of it and you are also reaping the reward of that. It is at this point, where you are starting to build your confidence. You are becoming more confident with the skills even though you are **actually** applying them in a very measured and thought-out process. It is not unconscious, it is still conscious at this point. Therefore, as a practitioner, you are concentrating very hard on applying skills and making sure that you are applying them in the right sort of way. It is very much about being more aware of what you are doing and what is important to move from level 3 to level 4 is to look for opportunities where you can practice your skills, develop your skills in new ways in order to move away from the consciously skilled component to the unconsciously skilled aspect of the model.

The Importance of Self-Awareness: Learning to Learn

Learning to learn

- The ability to be able to pursue learning and to persist in learning throughout one's career, even when learning can be very difficult. In this regard, it is very much about the individual knowing what works for them as a learner, their different approach to learning.
- To organize one's own learning and also being able to organize the way they learn and what they learn.
- It is very much also about effective management of time and also of things like motivation and approaches both individually and in groups that you bring as a learner to a learning situation.

Learning Perspectives

In this regard, therefore, we can consider several perspectives about learning. We can think about their **orientation** that we bring to learning or in other words, is learning purely an **individual approach** or is it about learning within a **group** or learning with it within a team and it can also be much wider than that it could be learning within and for the **organization** meeting the needs of the organization as much as meeting the needs of the individual.

Another perspective is the actual **approach to learning** that we take as reflective practitioners. Are we focusing on the **causes of situations**? or are we actually focusing on **subjects**? Or are the subjects that we are interested in not so important? so, we reject those and just focus on the situations so the perspectives that we have to guide the learning process. We can look at those perspectives diagrammatically.

The Importance of Self-Awareness: The Jo-Hari Window (Self-Evaluation in Practice)

The Jo-Hari Window is a communication model that is used by people to develop a better understanding not only of themselves but also of other people that they work with. “Johari” comes from the two people who originated the model **in 1955**: Joseph Luft and Harry Ingham. The model has developed a long time ago but it is a very popular model for communication and therefore for learning,

The Jo-Hari Window Model

In **the Jo-Hari Window** there are four quadrants. The top left quadrant is called the **Open Area (Quadrant 1)**. This area represents for the reflective practitioner the things that you know about yourself and the things the other people know about you as well. This includes things like your behaviour, your knowledge, the skills that you have, and the attitudes that you have as a professional in a way it is almost like your public history.

The next quadrant area 2 is known as **Blind Area**. This area represents the things that you are not aware of but are known by others in a way this is like being unconsciously incompetent. And this can involve simple information that you do not know or it could be even deeper issues that you do not know about yourself but other people do know.

The third quadrant is known as the **Hidden Area**. It is hidden because it represents the things that you know about yourself but the others do not know. This is the private, closed-off area that you keep secret to yourself and the others do not know.

The fourth quadrant is known as the Unknown Area. This area represents all those things that are unknown to you and which also unknown to other people.

So, we can see we have these four quadrants in this model quadrants which relate to how much we know about ourselves and how much other people know about us. And what we are always trying to do in terms of

developing our teacher identity is to raise our own awareness about our skills and knowledge about ourselves and also helping others to become more aware of those things in us too.

The Importance of Self-Awareness: Using the Jo-Hari Window in the Team Context

Feedback:

The important thing in terms of a new teacher joining a school where they are open-air is smaller and will be the importance of feedback because this will be vital in terms of reducing the size of the blind area and increasing the size of the open area. And where feedback becomes significant is in terms of helping an individual understand where they fit in an organization, the spoken and unspoken rules, and the consequences of working there.

Feedback for group members is very much than about striving to help the whole group increase the size of the open area within this model. And therefore, feedback needs to be constructive.

Any non-constructive feedback will actually reduce the size of the open area because the hidden area will start to increase. People will stop disclosing and because of the non-constructive feedback that they might get. So, feedback should always be constructive and therefore the size of the open area can expand vertically downwards reducing the hidden area and increasing the size of the open area for reflective practitioners. This is due to increased disclosure telling people about feelings and experiences the problems that you have the worries and concerns. These are all things which can help increase the open area and therefore get more support and reduce the hidden area.

From Theory to Practice: Theory, Pedagogy and Reflection

What we need to look at as reflective practitioner is how we translate theory into practice and how this is professionally and personally empowering for teachers. **Research shows us that:**

- Reflective practitioners are people who are regularly evaluating their approaches to teaching and the impact that this has on learning.

- Reflective practitioner is somebody who understands more about the positive impacts that good quality teaching has on learners and the quality of education that occurs in schools.
- Reflective practitioners are teachers who can draw upon a range of strategies and approaches to helping students learn.
- Reflective practitioners are teachers who can co-construct learning with students, with the people that they are working with in a way that is mutually empowering.

Therefore, what we need to do is think about what this means for the reflective practitioner. What does critical reflection actually involve? Well, it really means that it involves understanding our own learning and understanding ourselves as learners. So that we can bring this to the pedagogy that we use the teaching approaches that we apply in the classroom. **These are relevant to several areas.**

- One particular area is that the reflective practitioner understands children and their needs as learners.
- The reflective practitioner also someone who can build effective and supportive relationships within the context that they work.
- The reflective practitioner is also somebody who can establish learning environments that are flexible and adaptable.
- The reflective practitioner is somebody who can explore what children are learning, what they are interested in learning, and also how they are learning.

This means that teachers hold multiple roles and certainly the teacher as a reflective practitioner is somebody who takes on these many roles within the classroom. And therefore, the reflective practitioner is someone whom we need to look at through various lenses because of the different roles that they take on.

This raises several questions for us as reflective practitioners. Questions such as:

- How do we build and manage relationships with people, relationships with colleagues, relationships with management, and also relationships with students?

- How we support and guide students' learning? This is a significant question for the reflective practitioner because what we are always doing is questioning our approaches in order to reinforce and develop them.
- How well do we plan for learning and how good is our preparation to maximize the effectiveness of what we do.
- How would we describe ourselves as a learner part of being a good teacher is understanding what makes us good learners?

From Theory to Practice: Application

“It is because professionals face complex and unpredictable situations that they need a specialized body of knowledge; if they are to apply that knowledge, they need autonomy to make their own judgments. Given that autonomy, it is essential that they act with responsibility; collectively they need to develop appropriate professional values”

Furlong et.al (2000)

From Theory to Practice: RP and Professionalism

Autonomy

Well, one thing we can identify as an attribute of being a professional is our autonomy as professionals; as learners within the complex environment of teaching. There is a conflict between the unpredictable nature of the context we teach in & centralisation of control in education. This illustrates the fact that there is a conflict for us as reflective practitioners between the work we are doing and the nature of working in an environment that tends to be quite rigid and quite strict. By developing autonomy, the reflective practitioner develops becomes more empowered as a person and as a professional.

Professional Knowledge

Another aspect of professionalism is the knowledge that we bring to the role of teaching. Their professional knowledge that we have and perhaps more than in any other profession. This is a complex issue because it

has two strands. Of course, a teacher has to have **subject knowledge**. For example, a math teacher must be very knowledgeable about mathematics but the reflective practitioner is awesome also somebody who knows how to teach their particular subject, the **pedagogical knowledge**. So, these two things come together to actually make a professional knowledge and this is an important aspect of professionalism.

Responsibility

Also, an important aspect of professionalism is the accountability that we have as teachers because we are teaching to other people so we have the responsibility to the families and to the children, the people that we work with. In terms of our role and of course, there is in this respect a responsibility to the profession because we all take a particular ethical moral values approach to what we are doing. We cannot do certain things in the classroom, certain things we can do and we need to work within certain ethical and moral frameworks.

Rigour and Relevance

And what all of this means for the reflective practitioner is that we need to develop a particular rigor and relevance in terms of our professional role. This is something that we bring with us into the classroom. We bring with us into the school environment. It's something that underpins the approach to reflective practice. The real issue here is that teaching especially in the modern era is very complex. It's very messy very undefined and therefore the reflective practitioner as a professional is somebody who is working within this very unstable **and** messy framework. Therefore, if the reflective practitioner is somebody who is simply remaining fixed to theoretical backgrounds, then we are not being responsive and losing that particular rigor and relevance that we need to work effectively with the students.

From Theory to Practice: Good Professional Standing

Moving from theory to practice for the reflective practitioner is about developing a good professional standing becoming seen as a high-performing professional. This means that the reflective practitioner has particular **commitments**.

Commitment to Learners:

One particular commitment is to the learner of course and perhaps this is the starting point for reflective practice. What the teacher must do is maintain professional relationships and take on a professional role at all times with learners. It is simply because of the learners, their families of learners, the people that we're working for, and trust learning to us as teachers. And therefore, there has to be a certain level of respect that we pay to the learners as reflective practitioners.

In terms of the commitment to learners, we also need to acknowledge be aware of and communicate our understanding that all the people we work with are unique individuals. They all have particular learning needs and interests and therefore the reflective practitioner commits to recognizing this and attending to this in the work that they do.

And also, in terms of a commitment to learners, what we need to be aware of as reflective practitioners is that we have a role in terms of motivating students and also encouraging and supporting our students in terms of learning. Learning real learning is actually a very difficult thing for anyone to do and therefore part of our role is to help, support, and therefore motivate students in terms of their roles as learners.

Commitment to Colleagues and Others:

We also have a commitment to the people that we work with. We have a commitment to colleagues and we have commitments to the organizations that we work with and what this translates into is that the reflective practitioner is somebody who works within a community, a professional community, a community that not only supports growth for students but has a place in terms of intellectual, social, moral and ethical growth of everybody working within the community.

We also have a commitment to our colleagues in terms of promoting professional relationships and the collegiality that we have with others, sharing knowledge, and giving feedback to each other on the work that we're doing.

And in terms of having a commitment to colleagues, we need to work as people who will cooperate effectively with others, spend time supporting, sharing resources, and learning from others, and it through that actually raising the standard of the profession.

And along with this particular commitment is that we need to understand that we have a professional role with parents of children families who are bringing their children into the school and therefore we need to ensure that relationships with parents, in their capacity as partners in the educational process, are characterized by respect and trust.

Commitment to colleagues is also concerning a confidential role. As teachers, we have the opportunity to access information. We develop information about the people that we work with and we should see this as confidential and not for the public domain. We should respect confidential information relating to students or colleagues gained in the course of professional practice, unless the well-being of an individual requires disclosure.

Commitment to the profession:

And of course, we have a commitment to the profession as a whole. We need to ensure as reflective practitioners that we are contributing to the profession that we're contributing to review and revision of policies and practice with a view to optimising the opportunities for students or addressing identifies individual or institutional needs.

And in terms of a commitment to the profession, we need to hold the profession in very high standing. We need to maintain its integrity, value it, and assume responsibility for their ongoing professional development needs as an essential expression of their professionalism.

From Theory to Practice: Personal Challenges to Effective Reflective Practice

Building more professional and personal empowerment for the reflective practitioner is about moving from theory to practice. There are certain challenges that we can face as reflective practitioners. For example:

- we can have expectations that are unrealistic or in a way misplaced in terms of what we expect from ourselves and colleagues and from the students that we work with.

- Another challenge is that early on in our training we can develop our notions about what education is about, about teaching and learning as a whole. These notions can begin to interfere with our work in the classroom if we are not being flexible and adaptable enough.
- There can be misconceptions or misunderstandings and from what we have learned from theory when we are confronted by the reality of work in the classroom.
- There can also of course be personal preferences which can become a challenge in other words the way we like our classroom to look the way we like students to behave may not be what we experience and of course.
- There are also things like personal egotism and pride which can get in the way of being a reflective practitioner. These can actually become barriers to development.
- We all have we all hold our own beliefs about education and therefore the philosophy that we have about education can be something that can become a challenge for us once we are confronted by the work that we are engaged with.
- And of course, not least is life can actually get in the way. We all have families, we all have a home life, we have friends, and trying to balance the demands that education as the work we have as professionals as teachers can be challenged by the commitment. The responsibilities that we have in our life in general.

Over-coming:

So, the way to try to overcome these things is very personal to each teacher and really, it is about trying to find your own strategy for overcoming particular barriers.

- **Misplaced expectation:** In terms of expectations that may not be realized the thing really to understand is that you need to be honest with yourself about what is possible what is not possible. This can be very difficult because it does require you to be very open-minded and perhaps a good approach would be to have a colleague as a critical friend and with who you discuss particular expectations. They can give you a different perspective on the reality of the work that you are doing.

- **Pre-conceived notions:** They can of course be the preconceived ideas about the way we should be teaching, the way students should be learning the way to try to overcome these is perhaps to refer back to the literature, refer to research, develop a broader understanding of what is possible and again share these ideas and explorations with colleagues is a good way to actually develop a better understanding of the notions.
- **Misconceptions** about teaching and learning a good thing to do is perhaps develop and for yourself will develop with colleagues a set of checklists that you can use to guide what you are actually doing.
- **Personal preferences:** In terms of personal preferences, we all have our own style as a teacher and really what you need to do again is be honest with yourself as a practitioner and to realize that what you may need to do is to change your approach and to change your preferences to meet the needs of the learners that will require you to be quite flexible and adaptable in your approach.
- **Egotism:** If you have a personal egotism this can get in the way of learning and what you need to do is try to move away from that a good thing to do is to observe other teachers and to see how they teach. Therefore, to get a bigger perspective on the role that we take on.
- **Pride:** If you develop a certain level of pride this can become a barrier to reflective practice and also be empowered as a professional. The key to this is to have somebody you can confide in, somebody you can talk to about your professional role to stay grounded, to stay realistic about what you are doing.
- **Educational philosophy:** Your personal educational philosophy is something that can become a hindrance. This can be overcome simply by sharing understandings with colleagues about philosophies of teaching and learning, about personal values and beliefs along the lines of education.
- **Life** is perhaps the biggest barrier that can get in the way the key thing is not to try too hard remember that teaching is ongoing. You cannot do everything in your life and sometimes you need to find a better balance.

From Theory to Practice: Questioning

Developing a framework for personal and professional empowerment is about moving theory about reflective practice and to the practical application of reflective practice. A key requisite for doing this is really taking on a questioning role for ourselves as teachers. It is, therefore, useful to explore some key questions that we can relate to our professional role, and through this questioning approach, we can start to develop a sense of personal empowerment. **It helps you to move from theory and discussion, to practice and application.**

Be Backward-looking:

So, one set of questions that we can explore our questions which get us to look backward, look into the past, look at what has happened in the classroom. Questions like how much did you know about the subject before you started teaching it? Questions like have you done similar kinds of work in the past? and also questions like what problems did you encounter when you were teaching these subjects previously? Backward-looking questions provide a foundation for us to explore current experiences. They provide a foundation from which we can build.

Be Inward-looking:

After we have engaged with some questions which get us to look backward, look historically at our practice. We can consider some questions which help us look inward. Inward-looking questions are those which help us understand our reactions, our responses to experiences that we have in the classroom. Questions like what did you or do you currently find frustrating about work? Did you meet particular standards for teaching when you were working in the classroom? and perhaps one of the most important is how do you currently feel about the work that you are doing? By looking inward, the reflective practitioner is taking chance to contextualize the backward-looking responses. And thereby, understanding within different this particular framework how we are working as reflective practitioners.

Be Outward-looking:

So, we have backward-looking questions and we have inward-looking questions and we also have outward-looking questions. This is where we are looking at the context that we are working in as reflective practitioners and trying to develop an understanding about where we fit into the jigsaw picture that is the whole of

education whether that is just within the school, the educational environment that we are working in or whether it's a bigger picture nationally but there are questions that we can ask here for an example: did we teach in the way that other people are teaching in the school? how do we compare with others? and questions like in what ways there are approaches to teaching similar to others? how are they different? and what grade would you give yourself concerning others if you are going to try and measure the work that you are doing?

Be Forward-looking:

So, what we need to think about is how we can put these questions, these categories of questions into a form that we can operationalize. We have a foundation in backward-looking questions and we analyze our role with inward-looking questions and outward-looking questions. So, what we need to do is ask forward-looking questions. In other words, what will we do next? what changes do we need to make? how will we operationalize things of which we are learning? And another question is, what will I change for the next lesson? It is a particular forward-looking question that helps us put into action the learning that we require from the previous types of questioning.

Frames for Learning: Frames, Perceptions and Interpretation

Frames 1

- Frames help us perceive, understand and describe situations.

What are Frames?

- Frames are particular ways of making meaning of the complexity of situations.
- Frames are structures of beliefs, perceptions, values and appreciations about a situation, most of them underlying conscience.
- From the multiple components of a situation, we select a particular set with which we try to understand the situation and to transform it.

The Process of Framing

- The starting point is a vague image of a reality named by Jon Dewey as a “problematic Situation”.
- This situation is named and framed selecting a few salient /selected features and relations multiple elements of the complex reality.
- These elements receive a coherent organization, and are used for describing what is correct and what is wrong with the situation or what is the problem.
- The direction for the transformation of the situation will come from the elements established in the frame.

Rhetorical and Action Frames

There are two main kinds of frames:

- Rhetorical Frames, and
 - Action Frames
- Rhetorical frames are related with espoused beliefs, and are used for arguing about a situation.
- Action frames are related to policies in use. They are used for dealing with the situation.

Frames and Metaphors

- Frames frequently use metaphors as a way to develop and/or to communicate their particular logic.
- Metaphors are of great utility for explaining and for thinking with a particular frame.
- Based on a metaphor, the actions look more familiar and consistent with the requirements of change of the situation.
- There is a double-sided interaction between frames and interests.
- The teacher’s interests influence the way they frame the situation.
- But the frames of the teachers also shape the way they perceive their interests.
- These both relations tend to be tacit and invisible for the teachers.

Circularity of Frames

- In conflicting frames there is no way of falsifying a frame with valid data.
- When we arrive to situations where different frames are resistant to critique, we are in a situation that is not normal.
- Frames are mostly tacit.
 - “The frames that shape policies are usually tacit, which means that we tend to argue from our tacit frames to our explicit policy positions”.
 - “.... we are usually unaware of their roles (of the frames) in organizing our actions, thoughts and perceptions” (Donald Schon --- Frame Reflection)

Frames for Learning: Frames, Perceptions, and Interpretation

Frames 2

- In most cases, we are not working with problems, but with messy situations.
- The process of going from the messy situation to the problem is the process of framing.
- The names we give things are important – they determine how we will work with them in our mind.
- Then we organize them and we describe what the problem is.
- In the same way that Schon talks about espoused theories and theories in use, he talks about rhetorical frames (espoused policies) and action frames (policies in use).
- Rhetorical frames are weapons used to attack other frames and convince people that our frame is a good one.
- The frame that wins is the one that can expose the weaknesses in other frames, while hiding its own weaknesses.
- Frames work with metaphors.
- When we create a frame for a situation, we always use a metaphor.
- Frames are not things that emerge from scratch, they are always connected to a metaphor.

- Changing the metaphor is a very powerful way of changing the frame.
- The way we frame a problem affects how we perceive our interests.
- It's difficult to discover the limits of our thinking – we always believe our framing is consistent. It usually is not possible to convince someone that their frame is inconsistent by giving them more information, since that information will be processed into the existing frame
- The frame is able to digest information.
- So basically, more data won't ever change frames, and we need to engage people at the level of their assumptions.
- The frame gives meaning to the information by its own logic. Most frames are tacit, we are not aware of the way we frame situations or of the way other people are framing the same situation.
- The way to solve this difficulty is to try to make the frames explicit.
- Finally, when we are discussing other frames, we need to bring their meaning to own frame.

Try this activity in framing a problem.

1. Define the problem?
2. Why is it a problem? (This is about your reasoning in defining the problem)
3. What are the factors that contribute to problem? We can talk here about causes, context, conditions that allow the problem to exist.
4. What do you see as the impact of the problem?
5. Analyze for consistency and completeness.

Frames for Learning : Analyzing Frames 1

Discovering Underlying Assumptions and Values in Frames

Look at the following example:

Naming and Framing the Education Crisis

- **No Licensure:** We need to remove the inconsistencies between different education providers.
- **Quality of Teachers' Education:** We need to ensure that teachers are being prepared according to uniform standards.
- **Common Inspection Framework:** We need to have a system of checking the quality of education across all providers.

Assumptions

- Assumptions are belief about the world
 - 90% of students are happy in our school.
 - Education improves the quality of life.
- The first type of assumption is situational.
- The second type of assumption is theoretical.
- If a belief is tacit in our thinking, we name it as “assumption”, once it is made explicit it becomes a “hypothesis”.

What are Values?

- Values are criteria that we use for deciding if a situation is desirable or undesirable, or if an idea is a good one or a bad one:
 - Being secure or insecure (security)
 - Being treated fairly or unfairly (justice)
- Every time we frame or evaluate situations, we apply values. Frequently values are tacit.
- In the example “**Quality of Teachers' Education:** We need to ensure that teachers are being prepared according to uniform standard” the quality of teaching is a value.
- In the example “**Common inspection Framework:** We need to have system of checking the quality of education across all providers” how schools provide education is a value.

Frames, Assumption and the “Thinking Box”

- Every time we frame a situation, we base elements of our frame in belief about the world; but we are not aware of them.
- The discovery of assumption is easier when we are “out of the box”.
- People who disagree with our frames are more acute for identifying our assumptions.

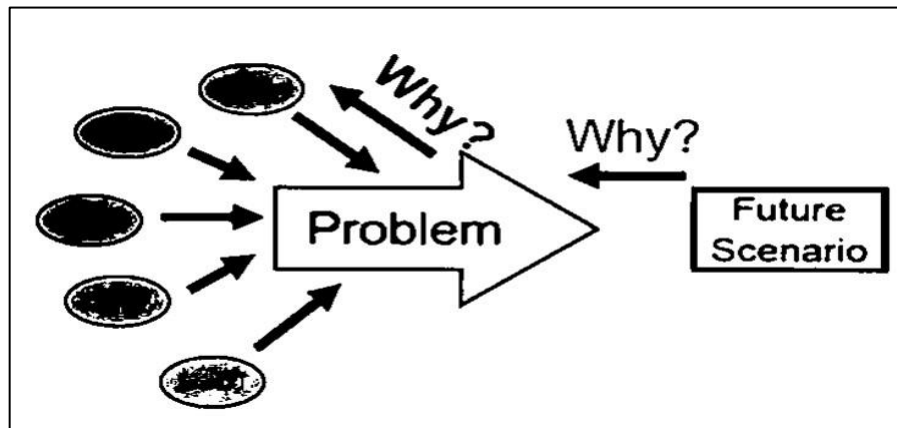
Frames for Learning: Analyzing Frames 2**Discovering Assumptions**

- Assumptions are beliefs that are behind the causal relation of our frames.
 - If “assumption A”. then “B” generates “C”.
- A possibility for discovering assumption is to go through the components and the causal relations of the frame asking “why” they are stated in that particular way.
- The assumptions should be falsifiable. It means that it should be possible to get information for verifying if an assumption is real or not.

Exploring Assumptions

- If a situation is framed as a problem generated by a group of factors and a scenario that is resulted from the problem, then the exploration of assumptions may go back through the causal relations of the frame. Let’s look at this in diagram form:

Exploring Assumptions

**Questions for Exploring Assumptions**

- Why the scenario or one component of it, is generated by the problem or a component of it?
- Why the factor A contributes to the problem?
- Which conditions are required for the work of causal relations?
- Why do you think it is a problem?
- Which values are threatened by the problem or by the scenario?
- Which motivations are behind the way the situation was framed?

Frames for learning: Interpolating Concepts and Ideas**Theories –in-use- are means:**

- For exploring concepts
- Developing ideas

Schema of Theory and Action

- A theory of action is a set of interconnected propositions that have the same referent (the subject of the theory).

- For example:
- In situation of S, if you want to achieve consequence C, under assumptions a1, a2, a3, and, do action A.
- Theories are vehicles for explanation, prediction or control.

Theory-in-use

- Agyns an Schon say that theory-in-use is:
“...existing theories people have that already determine practice. We call these operational theories of action theories-in-use to distinguish them from the espoused theories that are used to describe and justify behavior.”
- Agries and Schon say that theories-in-use:
“.... help teachers remain blind to the actual degree of their ineffectiveness.”

Theory-in-use, Assumptions and Everyday Life

“.... the full set of assumptions about human behavior that function in the theories-in-use constitutes a psychology of everyday life.”

- There is a theory –in-use of building design (for architects).
- There is a theory in –use for diagnosis and treatment of disease (for physicians).
- Theories in-use are means for getting what we want.

How to make Theory-in-use visible

- We can construct someone’s theory in-use from observations of behavior.
- This includes recognizing general characteristics:
 - The governing variables
 - The action strategies
 - The principal assumptions

Clues for discovering the Theory-In-Use

- Reconstruct the espoused theory and /or the action strategy.
- Look for the assumptions through the “Chain of Whys”.
- Look for the relations among the assumptions.
- Discover the governing variables of Espoused theory and /or the action.
- Construct the theory in-use based on these elements.

Requirements for Discovering the Theory-in-use

- Internal commitment to discovery
- Public testing of theory, assumptions and data
- Clear distinction between the self and assumptions (non-defensiveness)
- Look for valid information
- Aptitude for reframing

Promoting PR through Inductive Action Planning

- As a teacher you will be learning every day.
- Knowing what you have learned or what you want to learn from practice and experience needs some planning
- This will be helped by **inductive action planning**.

Essentially this involves:

- Selecting a specific target (e.g., effective rules and routines in class, widening your range of strategies, effective use of ICT etc.)
- Plan a program of discussion, observation and training over a fixed period.
- Evaluate your effectiveness
- Reflect on your own development and select further targets.

Inductive action planning is therefore about:

- Identifying opportunities for you to discuss, reinforce and develop your practice.

- Helping you decide what information/data of your planning and practice you can use as evidence of your development.

An inductive Action Plan will be one that:

- Builds on your career profile
- is precise
- Answer the following questions:
 - What is my priority and what do I want to achieve?
 - What am I going to do and when?
 - Who/ what will help me?
 - How will I know whether what I have done has been worthwhile?
 - How will I collect information?

In detail, an inductive action plan will:

- Identify your priorities
- At this stage of the year what precisely do I want to be able to do with greater confidence and more effectively?
- E.g., Motivate the class?
- Plan, manage and evaluate the teaching of reading...

In detail, an inductive action plan will:

- Identify what you will do and when
- E.g., I shall discuss school policy and expectation with my principal, I can attend a course which will help me think more deeply about my practice; I shall see school policy in practice by observing colleagues and discussing issues with them.

In detail, an inductive action plan will:

- Identify who can support you
- E.g., I shall enlist the support of my head of department, I shall engage in some background research.

In detail, an inductive action plan will:

- Identify Success Criteria by identifying what has been worthwhile -
- My planning now demonstrates...
- Through being observed and discussing the lesson I am clearer about my strengths and priority areas for development in....

In detail, an inductive action plan will:

- Identify how you can demonstrate your development -
- I have examples of the new lesson planning that I am doing...
- I have examples of the resources I'm using...
- A copy of my appraisal report...

Inductive action planning:

- Responds to, and support, your day-to-day teaching activity.
- It is directed by you.

Frames for learning

Reframing for Resolving Difficult Controversies: The Reframing Matrix 1

The Reframing Matrix 1

- Things look different when you change the perspective.
- When you are stuck on a problem it helps to look at it from another perspective.
- This can be all that you need to do to come up with a great solution.
- However, it is sometimes difficult to think what these perspectives might be.
- Using a reframing matrix relies on the fact that different people with different experience are likely to approach problems in different ways.
- Reframing helps to put you in the minds of different people- imagine how they would face these problems and explore their solutions.

Step 1: Draw the Grid

- Start by drawing a simple four-square grid.
- Leave a space in the middle to describe your problem.
- The boxes are for your different perspectives.

Step 2: Decide on perspectives

- Decide on 4 different perspectives to use in your matrix.
- Use the 4P's approach:

The 4P's

- **The Product Perspective**

- Is there something wrong with your teaching?
- Does it respond to the needs of your learners?
- Do you adapt your teaching to the circumstances in the classroom?

- **The Planning Perspective**

- Is your planning at fault?
- Could you improve your planning and preparation?
- How good are your resources?
- What resources are you using?

- **The Potential Perspective**

- How would you improve the learning outcome?
- If the students' grades were to go up how would that affect the problem?
- How are others doing the same work as you?

The People Perspective

- What are the people impacts and people implications of the problem?
- What do people involved with the problem think?

- Why are students not happy?

- These are just some of the questions that you can ask as you look at your problem using these four perspectives.

Reframing 2: The Professions Approach

- This approach helps you to look viewpoint of different specialists, or stakeholders.
- For example, the way a medical doctor would look at a problem is different to what a civic engineer might use.
- This approach is particularly useful when you are trying to solve a problem which involves many different types of people.
- It is useful when you need to 'step away' from your usual way of thinking so that you can be more creative.

Step 1: Brainstorm Factors

- Brainstorm the factors related to your problems from the four perspectives viewpoints:
 - Product perspective
 - Planning Perspective
 - Potential perspective
 - People perspective

Fill out the matrix.

Example of Reframing

- In the following example you will see how a teacher has used th 4ps approach to explore problems in class.

Reframing for Resolving Difficult Controversies

<u>Product Perspective</u> The subject is complex The subject needs some introduction in previous classes The subject is new in the school	<u>Planning Perspective</u> My planning follows a rigid pattern I use the same resources in every lesson I plan to get through a great deal of information in each lesson
Students are not interested!	
<u>Potential Perspective</u> The students are very comfortable with technology Students like to talk a lot and guided discussion would be a useful approach to exploring the subject	<u>People Perspective</u> Students seem generally bored in school Students do not see the need to study Other teachers are facing similar issues

In this example, we can see how a teacher might use the 4 P's approach and the professions perspective in order to reframe their perspective of an issue in order to resolve it. For example, the issue that you might face or any teacher might face is that students are not interested in the content of a lesson. So, by looking at it from the **product perspective**, if we can actually start to deconstruct issues and which students might be faced with in terms of the particular subject. For example, it might be that aspects of the subject are quite complex and difficult to understand. By looking at it from a **planning perspective**, we can then start to think about how we're actually working with students in terms of the subject content and for example, are we planning our lessons in very rigid and strict fixed ways rather than being much more diverse and flexible about it. By looking at the problem from a **people perspective**, we can actually start to understand the problem from other people's point of view, from our students' point of view and for example, it might be that the students don't actually see the need for certain aspects of a subject and in terms of their future goals or indeed their future life. And of course, we can look at a particular problem from a **potential perspective**, it could be for example that the students especially nowadays may be very much inclined to the use of technology in

terms of learning. And, this can help us as teachers to find new ways of working with students and therefore, helping them become more interested in terms of that learning.

Reframing for Resolving Difficult Controversies: Problem Statements

Problem Statements

- The first step is to define the problem statement
- At which level should we formulate the problem, and how widely?
- If we go too deeply and try to look for final causes of everything, we may reach conclusions that are not useful for the problem.
- Select a level where you have enough knowledge about the problem and then act in a timely way.
- As reflective practitioner we should always be working with ‘unsatisfactory’ understanding of the problem.
- The reflective practitioner works with ‘unsatisfactory’ because s/he has to combine the action with improving understanding.
- What method you actually use depends on the knowledge you have, the conditions and other factors (e.g., other people).
- You need to design your own cognitive strategy in each case.

The reflective practitioner has two levels:

- The action level
- The cognitive level
- Consequently, we need to combine knowledge from our own experience with knowledge from other people’s experience.

And in terms of cognitive strategies, the reflective practitioner has two key levels that we can work at. One is the **action level** which is very practical and one is the **cognitive level** where we're thinking and reflecting about a particular problem and what we need to do as reflective practitioners therefore, is to find a

way with a particular problem statement to combine the action level and the cognitive level together so that we have a very full and deepening understanding of the problem.

- Conceptual innovation
- The connections we have in our brains are related to the concepts we use.
- When we change our concepts, we create new areas of relations in our brains.

Therefore, our concepts can change, can evolve, and also can disappear with experience and with thinking through problems.

- Concepts are abstractions of situations – the patterns that emerge when we have seen many situations.
- Without concepts we cannot perceive reality.
- With concepts we can think more deeply about reality.

- With concepts we can:
 - Monitor situations.
 - Understand new situations by extensions of old ones.
 - Work to extend the concepts we are using.

So, conceptual innovation then is very much about working with concepts and their concepts in terms of how we're understanding situations, how we're perceiving situation, how we're monitoring situations and how we're understanding how others are working and experiencing those situations.

- The reflective practitioner needs to be comfortable with grasping and internalizing new concepts.

Reframing for Resolving Difficult Controversies: Concept-based Teaching and Learning

Concept-based Teaching and Learning

Concept –Based Teaching and Learning:

- Enables the ability to transfer knowledge
- Promotes emotional engagement and motivation

Concept –Based Teaching and Learning:

- Thinking which draws on critical, creative, reflective and conceptual abilities.
- Facilitates ‘synergistic thinking’ (factual and conceptual levels of thinking).
- Requires deeper intellectual processing as you relate facts to key concepts.

Concept –Based Teaching and Learning:

- Develops conceptual structures in the brain to relate new knowledge and to illuminate the connections and pattern of knowledge.

Concept –Based Teaching and Learning:

- Facilitates the transfer of knowledge at the conceptual level.
- Provides opportunities for personal meaning making through processes of thinking, creating and reflecting.

Concept –Based Teaching and Learning:

- Motivation for learning which recognizes that emotional engagement is essential to motivation.
- Increases motivation by thinking through the facts, through a personally engaging key concept.

Concept –Based Teaching and Learning:

- Values and respects the thinking of the individual by ‘drawing understandings from’ rather than ‘telling understanding to’.

Concept –Based Teaching and Learning:

- Encourages constructivist learning experiences that are relevant and important.

Concept –Based Teaching and Learning:

- Values collaborative thinking, discussions and problem-solving with the belief that the social construction of meaning leads to quality of teaching and motivation.

Reframing for Resolving Difficult Controversies: Benefits of Concept-based Teaching and Learning

Benefits of Concept-based Teaching and Learning

The benefits correspond directly with the profile of the reflective practitioner:

The reflective practitioner as enquirer:

- A teacher who questions, investigates, explores and discovers.

The reflective practitioner as knowledgeable:

- A teacher who has thorough grasp on their subject and a strong approach of its pedagogy.

The reflective practitioner as thinker:

- A teacher who ponders potential reasons, causes, effects and outcomes.

The reflective practitioner as communicator:

- A teacher who is able and confident enough to express to others their problems, issues and concern, and who is able to listen to feedback, advice and criticism.

The reflective practitioner as open-minded:

- A teacher who holds onto opinion, judgment and conclusion even with multiple sources of information.

The reflective practitioner as reflective:

- A teacher who sees reflection not as a mirror but as a window onto new insights, perspectives and horizons.

Reframing for Resolving Difficult Controversies: Pedagogical Shifts

Pedagogical Shifts

The reflective practitioner relies on:

- Synergistic thinking:
 - ...is essential to intellectual development.
 - ...is a cognitive interplay between the factual and cognitive levels of mental processing.
 - ...assumes that thinking our this factual / conceptual interaction can be shallow.

The Reflective Practitioner is able to:

- Transfer of knowledge and skills.
- Facts do not transfer-they are locked in time, place and situation.
- Knowledge transfer at the conceptual level as concepts, generations and principles
- The ability to use the conceptual level of thinking to relate new knowledge to prior knowledge.

The Reflective Practitioner is skilled in the:

- Social construction of meaning.
- Quality thinking is hard work.
- Reflective thinking requires collaboration to enhance thinking and problem-solving.
- Different minds working together provide support and generate new ideas and solutions.

These shifts in pedagogical thinking lead the reflective practitioner to a number of key components for reflection:

- Focus on the learner (students learning strategies, their interests, their developmental readiness).
- Focus on the teacher (teachers' anxiety, teaching styles, teachers' affective factors/personality).
- Focus on practical aspects (journal writing, surveys and questionnaires, observation, etc.).
- Focus on teachers' cognitive development (during small-scale classroom research projects, attending conferences and workshops).
- Focus on critical/conceptual aspects (teachers' ability to intellectualize, teachers' practice).
- Focus on moral/ethical parameters (listening to students, finding ways to support and challenge students).

Reframing for Strategic Creativity: Framing in Four Steps

- Reframing in 4 steps
- The approach to reframing is a four-step process

Step three is a questioning process and step four is the actual production of the new frame for your reflective practice.

- Reframing uncovers, challenges and overturns assumptions.
- This generates new ideas.
- Allows for innovation.

Reframing for Strategic Creativity: 3-Dimensional Instruction

- The goal of reflective practice is 3- dimensional instruction

This includes:

- Increasing conceptual understanding supported by factual knowledge and skills and the transfer of knowledge across multiple contexts.
- Facilitating student enquiry into important interdisciplinary and disciplinary topics and issues using one or two key concepts.
- Instruction and learning experiences that use concepts along with factual content to ensure synergistic thinking.
- The teacher encouraging group work to facilities shared enquiry collaboration, synergistic thinking and problem – solving across contexts.
- The teacher uses inductive teaching to draw the statement of conceptual understanding from students.
- The teacher focuses on student thinking and understanding and each teacher is aware of each student's ability to think synergistically.

Reframing for Strategic Creativity: Conceptual Learning

- What is conceptual learning?

- Conceptual learning is a process by which the reflective practitioner learns how to organize information in logical mental structures.
- Conceptual learning focuses on learning organizing principles the pockets in which the mind organizes facts into ideas.
- Conceptual learning is a catalyst for challenging the reflective practitioner to think at more advance levels.
- Focuses on broad categories of problems
- Fosters systematic observations
- Fosters understanding of relationship among events or conditions that impact situation
- Focus on teaching actions

Tanner's Model of Reasoning

- **Noticing**
- **Interpreting**
- **Responding**
- **Reflecting**

We can talk about a model to help us work conceptually in terms of our learning and the model is called **Tanner's Model of Reasoning**. It looks quite similar to the models of reflective practice that we've been speaking about. So, this is a very convenient model for us to think of in terms of conceptual learning. The first step is **noticing** or in other words describing the experience that we're having. The next step would be to **interpret** or in other words begin to analyze that experience. The third step is what is our **response** to the experience, the emotions, the feelings which are developed, and then this leads to **reflecting** or in other words how we're going to work to resolve those particular issues.

Reframing for Strategic Creativity: Epistemic Stances

Reframing for Strategic Creativity

Reflective practitioner can take different perspectives on their roles and in the discussion around reflective practice we talk about epistemic stances

Epistemic Stances.

- These are ways of knowing and reflection
- These processes are developmental
- The quality of reflection changes as we mature and new epistemic stances emerge

Categorization of Epistemic Stances

Epistemic stances can be categorized into 4 stages

- Absolute knowing
- Transitional knowing
- Independent knowing
- Contextual knowing

We can categorize epistemic stances into four key areas. The first key area is Absolute Knowing.

Two other areas are Transitional Knowing and Independent Knowing and the fourth is

Contextual Knowing. There are four different stances; we can take in terms of the reflective role. **Absolute knowing**

- The teacher who seeks to learn by receiving knowledge from others (experts)

Transitional knowing and independent knowing

- These are intermediate stages

The second and third stage is Transitional knowing and Independent Knowing. The intermediate stage is they move the reflective practitioner from the absolute knowing to contextual knowing. Transitional and

Independent Knowing are simply part way phases that the reflective practitioner is working with, beginning to take facts and figures from others and internalize understanding and begin to own particular skills and knowledge in relation to their own role in the classroom.

Contextual Knowing

- The teacher judges all information on the basis of evidence within context
- Highly critical in pursuing understanding
- Examine both data and own/ theirs' perceptions and values

These processes lead the reflective practitioner to Contextual Knowing, this is where real quality contributions too arose as teachers in the classroom begin to manifest themselves, where teachers exchanges and perspectives exchanged and used productively by reflective practitioner.

- Implications
- Ways of knowing impact the ability to reflect
- More sophisticated types of reflection lead to multiple solutions
- Absolute knowers only find and accept information from some authority.

Reframing for Strategic Creativity: Taxonomy of Reflecting Thought

Taxonomy of Reflective Thought

An aspect of reflective practice we can consider at this point is how we look at reflect thought, how we define reflective thought and, in this way, can begin to look at taxonomies of reflective thought.

Taxonomy

- To examine reflective thinking, researchers and theorists have developed taxonomies to explain qualitative aspects of reflection and have identified critical reflection as the deepest level of reflectivity.

Taxonomy of reflective thought

- For teachers critical reflection is the type of reflective thought most closely associated with Dewey's definition, explained earlier and involves reflection on the teaching practices as they relate to moral and ethical issues in society.

So, for teachers, the critical reflection is type of reflection which is most closely associated with models of teaching and reflection which we have looked at in the past. Some of the foundational models we are looking at for example Dewey and Schon, both define processes of reflection which taxonomies tend to reflect quite closely. Essentially taxonomies will look at aspects of teaching and aspect of knowing but within a contextual frame and that contextual frame can be the moral and ethical considerations, the teacher has to confront in the classroom.

- The taxonomy categorizes thinking into levels explaining the different qualitative aspects of the process.

Three levels of reflective thought:

- Non-reflective action
- Reflective action
- Premise reflection

Reframing for Strategic Creativity: The Ripples Model

The Ripples Model

- The Ripples model can be used by the reflective practitioner to consider how to consider how their teaching addresses certain questions:
- How do you enhance your learner desire to learn?
- How do you help learner develop ownership of the need to learn?

- Do you help learner learn by techniques such as practical activities, trial and error tasks and repetitive activities?

Other questions the Ripples Model can be used for in terms of the strategies that we are using for teaching? Are the learners learning through practical activities through processes where they can directly engage in knowledge and skill development? And learn inductively as well as deductively in the classroom.

- Do you ensure learners receive quick and useful feedback, both from you and from their peers?
- What activities do you provide where learners can make sense of what they have learned?

In the Ripples Model here we can see how these questions can come together in a particular taxonomy. It's called Ripples Model because it is like ripples when you throw stone in water. At the center is what is known wanting our needing aspect of reflective practice process which is focused as what the learners want for learning? What the students need in order to learn? This has a consequence of what we do with them, how we work with them. The strategies and approaches we are using for teaching in the classroom. From this doing in the classroom the reflective practitioner begins to make sense of teaching and learning, starts to create conceptual frames for the approaches which are being employed. From making sense of doing in the classroom, there is the system of feedback or in other words the reflective process that takes place where the information is brought back internalize by the teacher and therefore inform the next level which is teaching the actual process of developing the approaches and strategies. In the Ripples Model what's important to realize is the final stage, the final level is assessing the effect of our teaching. What is actually the benefit or outcome of what we are been doing with the students?

- This model is not a cycle- it is a progression
- Learning by doing is at the Centre
- True learning and understanding states from the center and radiates outwards like ripple on water encompassing other aspects of learning.